

A Persistent Ni(0)–Pentalene Complex with High Fluxionality: Does Stronger Anti-aromaticity Promote Metal–Ligand Interactions?

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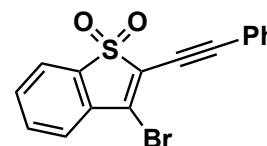
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1. Synthetic Methods and Characterization

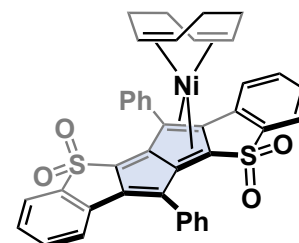
General. The names of the molecules in the manuscript do not necessarily follow IUPAC recommendations to guide the reader. Melting points (mp) were determined with a Yanaco MP-S3 instrument. ^1H and $^{13}\text{C}\{^1\text{H}\}$ nuclear magnetic resonance (NMR) spectra were recorded with a Bruker AVANCE III spectrometer (500 MHz for ^1H and 125 MHz for ^{13}C) in chloroform-*d* (CDCl_3), dichloromethane-*d*₂ (CD_2Cl_2), or 1,1,2,2-tetrachloroethane-*d*₂ (TCE-*d*₂). The chemical shifts in ^1H NMR spectra are reported in δ ppm using the residual protons of the solvents, CHCl_3 7.26 ppm, CH_2Cl_2 5.32 ppm, and benzene 7.16 ppm as an internal standard, and those in $^{13}\text{C}\{^1\text{H}\}$ NMR spectra are reported using the solvent signals of CD_2Cl_2 53.84 ppm as an internal standard. High-Resolution mass spectra (HRMS) were measured with a Bruker timsTOF (IMSQTOF) system with the ionization method of ESI or APCI. Thin-layer chromatography (TLC) was performed on plates coated with 0.25 mm thickness of silica-gel 60F₂₅₄ (Merck). Column chromatography was performed with a Biotage Isolera Selekt instrument using normal-phase silica-gel Wakosil® HC-N (FUJIFILM Wako Chemicals). Recycling preparative gel permeation chromatography (GPC) was performed with a LaboACE LC-5060 (Japan Analytical Industry) equipped with a polystyrene gel column (JAIGEL-2HR-40, Japan Analytical Industry) using CHCl_3 as an eluent. Anhydrous toluene and CH_2Cl_2 were purchased from Kanto Chemicals and further purified by Glass Contour Solvent Systems. Anhydrous benzene was purchased from FUJIFILM Wako Pure Chemical Corporation and further dried over molecular sieves 4A 1/16 purchased from FUJIFILM Wako Pure Chemical Corporation and degassed by freeze-pump-thaw cycles. Anhydrous 1,4-dioxane was purchased from FUJIFILM Wako Pure Chemical Corporation and were used as received. Deoxygenated anhydrous hexane, CH_2Cl_2 , and tetrahydrofuran (THF) were purchased from FUJIFILM Wako Pure Chemical Corporation and were used as received for experiments inside an argon-filled glovebox. The literature-known 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene were prepared according to the literature methods.^{S1}

3-Bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide. To a solution of 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene (890 mg, 2.85 mmol) in CH_2Cl_2 (50 mL) was added 3-chloroperoxybenzoic acid (*m*CPBA, contains *ca.* 30% water, 2.10 g, 3.0 equiv.) at room temperature, and stirred for 21 h at the same temperature. The mixture was quenched by saturated aqueous Na_2SO_3 solution and extracted with CH_2Cl_2 (15 mL $\times$ 3). The combined organic layers were dried over anhydrous Na_2SO_4 . After filtration, the filtrate was concentrated under reduced pressure, and the residue was purified by silica-gel column chromatography (hexane/ CH_2Cl_2 = 9/1 to 1/1 as eluent). The resulting solids were triturated with ethanol and then hexane to afford 668 mg of 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide as yellow crystals (1.94 mmol, 68% yield). Mp: 170.2–170.9 °C. ^1H NMR (500 MHz, CDCl_3): δ 7.74 (br dd, 1H), 7.69 (ddd, J = 7.6 Hz, 7.3 Hz, 1.2 Hz, 1H), 7.67–7.59 (m, 4H), 7.48–7.37 (m, 3H). $^{13}\text{C}\{^1\text{H}\}$ NMR (125 MHz, acetone-*d*₆): δ 136.67, 135.57, 133.43, 132.90, 131.50, 131.23, 130.08, 129.90, 128.43, 125.98, 122.40, 121.71, 106.89, 77.04. HRMS (ESI): m/z calcd. For $\text{C}_{16}\text{H}_{13}^{79}\text{BrNO}_2\text{S}$: 361.9845 ($[M+\text{NH}_4]^+$) and $\text{C}_{16}\text{H}_9^{79}\text{BrNaO}_2\text{S}$: 366.9399 ($[M+\text{Na}]^+$); found. 361.9845 and 366.9398.



(1,5-Cyclooctadiene)(4,8-diphenylpentaleno[2,1-*b*:5,4'-*b'*]bis[1]benzothiophene-1,1-dioxide)nickel ([Ni(cod)(1)]).

Two flame-dried 100 mL 2-neck flasks were charged with 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide (total 1.72 g, 5.00 mmol; 861 mg per flask) and Ni(cod)₂ (total 1.66 g, 6.02 mmol, 1.2 equiv.; 828 mg per flask) in an argon-filled glove box. After taking the flask out from the glovebox, anhydrous toluene (total 50 mL; 25 mL per flask) was added into the reaction vessel, and the reaction suspension was immediately heated to 80 °C for 17 h. After cooling to room temperature, the mixtures were combined and diluted with CH₂Cl₂, then filtered through a plug of cotton.

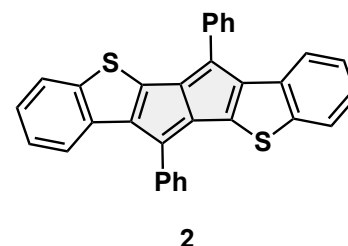


[Ni(cod)(1)]

After removal of the solvent under reduced pressure, the resulting dark green solids were subjected to silica-gel column chromatography (hexane only to CH₂Cl₂ only, *R_f* = 0.34 (hexane/ethyl acetate = 1/1)), and further purified by trituration with THF followed by the recrystallization from deoxidized CH₂Cl₂/THF in an argon-filled glovebox to give 16.6 mg of [Ni(cod)(1)] as dark green crystals (0.0238 mmol, 1% yield). Mp: no clear melting point was observed. The color of the compound irreversibly changed from dark green to brown at 173 °C without melting. ¹H NMR (500 MHz, CD₂Cl₂): δ 8.25 (br, 4H), 7.87 (d, *J* = 7.1 Hz, 2H), 7.65 (t, *J* = 7.4 Hz, 4H), 7.61–7.56 (m, 4H), 7.50–7.42 (m, 4H), 4.84 (td, *J* = 8.3 Hz, 5.5 Hz, 2H), 3.50 (td, *J* = 8.0 Hz, 3.9 Hz, 2H), 2.87–2.78 (m, 2H), 2.69–2.60 (m, 2H), 2.28–2.20 (m, 2H), 2.17–2.08 (m, 2H). ¹³C{¹H} NMR spectrum was not obtained due to its poor solubility. HRMS (ESI): *m/z* calcd. for C₄₀H₃₁NiO₄S₂: 697.1012 ([*M*+H]⁺) and C₄₀H₃₀NaNiO₄S₂: 719.0831 ([*M*+Na]⁺); found. 697.1009 and 719.0828.

4,8-Diphenylpentaleno[2,1-*b*:5,4'-*b'*]bis[1]benzothiophene (2). The synthesis

was conducted according to the reported procedure.^{S2} To four 20 mL microwave vials, 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene (total 1.25 g, 4.00 mmol; 313 mg per vial), Pd(OAc)₂ (total 90.7 mg, 0.40 mmol, 10 mol%; 22.6 mg per vial), Bu₄NOAc (total 1.22 g, 4.0 mmol, 1.0 equiv.; 305 mg per vial), Bu₄NI (total 1.48 g, 4.0 mmol, 1.0 equiv.; 369 mg per vial) and anhydrous 1,4-dioxane (total 40 mL; 10



2

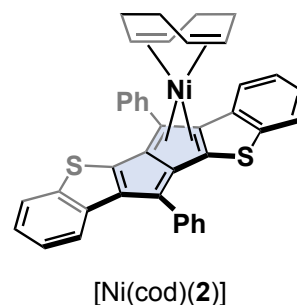
mL for each vial) were added separately, and these were stirred at 150 °C for 4 h. After cooling to room temperature, all the reaction mixtures were combined and diluted with CH₂Cl₂, filtered through celite and evaporated. The residual solids were triturated with CH₂Cl₂ and further purified by recrystallization (toluene, 100 °C) to give 186 mg of 2 as brown solids (0.398 mmol, 10% yield). Mp: >300 °C. ¹H NMR (500 MHz, CD₂Cl₂): δ 7.64 (dd, *J* = 8.1 Hz, 1.7 Hz, 4H), 7.57–7.49 (m, 8H), 7.28 (d, *J* = 8.3 Hz, 2H), 7.11 (t, *J* = 7.5 Hz, 2H), 7.03 (t, *J* = 7.3 Hz, 2H). ¹³C{¹H} NMR spectrum was not obtained due to its poor solubility. HRMS (APCI): *m/z* calcd. for C₃₂H₁₉S₂: 467.0923 ([*M*+H]⁺); found. 467.0922.

NMR observation of (1,5-Cyclooctadiene)(4,8-diphenylpentaleno[2,1-*b*:5,4'-*b'*]bis[1]benzothiophene)-nickel ([Ni(cod)(2)]). To a suspension of **2** (8.9 mg, 19 μ mol) in C₆D₆ (4.5 mL) in a vial, a

solution of Ni(cod)₂ in degassed benzene (40 mM, 0.50 mL, 1.0 equiv.) was added at room temperature in an argon-filled glovebox. The reaction mixture was stirred at the same temperature, and the portions of the mixture (0.60 mL each) were subjected to the ¹H NMR measurements after 1 h, 2 h, 4 h, and 8 h.

¹H NMR (500 MHz, CD₂Cl₂): δ 7.98–7.65 (br, 4H), 7.83 (d, *J* = 7.4 Hz, 2H), 7.47 (br, 6H), 7.27 (br, 2H), 7.22 (br, 2H), 4.06 (td, *J* = 8.6 Hz, 5.2 Hz, 2H), 3.16 (td, *J* = 7.8 Hz, 3.6 Hz, 2H), 2.32–2.21 (m, 2H), 2.20–2.08 (m, 2H), 1.95–1.82 (m, 4H), two protons

were not observed due to line broadening. ¹³C{¹H} NMR spectrum could not be obtained in satisfactory resolution due to its poor stability. HRMS (APCI): *m/z* calcd. for C₄₀H₃₀NiS₂: 632.1137 ([*M*]⁺) and C₄₀H₃₁NiS₂: 633.1215 ([*M*+H]⁺); found. 632.1131, 632.1200. Sample preparation for XAS measurements was summarized in Section 3.



2. X-Ray Crystallographic Analysis

General. Single-crystal X-ray diffraction data for [Ni(cod)(1)] and **2** were collected on a Rigaku XtaLAB AFC10 diffractometer equipped with FR-X generator, Varimax optics, and PILATUS 200K photon counting detector with MoK α radiation ($\lambda = 0.71073$ Å). The structure was solved by direct methods (SHELXT 2018/2)^{S3} and refined by the full-matrix least-squares on F^2 (SHELXL-2018/3)^{S4}. All hydrogen atoms were placed using AFIX instructions, while all non-hydrogen atoms were refined anisotropically. Detailed measurement conditions and crystal data for these compounds are described in the supplementary information, see Figure S1, Tables S1 and S2.

X-ray Crystallographic Analysis of ([Ni(cod)(1)]). Green needle-shaped single crystals were grown by slow diffusion of hexane into a solution of [Ni(cod)(1)] in CHCl₃. Intensity data were collected at 133 K on a Rigaku XtaLAB AFC10 diffractometer equipped with FR-X generator, Varimax optics, and PILATUS 200K photon counting detector with MoK α radiation ($\lambda = 0.71073$ Å). A total of 56249 reflections were measured with the maximum 2θ angle of 56.4°, of which 7611 were independent reflections ($R_{\text{int}} = 0.0260$). The crystal data are as follows: C₄₁H₃₁Cl₃NiO₄S₂; FW = 816.84, crystal size = 0.20 × 0.20 × 0.03 mm³, monoclinic, $P2_1/c$ (#14), $a = 13.2160(2)$ Å, $b = 17.4842(3)$ Å, $c = 14.9906(3)$ Å, $\beta = 96.722(2)^\circ$, $V = 3440.08(11)$ Å³, $Z = 4$, $D_c = 1.577$ g cm⁻³, $\mu = 0.964$ mm⁻¹, $R_1 = 0.0382$ ($I > 2\sigma(I)$), $wR_2 = 0.1040$ (all data), GOF = 1.039. CCDC 2451621 contains the supplementary crystallographic data for this compound. This data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at <https://www.ccdc.cam.ac.uk/structures/>.

X-ray Crystallographic Analysis of 2. Brown needle-shape single crystals were grown by slow diffusion of EtOH into a solution of **2** in THF. Intensity data were collected at 133 K on a Rigaku XtaLAB AFC10 diffractometer equipped with FR-X generator, Varimax optics, and PILATUS 200K photon counting detector with MoK α radiation ($\lambda = 0.71073$ Å). A total of 30407 reflections were measured with the maximum 2θ angle of 56.0°, of which 4574 were independent reflections ($R_{\text{int}} = 0.0194$). The crystal data are as follows: C₃₂H₁₈S₂; FW = 466.58, crystal size = 0.20 × 0.20 × 0.03 mm³, triclinic, $P-1$ (#2), $a = 7.3928(1)$ Å, $b = 11.1464(2)$ Å, $c = 13.2858(2)$ Å, $\alpha = 77.244(1)^\circ$, $\beta = 84.052(2)^\circ$, $\gamma = 89.254(2)^\circ$, $V = 1061.97(2)$ Å³, $Z = 2$, $D_c = 1.459$ g cm⁻³, $\mu = 0.272$ mm⁻¹, $R_1 = 0.0295$ ($I > 2\sigma(I)$), $wR_2 = 0.0828$ (all data), GOF = 1.040. CCDC 2451622 contains the supplementary crystallographic data for this compound. This data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at <https://www.ccdc.cam.ac.uk/structures/>.

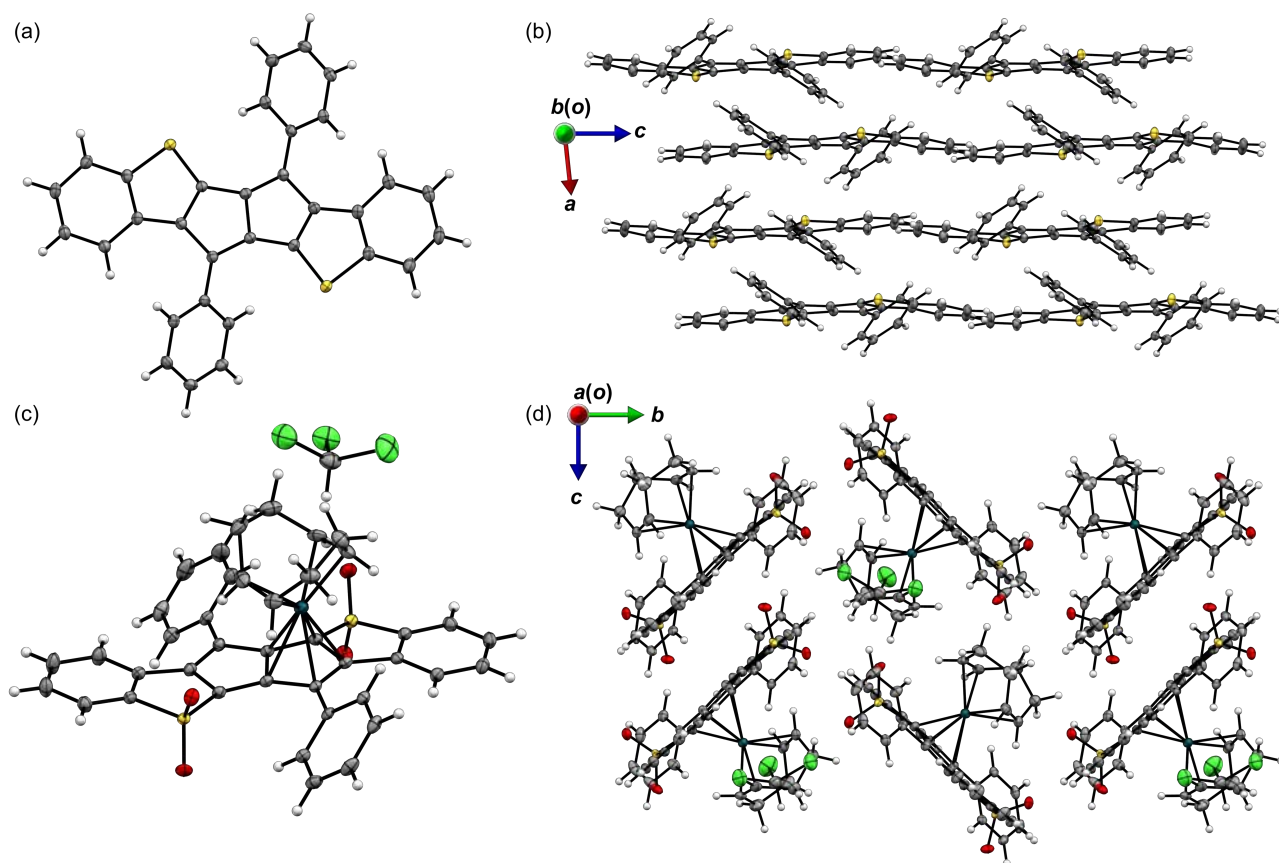
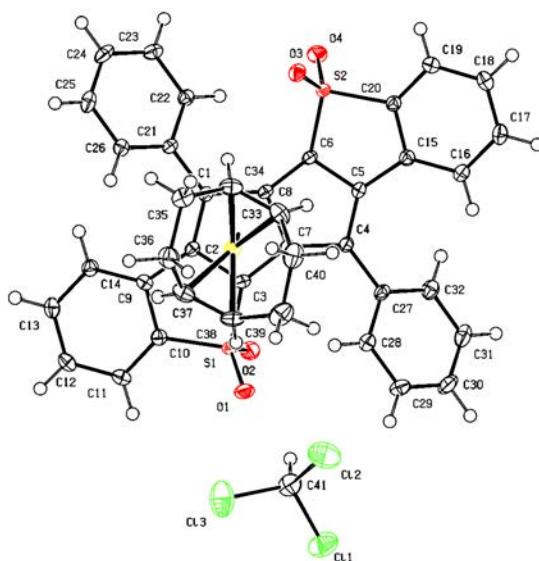
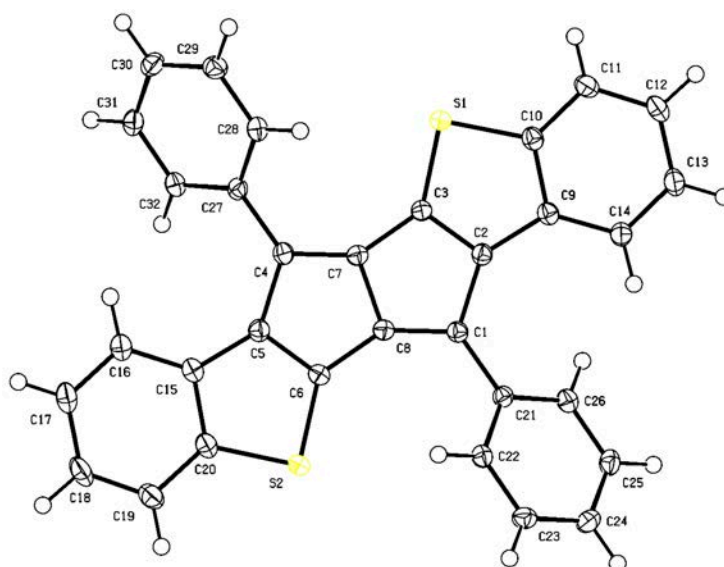


Figure S1. Crystal structures of **2** and $[\text{Ni}(\text{cod})(\mathbf{1})]$. Molecular structures are drawn in thermal ellipsoid plots (50% probability for thermal ellipsoids; grey: carbon; white: hydrogen; red: oxygen; yellow: sulfur; dark green: nickel; green: chlorine). (a, c) Asymmetric units. (c, d) Crystal packing structures.

Table S1. Selected bond lengths of [Ni(cod)(1)] determined by X-ray crystallographic analysis

Bond	Length / Å	Bond	Length / Å	Bond	Length / Å
Ni–C2	2.124(2)	C27–C32	1.398(3)	C15–C16	1.394(3)
Ni–C3	2.042(2)	C27–C28	1.393(3)	C14–C13	1.389(3)
Ni–C7	2.203(2)	C3–C7	1.404(3)	C11–C12	1.390(3)
Ni–C8	2.304(2)	C4–C7	1.446(3)	C26–C25	1.390(3)
Ni–C1	2.242(2)	C4–C5	1.384(3)	C13–C12	1.396(3)
Ni–C38	2.090(2)	C7–C8	1.459(3)	C19–C18	1.389(3)
Ni–C37	2.090(2)	C8–C1	1.468(3)	C23–C24	1.384(3)
Ni–C33	2.097(2)	C8–C6	1.379(3)	C25–C24	1.389(3)
Ni–C34	2.105(2)	C5–C15	1.465(3)	C37–C36	1.517(3)
S2–O3	1.445(2)	C5–C6	1.451(3)	C32–C31	1.389(3)
S2–O4	1.442(2)	C1–C21	1.471(3)	C28–C29	1.385(3)
S2–C20	1.772(2)	C9–C10	1.405(3)	C35–C34	1.489(4)
S2–C6	1.726(2)	C9–C14	1.393(3)	C35–C36	1.529(4)
S1–O1	1.439(2)	C22–C21	1.395(3)	C31–C30	1.375(4)
S1–O2	1.435(2)	C22–C23	1.393(3)	C33–C34	1.366(3)
S1–C3	1.759(2)	C10–C11	1.386(3)	C33–C40	1.499(3)
S1–C10	1.761(2)	C20–C15	1.403(3)	C18–C17	1.389(4)
C2–C3	1.470(3)	C20–C19	1.379(3)	C16–C17	1.391(4)
C2–C1	1.408(3)	C21–C26	1.397(3)	C30–C29	1.380(4)
C2–C9	1.469(3)	C38–C37	1.356(3)	C40–C39	1.529(4)
C27–C4	1.468(3)	C38–C39	1.506(4)		

Table S2. Selected bond lengths (in Å) of **2** determined by X-ray crystallographic analysis



Bond	Length / Å	Bond	Length / Å	Bond	Length / Å
S1–C3	1.707(1)	C27–C28	1.399(2)	C15–C20	1.416(2)
S1–C10	1.740(1)	C27–C32	1.401(2)	C15–C16	1.405(2)
S2–C6	1.715(1)	C7–C4	1.368(2)	C28–C29	1.385(2)
S2–C20	1.750(1)	C7–C3	1.450(2)	C11–C12	1.380(2)
C8–C7	1.455(2)	C3–C2	1.383(2)	C20–C19	1.397(2)
C8–C1	1.368(2)	C2–C9	1.439(2)	C14–C13	1.382(2)
C8–C6	1.453(2)	C2–C1	1.494(2)	C30–C31	1.388(2)
C21–C26	1.398(2)	C9–C10	1.418(2)	C30–C29	1.389(2)
C21–C1	1.465(2)	C9–C14	1.405(2)	C32–C31	1.386(2)
C21–C22	1.399(2)	C26–C25	1.384(2)	C12–C13	1.396(2)
C5–C4	1.485(2)	C23–C24	1.387(2)	C16–C17	1.381(2)
C5–C6	1.380(2)	C23–C22	1.386(2)	C19–C18	1.382(2)
C5–C15	1.433(2)	C24–C25	1.388(2)	C17–C18	1.394(2)
C27–C4	1.464(2)	C10–C11	1.396(2)		

3. Electrochemical and Photophysical Properties

Methods. Cyclic voltammetry (CV) measurements were recorded with an ALS/chi-660E electrochemical analyzer (BAS). The CV cell consisted of a glassy carbon electrode (ϕ 3 mm), a Pt wire counter electrode and an Ag/AgNO₃ reference electrode. The measurements were carried out under an argon atmosphere using a 1.0 mM sample solution in deoxygenated anhydrous CH₂Cl₂ containing 0.1 M tetrabutylammonium hexafluorophosphate ([Bu₄N][PF₆]) as a supporting electrolyte. The redox potentials were calibrated with a ferrocene/ferrocenium ion couple (Fc/Fc⁺). The result is shown in Figure S2a, and the data are summarized in Table S3.

Ultraviolet–visible–near-infrared (UV/vis/NIR) absorption spectra were measured with a SHIMADZU UV-3600i Plus spectrometer. Dilute sample solutions were prepared in spectroscopic-grade CH₂Cl₂ and placed in a standard 10 mm $\times$ 10 mm quartz cuvette. The solvent (CH₂Cl₂) was purified by vacuum distillation over calcium hydride and stored with 3 Å molecular sieves in an argon-filled glovebox. The result is shown in Figure S2b.

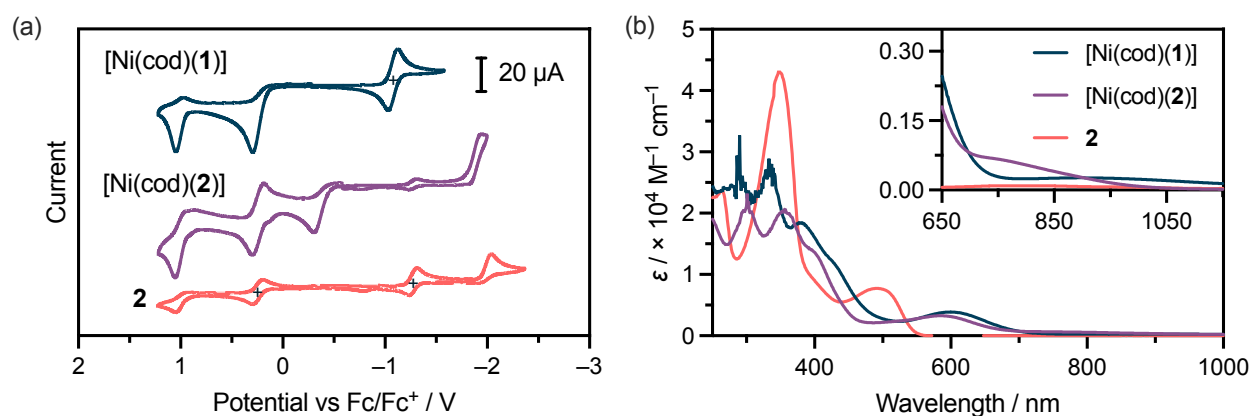


Figure S2. Cyclic voltammograms (a) and UV/Vis/NIR electronic absorption spectra (b) of [Ni(cod)(1)], [Ni(cod)(2)], and 2. Cyclic voltammograms were recorded at a scan rate of 0.05 V s⁻¹ in CH₂Cl₂ using [Bu₄N][PF₆] (0.1 M) as supporting electrolyte. All potentials are referenced against the Fc/Fc⁺ couple. The UV/Vis/NIR electronic absorption spectra of [Ni(cod)(1)] (1.76×10^{-4} M), [Ni(cod)(2)] (1.90×10^{-4} M), and 2 (8.74×10^{-5} M) were recorded in CH₂Cl₂ at room temperature. Inset: absorption spectra at 650–1200 nm region recorded at high concentration of [Ni(cod)(1)] (1.75×10^{-3} M), [Ni(cod)(2)] (1.40×10^{-3} M), and 2 (5.84×10^{-4} M) in CH₂Cl₂.

Table S3. Electrochemical data of [Ni(cod)(**1**)], [Ni(cod)(**2**)], and **2** in CH₂Cl₂

Cmpd.	$E_{\text{pa}} / \text{V}^{a,b}$ ($E_{1/2,\text{ox}} / \text{V}^{a,c}$)	$E_{\text{pc}} / \text{V}^{a,d}$ ($E_{1/2,\text{red}} / \text{V}^{a,c}$)	HOMO / eV ^e	LUMO / eV ^e	$\Delta E_{\text{H-L}} / \text{eV}^f$
[Ni(cod)(1)]	+0.30, +1.04	−1.12 (−1.08)	−4.96	−3.79	1.17
[Ni(cod)(2)]	−0.30	−1.94	−4.36	−3.00	1.36
2	+0.29 (+0.25), +1.04	−1.31 (−1.27), −2.03	−4.97	−3.60	1.37

^aIn CH₂Cl₂. ^bAnodic peak potential vs Fc/Fc⁺. ^cHalf-wave potential vs Fc/Fc⁺. ^dCathodic peak potential vs Fc/Fc⁺.

^eEstimated using the values of the onset potentials of the cyclic voltammograms based on the reported method.^{S1} ^fHOMO–LUMO energy gap estimated using the difference between the onset potentials.

4. Quantum Chemical Calculations

4.1. Computational Methods

The geometry optimizations were carried out using density functional theory (DFT) with B3LYP^{S5} or ω B97X-D^{S6} functionals and the 6-31G(d)^{S7-S11} for C, H, O, S, and LanL2DZ^{S12} for Ni basis set implemented in the Gaussian 16 Revision B.01^{S13} suite of programs with default thresholds. Vibrational frequency analyses at the same level of theory confirmed that the optimized geometries are energy minima, as indicated by the absence of imaginary frequencies. The computed geometries are given in **6. Appendix** section.

Harmonic oscillator model of aromaticity (HOMA) indices of the eight-membered ring periphery of pentalene frameworks were calculated from the bond lengths in the optimized geometries using the following equation:^{S14}

$$\text{HOMA} = 1 - \frac{\alpha}{n} \sum_i^n \left(R_{\text{opt}} - R_i \right)^2$$

where $\alpha = 257.7$ is an empirical normalization constant, R_{opt} and R_i represent the averaged bond length (1.388 Å) and i -th bond length (Å). The calculated HOMA indices were summarized in Table S5.

¹H chemical shifts were computed at B3LYP/6-311+G(2d,p) level of theory for the optimized geometries of **R**, **TS1**, **Int**, and **TS2** of [Ni(cod)(1)], employing the integral equation formalism variant of the polarizable continuum model (IEFPCM) to simulate the dichloromethane environment. The computed shifts were calibrated using a linear scaling procedure (intercept = -1.0553, slope = 31.8830) to enable direct comparison with experimental values.^{S15} ¹H NMR spectra were then simulated by broadening the scaled chemical shifts with a Lorentzian function (FWHM = 0.01 ppm).

For the Gibbs free energy calculations, the electronic energies (E_e) were evaluated at the significantly more accurate ω B97X-D/def2-TZVP^{S16} level of theory with a triple-zeta basis set, since E_e dominates the Gibbs free energy and, therefore, it is highly sensitive to the quality of the basis set.^{S17} The correlations to free energy were evaluated at the same level of geometry optimization. The results are partly summarized in Figures S3 and S10.

Nucleus-independent chemical shift (NICS)^{S18,S19} computations were performed at B97-2/def2-TZVP^{S20} level of theory using optimized geometries. For NICS(1)_{zz} computations, dummy atoms were placed 1 Å above the center of monocyclic rings. For 2D-NICS(1)_{zz} maps, a 2D grid of dummy atoms was generated 1 Å above the *XY* Cartesian plane with an interval of 0.2 Å. Input file preparation and result analysis for NICS calculations were carried out using the py.Aroma 4 program (version 4.2).^{S21}

The electron density of delocalized bonds (EDDB)^{S22} was computed using ω B97X-D^{S6} functionals and the 6-31G(d)^{S7-S11} (for C, H, O, S), and LanL2DZ^{S12} (for Ni) basis sets to study π electron delocalization along pentalene periphery. The results were summarized in Table S6 and Figure S9.

Natural atomic orbital analysis was carried out with the NBO 7.0 program^{S23} using ω B97X-D functional with 6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) basis sets implemented in the Gaussian 16 suite of programs. The NAOMO keyword is used to generate the molecular orbital coefficient matrix in the NAO basis for examining the Ni-pentalene orbital interactions (Table S7).

To assess the coordinative interactions between Ni and the ligand of interest, the interaction energy (ΔE_{int}) was decomposed into the following five components; ΔE_{elstat} , ΔE_{Pauli} , ΔE_{disp} , ΔE_{pol} , and ΔE_{CT} , which signify the permanent electrostatics, Pauli repulsion, dispersion, polarization, and charge-transfer (CT) terms, respectively.

$$\Delta E_{\text{int}} = \Delta E_{\text{Frozen}} + \Delta E_{\text{pol}} + \Delta E_{\text{CT}}$$

$$\Delta E_{\text{Frozen}} = \Delta E_{\text{elstat}} + \Delta E_{\text{Pauli}} + \Delta E_{\text{disp}}$$

The energy decomposition analysis (EDA) was carried out using the second-generation energy decomposition analysis based on the absolutely localized molecular orbitals (ALMO-EDA)^{S24} method at the ω B97X-D/def2-TZVP level using the Q-Chem 5.0.0.^{S25} The CT term was further decomposed to forward CT (ΔE_{CTf}) and backward CT (ΔE_{CTb}) terms by variational forward-backward charge-transfer analysis (CTA-VFB)^{S26} method using Q-Chem 5.3.2. The geometry optimization and vibrational frequency calculations for each one-way CT state (forward and backward) were carried out at the ω B97X-D/def2-SVP level in gas phase. Based on the calculated one-way CT coordinates, a single-point calculation for CTA-VFB analysis was performed at the ω B97X-D/def2-TZVP level. Examples of input file descriptions for EDA and CTA-VFB analyses are given in Methods 1 and 2, respectively. The results thus obtained were summarized in Figures 6, S12, and Table S8.

For the characterization of the fluxional behavior of [Ni(cod)(1)], geometry optimizations, frequency analyses, and IRC calculations were performed at ω B97X-D/6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) level of theory, electronic energies were evaluated at ω B97X-D/def2-TZVP level of theory. Saddle points corresponding to transition states **TS1** and **TS2** were optimized using QST3 algorithm implemented in Gaussian 16.^{S13} Each structure exhibited a single imaginary frequency, which was confirmed to correspond to the expected reaction coordinate by IRC calculation^{S27} (Figure S13) and vibrational mode analysis (Figure S14). IRC computations were performed using a local quadratic approximation (LQA)^{S28,S29} for the predictor step.

4.2. Comparison of Geometries and Thermodynamic Stability between [Ni(cod)(1)] and [Ni(cod)(2)]

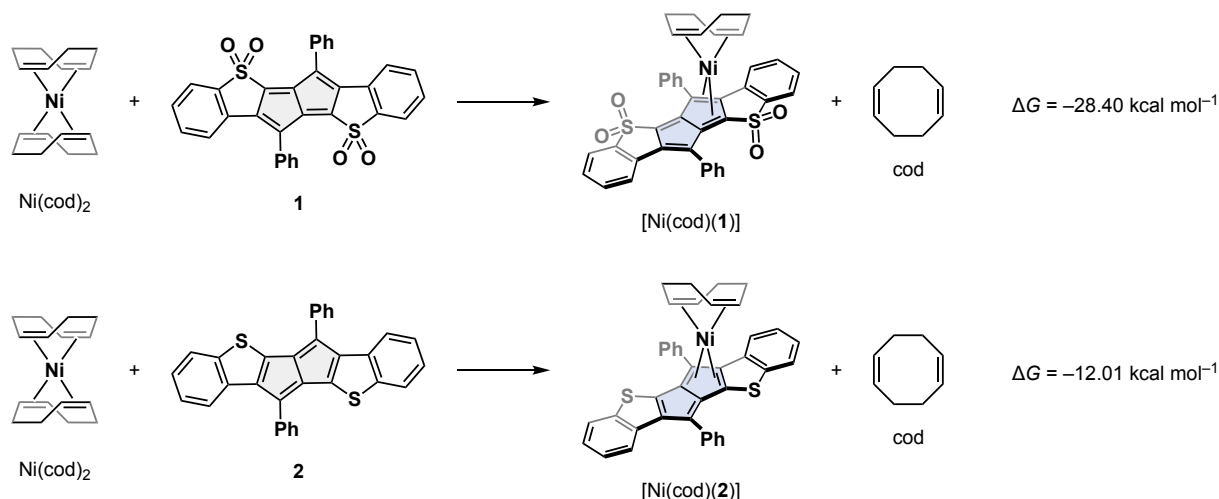


Figure S3. Comparison of thermodynamic stability of [Ni(cod)(1)] and [Ni(cod)(2)] based on the isodesmic reactions of ligand exchange between Ni(cod)_2 and pentalene **1** or **2** at ω B97X-D/def2-TZVP level of theory.

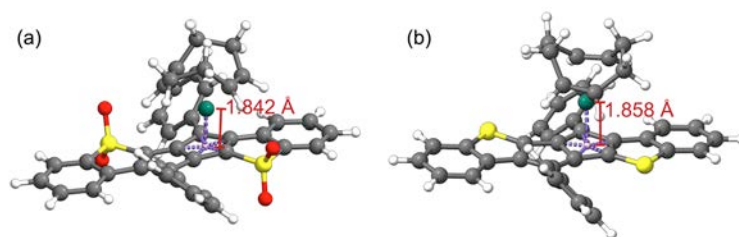
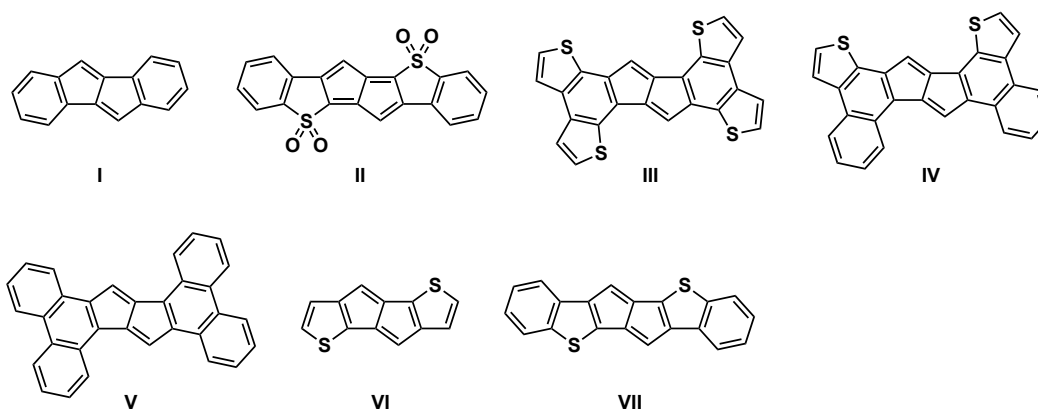


Figure S4. Optimized geometries and the distance between Ni and the ring center of the coordinated five-membered ring for [Ni(cod)(**1**)] (a) and [Ni(cod)(**2**)] (b).

4.3. Correlation of LUMO Energy Levels and Paratropicity Strengths of a Series of Pentalene Derivatives

Table S4. LUMO energy levels and NICS(1)_{zz} values for arene-fused pentalenes **I–VII**



Compound	I	II	III	IV	V	VI	VII
Kohn–Sham LUMO level / eV ^a	−2.27	−3.82	−2.66	−2.72	−2.71	−2.51	−2.77
NICS(1) _{zz} / ppm ^b	25.8	42.7	45.3	51.5	55.2	56.5	71.1

^aCalculated at the B3LYP/6-31G(d) level of theory. ^bCalculated at the B3LYP/6-311+G(2d,p)//B3LYP/6-31G(d) level of theory.

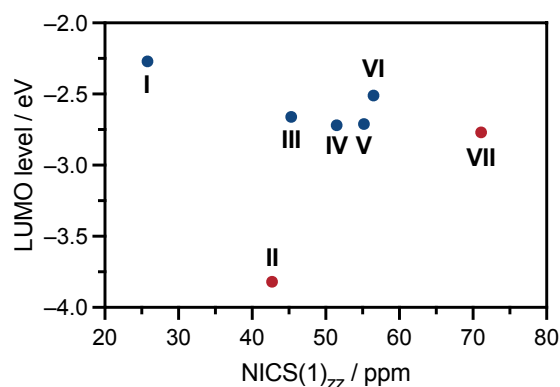


Figure S5. Plot of Kohn–Sham LUMO energy levels against NICS(1)_{zz} for fused pentalenes **I–VII**. **II** and **VII**, which share the same backbone as **1** and **2**, are marked in red.

4.4. Bond Alternation of Pentalene Cores in **1**, **2**, [Ni(cod)(**1**)], and [Ni(cod)(**2**)]

Two isomers of fused pentalene derivatives can be considered, differing in the fusion site at either double bonds (*a/a'*) or single bonds (*b/b'*) (Supplementary Fig. 6a). To evaluate the bond alternation pattern in the pentalene cores, computational investigations were carried out for compounds **1** and **2**, their bond-shifted isomers **1'** and **2'**, and the corresponding Ni complexes. Geometry optimizations of **1**, **2**, **1'**, **2'**, and the corresponding Ni complexes were performed at the ω B97X-D/6-31G(d) for C, H, O, and S and LanL2DZ for Ni, and the results were summarized in Table S5.

X-ray crystallographic data confirmed the solid-state geometries of [Ni(cod)(**1**)] and **2**, consistent with previous bond-length analyses. In the absence of crystallographic data for **1** and [Ni(cod)(**2**)], their gas-phase geometries were determined from computed electronic energies (E_e ; Figure S6b). Compound **1** was calculated to be more stable than **1'** by 1.13 kcal mol⁻¹, and [Ni(cod)(**2**)] was more stable than [Ni(cod)(**2'**)] by 1.91 kcal mol⁻¹. These results are consistent with the single-crystal geometries. Therefore, the discussion in this study is based on the most stable isomers: **1**, **2**, and their corresponding complexes [Ni(cod)(**1**)] and [Ni(cod)(**2**)].

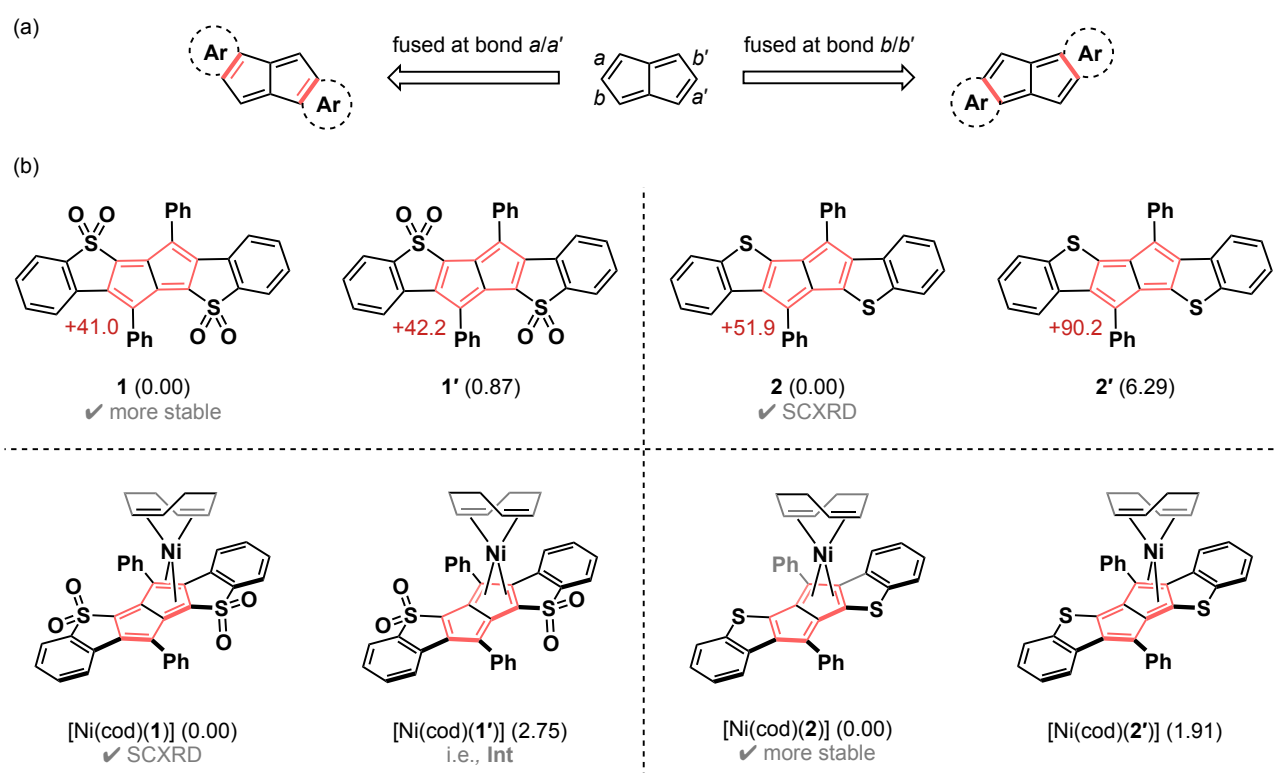
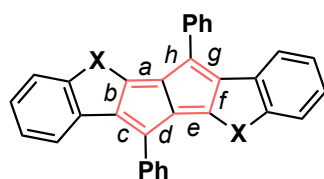
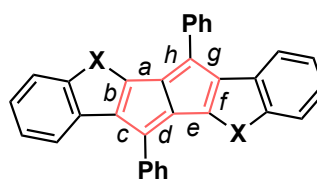


Figure S6. (a) Possible fusion pattern on arene-fused pentalenes. (b) Structures of computed geometries. [Ni(cod)(**1'**)] is identical to **Int** in Figure 7. Values in parentheses show relative electronic energies (E_e in kcal mol⁻¹) to the investigated geometries calculated at ω B97X-D/6-31G(d) for C, H, O, and S and LanL2DZ for Ni. NICS(1)_{zz} values of **1**, **1'**, **2** and **2'** at the pentalene five-membered ring centers calculated at B97-2/def2-TZVP// ω B97X-D/6-31G(d) are marked in red.

Table S5. Selected bond lengths (in Å) of the optimized geometries for [Ni(cod)(**1**)], [Ni(cod)(**1'**)], [Ni(cod)(**2**)], and [Ni(cod)(**2'**)]



X = SO₂: 1, [Ni(cod)(1)]
S: 2', [Ni(cod)(2')]



X = SO₂: 1', [Ni(cod)(1')]
S: 2, [Ni(cod)(2)]

Bond	1	1'	[Ni(cod)(1)] ^a	[Ni(cod)(1')] ^a	2	2'	[Ni(cod)(2)] ^a	[Ni(cod)(2')] ^a
<i>a</i>	1.353	1.450	1.372	1.423	1.454	1.380	1.435	1.399
<i>b</i>	1.490	1.369	1.442	1.388	1.376	1.460	1.389	1.418
<i>c</i>	1.376	1.501	1.386	1.439	1.486	1.385	1.459	1.420
<i>d</i>	1.479	1.379	1.446	1.389	1.363	1.451	1.377	1.410
<i>e</i>	1.353	1.450	1.415	1.459	1.454	1.380	1.456	1.438
<i>f</i>	1.490	1.369	1.480	1.407	1.376	1.460	1.396	1.437
<i>g</i>	1.376	1.501	1.387	1.467	1.486	1.385	1.488	1.437
<i>h</i>	1.479	1.379	1.472	1.420	1.363	1.451	1.427	1.462
HOMA ^b	−0.295	−0.096	0.266	0.468	0.158	0.399	0.238	0.505

^aNi coordinates to the five-membered ring associated with bonds *e*–*h*. ^bThe values of harmonic oscillator model of aromaticity.

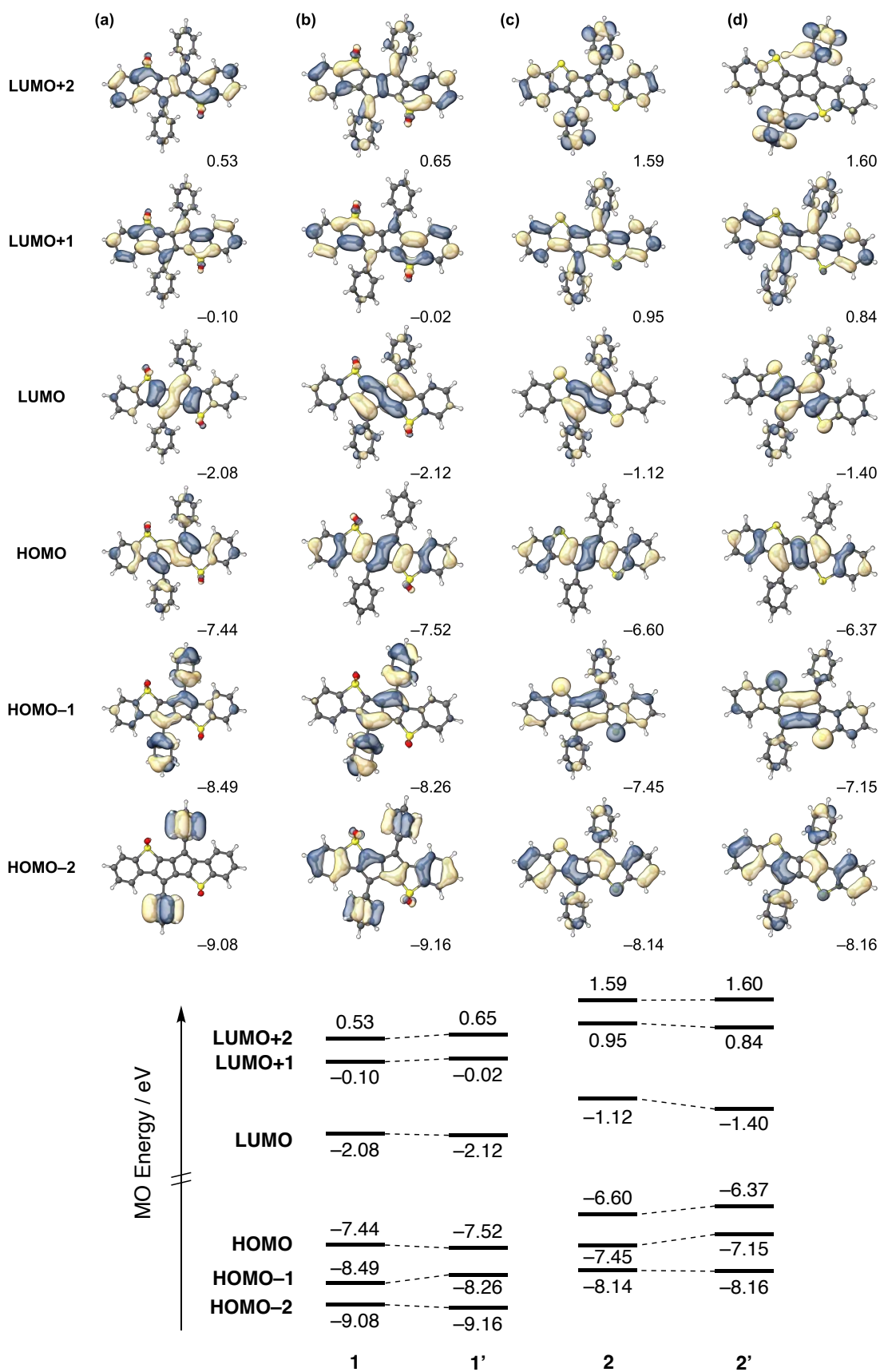


Figure S7. Kohn–Sham molecular orbitals and their orbital energy levels (isovalue = 0.03) of (a) **1**, (b) **1'**, (c) **2** and (d) **2'** calculated using ω B97X-D functionals with 6-31G(d) (for C, H, O, S) basis sets.

4.5. Change of (Anti)aromaticity of the Pentalene Substructure upon Ni-Coordination

To gain insight into the electronic structural changes upon complexation between Ni and pentalene, molecular orbital (MO) analyses were first carried out. Figure S8 shows the frontier and nearby orbitals of **1**, **2**, and their corresponding Ni complexes, [Ni(cod)(**1**)] and [Ni(cod)(**2**)]. In both complexes, contributions from the Ni 3d orbitals and the π orbitals of pentalene were observed.

To evaluate the extent of antiaromaticity alleviation of the pentalene moiety upon complex formation, the NICS(1)_{zz} values of the five-membered rings constituting the pentalene ligands were estimated, focusing on the ring farther from the Ni center. In [Ni(cod)(**2**)], the rings coordinated and uncoordinated to Ni exhibited large positive values of +6.8 and +4.3, respectively, whereas in [Ni(cod)(**1**)], the corresponding values were +3.5 and -2.0, indicating that the paratropic character of pentalene is more effectively suppressed in **1**, resulting in a slightly negative NICS value.

Considering the possible emergence of three-dimensional (3D) aromaticity similar to that known for metal-cyclopentadienyl complexes, further analysis was conducted based on the optimized geometry of [Ni(cod)(**1**)]. Natural bond orbital (NBO) analyses on [Ni(cod)(**1**)] were performed using ω B97X-D functional with 6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) basis sets. Based on the orbital composition analysis using natural atom orbital (NAO) basis, the molecular orbitals exhibiting significant bonding interactions between the Ni 3d orbital and C 2p orbitals of the five-membered ring coordinated to Ni were identified as HOMO-1, HOMO-8, and HOMO-41 (Figure S11, Table S7). These findings suggest that [Ni(cod)(**1**)] may possess 3D aromaticity arising from six interstitial electrons involving the Ni 3d and pentalene π orbitals.

To examine the manifestation of aromaticity upon complexation from the viewpoint of stabilization energy, dimethylated pentalene derivatives **1**^{Me} and **2**^{Me}, together with their structurally relevant nonaromatic hydrocarbons, the bis(pentafulvene) isomers **1**^{CH2} and **2**^{CH2}, were designed as model compounds of **1** and **2**. The aromatic stabilization in the corresponding Ni complexes was estimated by comparing the enthalpy changes (ΔH) of the homodesmotic reactions in which [Ni(cod)(**1**^{CH2})] or [Ni(cod)(**2**^{CH2})] is converted to [Ni(cod)(**1**^{Me})] or [Ni(cod)(**2**^{Me})] through ligand exchange with **1**^{Me} or **2**^{Me}, respectively. Geometry optimizations were performed at the ω B97X-D/6-31G(d) (for C, H, O, and S) and LanL2DZ (for Ni) levels of theory. The molecular structures of the model compounds and the calculated ΔH values are shown in Figure S10. Upon geometry optimization of [Ni(cod)(**1**^{CH2})] and [Ni(cod)(**2**^{CH2})], it was found that the structures in which the Ni atom is η^2 -coordinated to different C=C bonds are more stable. Therefore, the molecular structures illustrated in Figure S10 correspond to these more stable isomers. As a result, the calculated ΔH values are negative in both cases, indicating that the Ni-pentalene complexes are more stable than the corresponding Ni-pentafulvene complexes. Moreover, the degree of stabilization is greater for **1**^{Me} ($\Delta H = -10.54$ kcal mol⁻¹) than for **2**^{Me} ($\Delta H = -3.50$ kcal mol⁻¹), suggesting a larger contribution of aromatic stabilization in the [Ni(cod)(**1**)]. This result is consistent with the NICS(1)_{zz} values calculated for [Ni(cod)(**1**)] and [Ni(cod)(**2**)]; while the antiaromatic character of pentalene **2** is not completely compensated upon complexation with Ni in [Ni(cod)(**2**)], the aromatic character arising from six interstitial electrons in [Ni(cod)(**1**)] is supported also from an energetic standpoint.

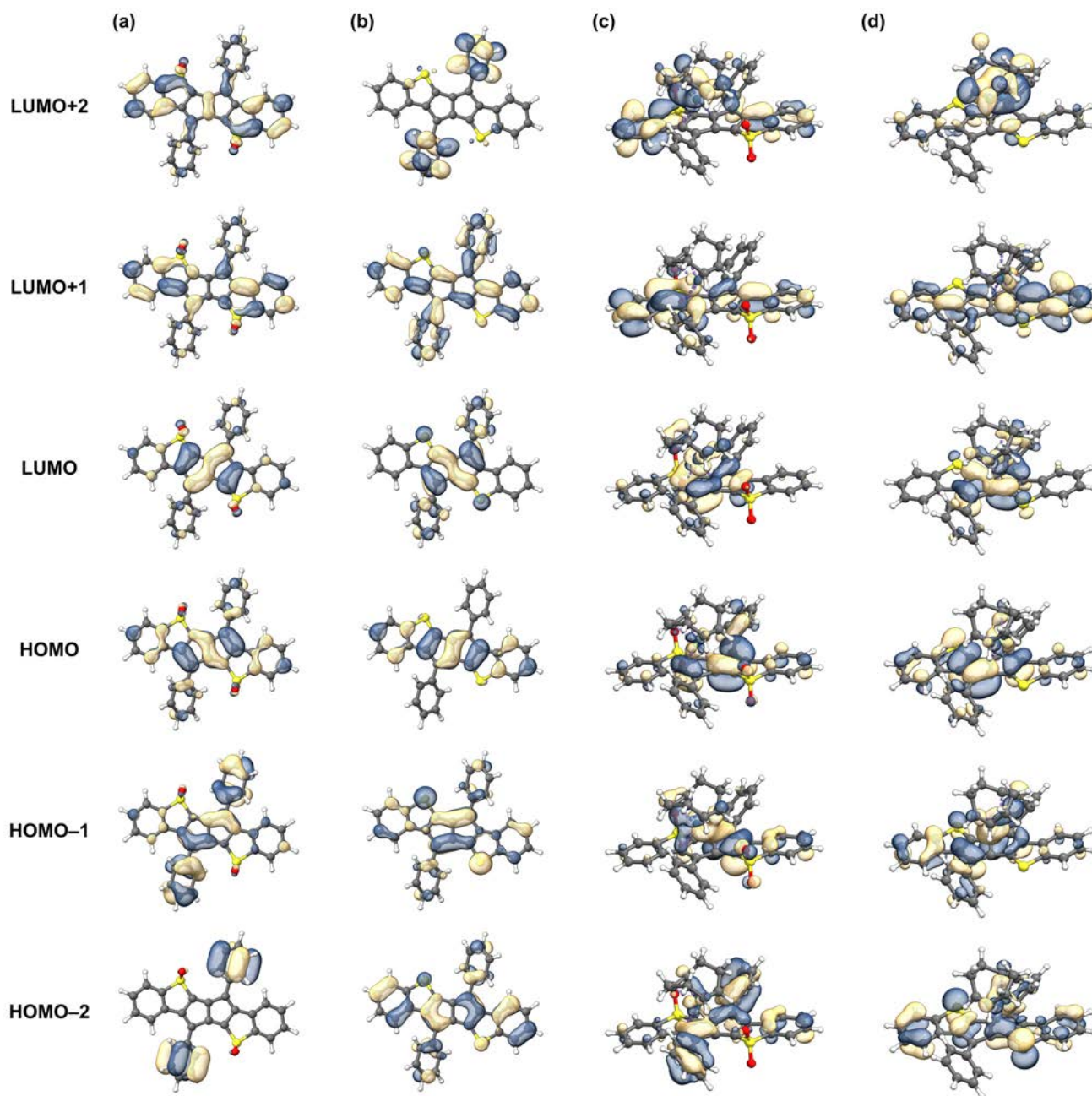


Figure S8. Kohn–Sham molecular orbitals (isovalue = 0.03) of (a) **1**, (b) **2**, (c) [Ni(cod)(**1**)], and (d) [Ni(cod)(**2**)] calculated using B3LYP (for **1** and **2**) and ω B97X-D (for [Ni(cod)(**1**)] and [Ni(cod)(**2**)]) functionals with 6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) basis sets.

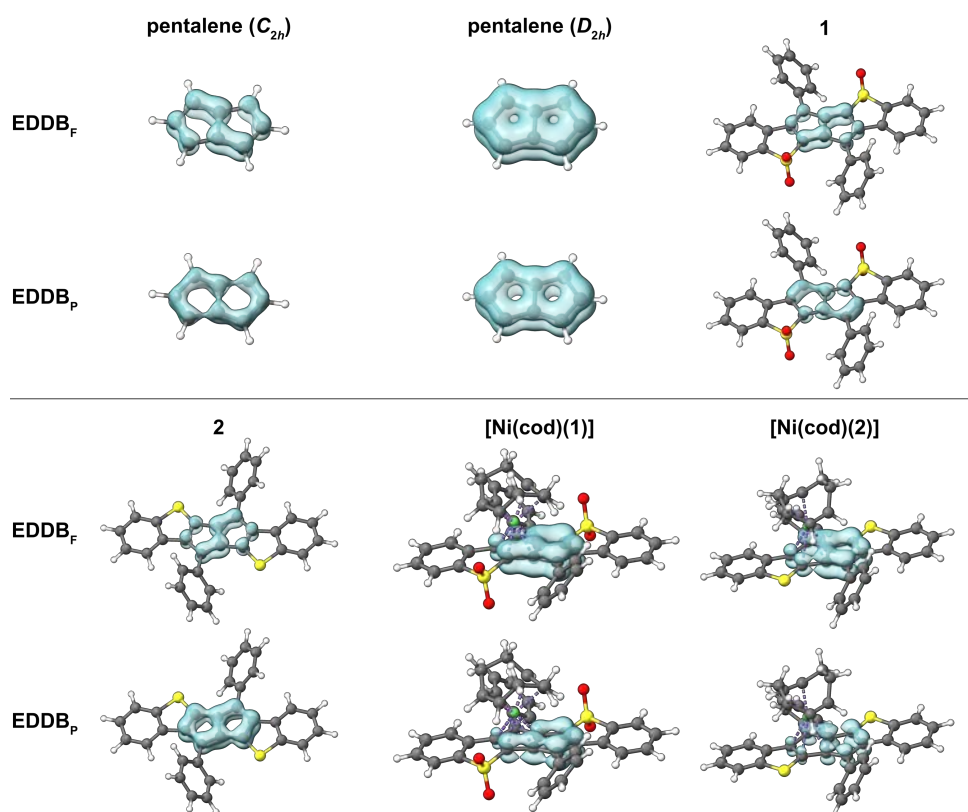
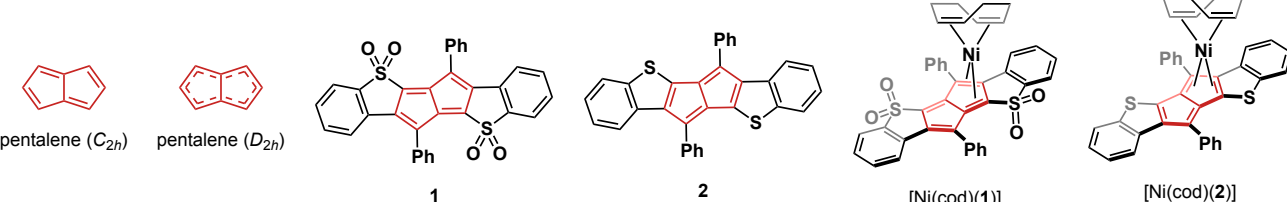


Figure S9. Representation of the EDDB_F and EDDB_P in pristine pentalene (in C_{2h} and D_{2h} symmetry), **1**, **2**, [Ni(cod)(**1**)], and [Ni(cod)(**2**)] (isovalue = 0.002 for EDDB_P of pentalene (C_{2h}), **1**, and **2**; isovalue = 0.01 for others).

Table S6. Population of total delocalized electrons and delocalized π -electrons over pentalene framework in pristine pentalene (in C_{2h} and D_{2h} symmetry), **1**, **2**, [Ni(cod)(**1**)], and [Ni(cod)(**2**)] estimated by EDDB method^a

				
	EDDB _F ^{b,d}	π -EDDB _F ^{c,d}	EDDB _P ^{b,e}	π -EDDB _P ^{c,e}
	(e ⁻ /atom)	(e ⁻ /atom)	(e ⁻ /atom)	(e ⁻ /atom)
pentalene (C_{2h})	2.3317 (0.2915)	1.8866 (0.3144)	0.3444 (0.0431)	0.1649 (0.0206)
pentalene (D_{2h})	7.5222 (0.9403)	7.0531 (0.8816)	5.2123 (0.6515)	4.9828 (0.6229)
1	1.7692 (0.2212)	1.3111 (0.1639)	0.3040 (0.0380)	0.1510 (0.0189)
2	1.8670 (0.2334)	1.4312 (0.1789)	0.5599 (0.0700)	0.3764 (0.0471)
[Ni(cod)(1)]	3.4236 (0.4279)	2.9373 (0.3672)	2.1193 (0.2649)	1.9188 (0.2399)
[Ni(cod)(2)]	2.8873 (0.3609)	2.4479 (0.3060)	1.7533 (0.1303)	1.5711 (0.1964)

^aCalculated at the ω B97X-D/6-31G(d) level of theory using the optimized geometries at the same level of theory. ^bEDDB indices for all electrons. ^cEDDB indices for π -electrons only. ^dComputed for the pentalene fragment, taking into account the cross-ring delocalization effects. ^eComputed along the specified path, i.e., pentalene fragment, without taking into account the cross-ring delocalization effects.

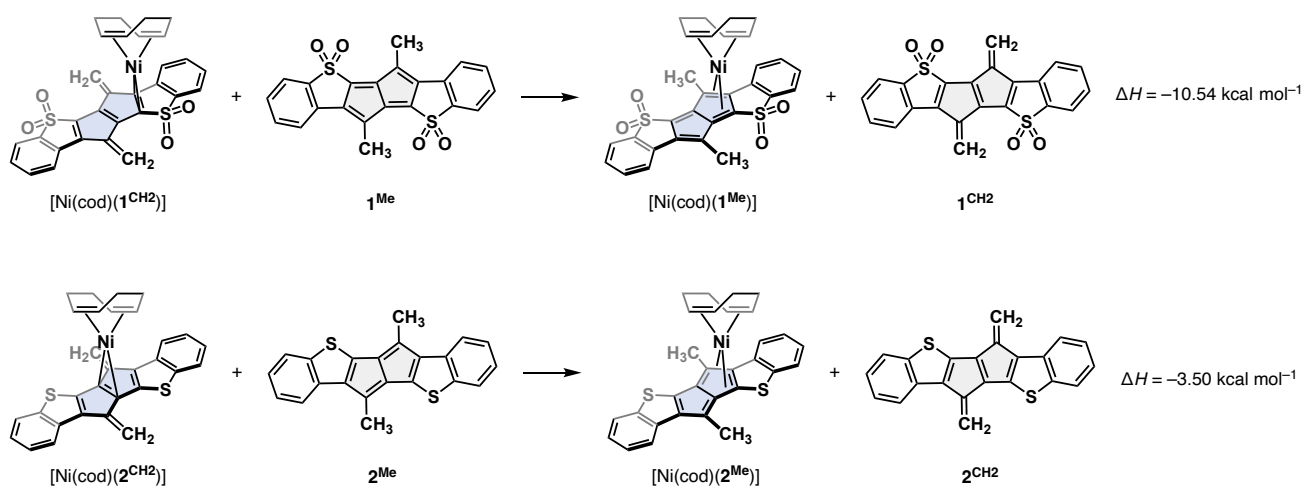


Figure S10. Model reactions for the stabilization of Ni-coordination in **1** and **2** compared to their nonaromatic analogs.

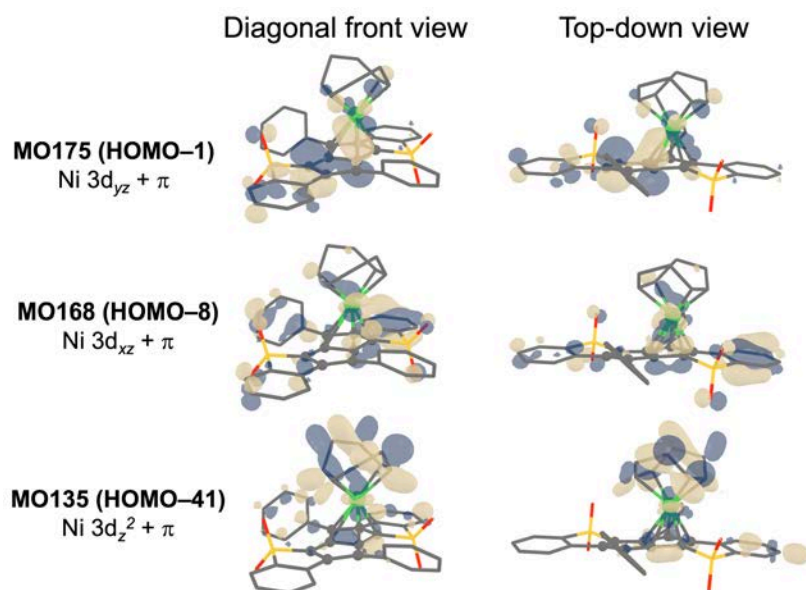


Figure S11. Selected molecular orbitals exhibit significant bonding interaction between Ni 3d and C 2p orbitals calculated using ω B97X-D functional with 6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) basis sets (isovalue = 0.04).

Table S7. Calculated contributions of selected atomic orbitals to molecular orbitals of [Ni(cod)(1)]^a

	MO	Ni 3d _{xy}	Ni 3d _{xz}	Ni 3d _{yz}	Ni 3d _{x²-y²}	Ni 3d _{z²}	Ni 4s	C22 2p _z	C23 2p _z	C24 2p _z	C25 2p _z	C26 2p _z	Σ _{occ} (Ni 3d)	Ni 3d _{yz} + 3d _{zx}	Σ _{occ} (C 2p _z)
LUMO	177	0.0%	-13.3%	14.4%	0.0%	-1.3%	0.1%	10.0%	-1.0%	-5.2%	-9.4%	4.2%	29.0%	27.7%	24.6%
HOMO	176	-1.6%	-3.3%	0.0%	0.0%	0.1%	0.0%	-2.8%	-1.0%	-9.2%	5.8%	8.5%	5.0%	3.3%	18.2%
HOMO-1	175	-0.5%	-1.8%	13.0%	-0.1%	-0.8%	0.0%	0.0%	16.6%	0.0%	0.0%	-2.9%	16.2%	14.8%	19.5%
HOMO-2	174	0.0%	7.0%	0.9%	-0.7%	0.0%	0.0%	1.2%	2.2%	-1.0%	-7.6%	-3.5%	8.6%	7.9%	14.5%
HOMO-3	173	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
HOMO-4	172	0.2%	0.1%	-0.1%	0.0%	-0.2%	0.0%	0.5%	0.0%	0.0%	0.0%	0.1%	0.7%	0.2%	0.7%
HOMO-5	171	3.0%	0.0%	0.2%	-0.2%	-1.0%	0.0%	0.0%	0.1%	2.7%	-0.2%	0.0%	4.3%	0.2%	0.2%
HOMO-6	170	0.9%	0.0%	0.1%	-0.2%	-0.1%	0.0%	-0.1%	0.0%	0.3%	0.0%	0.0%	1.4%	0.1%	0.1%
HOMO-7	169	0.1%	-1.4%	0.6%	-2.5%	1.3%	0.0%	-6.9%	0.0%	2.0%	-0.2%	-1.3%	5.8%	2.0%	8.4%
HOMO-8	168	0.5%	14.3%	7.3%	-2.6%	-3.9%	0.0%	3.3%	1.9%	-0.3%	-0.6%	0.0%	28.6%	21.6%	5.9%
HOMO-9	167	12.5%	-1.8%	-0.6%	-0.1%	-0.2%	0.3%	-1.2%	0.0%	0.0%	0.2%	-0.2%	15.2%	2.3%	1.5%
HOMO-10	166	40.4%	-0.1%	-0.4%	-0.7%	-7.7%	0.9%	-0.1%	0.0%	-1.0%	0.1%	0.0%	49.4%	0.6%	0.3%
HOMO-11	165	-2.1%	0.0%	0.0%	-0.8%	1.8%	-0.1%	0.0%	-0.1%	0.3%	0.0%	0.0%	4.7%	0.0%	0.1%
HOMO-12	164	1.8%	0.2%	0.6%	52.7%	0.3%	0.0%	-0.3%	1.0%	2.4%	0.0%	-0.1%	55.5%	0.8%	1.3%
HOMO-13	163	1.0%	7.8%	11.1%	-0.2%	23.7%	0.1%	-0.5%	-0.8%	-0.8%	-0.1%	-0.2%	43.9%	19.0%	1.6%
HOMO-14	162	0.2%	-0.6%	-0.9%	3.5%	4.1%	0.1%	0.7%	-0.1%	-4.0%	-3.8%	0.7%	9.3%	1.6%	5.3%
HOMO-15	161	0.1%	1.4%	0.5%	24.0%	-4.4%	0.0%	-0.2%	0.0%	-1.6%	-1.2%	0.5%	30.3%	1.9%	1.9%
HOMO-16	160	8.9%	-0.9%	-0.7%	-0.6%	-3.0%	-0.3%	0.1%	1.4%	0.0%	0.1%	0.7%	14.1%	1.6%	2.3%
HOMO-17	159	0.3%	14.1%	-7.3%	0.0%	-0.7%	0.0%	0.3%	-0.3%	-0.6%	-0.1%	0.4%	22.4%	21.3%	1.2%
HOMO-18	158	-0.1%	0.0%	-5.4%	0.0%	2.0%	0.1%	0.7%	0.4%	0.0%	-0.3%	-0.2%	7.6%	5.5%	1.5%
HOMO-19	157	10.1%	-1.9%	0.0%	0.0%	18.2%	-0.1%	0.0%	1.5%	0.0%	0.0%	-0.1%	30.3%	2.0%	1.6%
HOMO-20	156	8.5%	0.5%	1.1%	-0.2%	6.9%	-0.8%	0.0%	-0.8%	-0.3%	0.0%	0.1%	17.2%	1.6%	1.0%

Continued Table S7.

	MO	Ni 3d _{xy}	Ni 3d _{xz}	Ni 3d _{yz}	Ni 3d _{x²-y²}	Ni 3d _{z²}	Ni 4s	C22 2p _z	C23 2p _z	C24 2p _z	C25 2p _z	C26 2p _z	Σ _{occ} (Ni 3d)	Ni 3d _{yz} + 3d _{zx}	Σ _{occ} (C 2p _z)
HOMO-21	155	-0.8%	0.4%	-0.6%	-0.7%	-2.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	0.9%	0.0%
HOMO-22	154	0.0%	5.3%	8.2%	0.0%	-2.6%	0.0%	-5.7%	-3.3%	1.9%	7.2%	2.4%	16.2%	13.6%	18.6%
HOMO-23	153	0.2%	0.1%	0.0%	0.1%	0.2%	0.0%	0.0%	0.0%	-0.2%	0.0%	0.0%	0.5%	0.1%	0.0%
HOMO-24	152	-0.2%	0.0%	-0.3%	0.0%	0.0%	0.0%	1.5%	0.0%	-1.1%	-0.2%	0.5%	0.5%	0.3%	2.2%
HOMO-25	151	-0.5%	-3.4%	-0.7%	-0.1%	-0.6%	0.0%	0.0%	1.0%	0.2%	-0.9%	-2.0%	5.3%	4.2%	4.0%
HOMO-26	150	0.0%	0.0%	0.1%	0.0%	0.1%	0.0%	0.0%	-0.1%	0.1%	0.1%	0.1%	0.3%	0.1%	0.3%
HOMO-27	149	0.0%	0.5%	-0.4%	-0.2%	-1.0%	0.0%	0.2%	0.0%	-0.1%	-0.2%	0.0%	2.1%	0.9%	0.4%
HOMO-28	148	0.2%	0.0%	1.7%	0.0%	0.1%	0.1%	0.0%	-0.3%	0.0%	0.9%	2.5%	1.9%	1.7%	3.6%
HOMO-29	147	0.0%	0.4%	-0.3%	0.4%	0.0%	0.0%	0.1%	0.0%	-0.1%	-0.4%	0.0%	1.1%	0.7%	0.5%
HOMO-30	146	0.0%	0.0%	-1.2%	-0.1%	0.4%	0.0%	2.3%	0.0%	-1.8%	-0.8%	0.0%	1.7%	1.2%	3.1%
HOMO-31	145	-0.1%	0.0%	1.5%	0.0%	-0.2%	0.00%	-0.1%	-0.9%	-0.1%	1.9%	1.1%	1.8%	1.5%	4.0%
HOMO-32	144	0.0%	0.1%	0.9%	-0.1%	0.1%	0.08%	1.9%	-0.4%	-1.2%	0.5%	2.7%	1.2%	1.0%	5.5%
HOMO-33	143	-0.11%	0.39%	-2.52%	0.00%	0.02%	0.01%	0.64%	0.09%	0.00%	-0.03%	0.00%	3.03%	2.91%	0.75%
HOMO-34	142	-0.16%	-2.39%	2.51%	0.00%	-0.26%	0.00%	-0.65%	0.08%	1.02%	0.00%	-0.47%	5.31%	4.90%	1.21%
HOMO-35	141	0.00%	0.50%	-0.01%	-0.07%	0.03%	0.04%	0.46%	-0.16%	-0.03%	0.49%	1.13%	0.61%	0.51%	2.23%
HOMO-36	140	0.04%	-0.03%	0.21%	-0.06%	0.01%	-0.01%	-0.21%	-0.12%	0.00%	0.24%	0.00%	0.34%	0.24%	0.57%
HOMO-37	139	0.13%	0.52%	-0.14%	-0.03%	0.13%	-0.01%	0.52%	-0.02%	-0.01%	0.19%	0.59%	0.94%	0.66%	1.33%
HOMO-38	138	0.04%	-0.18%	0.38%	0.06%	0.26%	-0.02%	-0.21%	-0.03%	0.18%	1.57%	0.09%	0.91%	0.55%	1.90%
HOMO-39	137	-0.06%	0.25%	0.11%	0.00%	-1.22%	0.00%	-1.79%	-1.98%	-0.75%	-0.07%	-0.35%	1.64%	0.36%	4.20%
HOMO-40	136	-0.02%	-0.23%	0.07%	0.00%	-0.09%	0.06%	0.07%	0.02%	0.12%	0.19%	0.02%	0.40%	0.29%	0.30%
HOMO-41	135	0.00%	-0.08%	-0.08%	-0.24%	-2.20%	0.01%	-0.76%	-0.43%	-0.22%	-0.48%	-1.29%	2.61%	0.17%	2.96%
HOMO-42	134	-0.03%	0.00%	-0.07%	0.08%	0.33%	-0.02%	0.00%	0.30%	0.02%	-0.41%	-0.17%	0.51%	0.08%	0.88%

^a Calculated by natural atomic orbital analysis using ωB97X-D functional with 6-31G(d) (for C, H, O, S) and LanL2DZ (for Ni) basis sets.

4.6. Energy Decomposition Analyses for Ni–Ligand Interactions

Table S8. Summary of energy decomposition analysis^{a,b}

	ΔE_{Pauli}	ΔE_{elstat}	ΔE_{disp}	ΔE_{pol}	ΔE_{CT}	ΔE_{total}	ΔE_{CTf}	ΔE_{CTb}
Ni(cod)(1)	203.7	−109.6 (36.6%)	−35.7 (11.9%)	−32.1 (10.7%)	−121.9 (40.7%)	−95.5	−1.9	−98.2
Ni(cod)(2)	206.8	−111.7 (39.7%)	−35.0 (12.4%)	−32.3 (11.5%)	−102.6 (36.4%)	−74.7	−1.9	−86.6
Ni(cod)(dq)	185.9	−110.3 (42.8%)	−28.4 (11.0%)	−28.9 (11.2%)	−89.9 (34.9%)	−71.6	−2.9	−69.7
Ni(cod) ₂	187.4	−123.9 (47.9%)	−25.7 (10.0%)	−35.5 (13.7%)	−73.4 (28.4%)	−71.0	−3.9	−49.7

^aThe results of EDA and CTA-VFB calculations were expressed in kcal mol^{−1}. ^bThe contribution of the stabilization interaction is shown in parentheses.

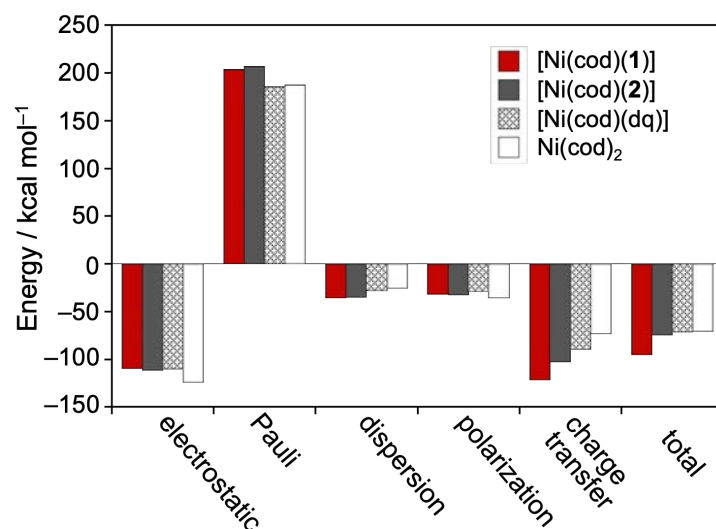


Figure S12. Bar plot graph for energy decomposition analysis.

Method S1: Example of Input File Description for EDA

```
$molecule
0 1
--
0 1
Ni fragment coordinate
--
0 1
Ligand fragment coordinates
$end

$rem
  JOBTYPED          eda
  EDA2              2
  METHOD            wB97X-D
  BASIS            def2-tzvp
  SYMMETRY          false
  MEM_TOTAL        8000
  MEM_STATIC        1000
  THRESH            14
  MAX_SCF_CYCLES    200
  SCF_CONVERGENCE    8
  SCF_PRINT_FRGM     true
  XC_GRID           000099000590
  NL_GRID           1
  FD_MAT_VEC_PROD    false
  EDA_CLS_DISP       true
$end

$rem
frgm scf_guess gwh
$end
```

Method S2: Example of Input File Description for CTA-VFB Analysis

```
$molecule
0 1
--
0 1
Ni fragment coordinates
--
0 1
Ligand fragment coordinates
$end

$rem
  JOBTYPED          opt
  METHOD            wb97x-d
  BASIS            def2-svp
  VFB_CTA           forward/backward  # input "forward" or "backward" to choose
  the one-way CT state
  THRESH            14
  SCF_CONVERGENCE    9
  SYMMETRY          FALSE
  SCF_ALGORITHM      DIIS
  IDERIV            1
```

```

    SCFMI_MODE      1
$end

@@@

$molecule
read
$end

$rem
    JOBTYPe        freq
    METHOD          wb97x-d
    BASIS          def2-svp
    VFB_CTA        forward/backward  # input "forward" or "backward" to choose
    the one-way CT state
    THRESH         14
    SCF_CONVERGENCE 9
    SYMMETRY       FALSE
    SCF_ALGORITHM  DIIS
    IDERIV         1
    SCFMI_MODE      1
$end

@@@

$molecule
read
$end

$rem
    JOBTYPe        sp
    METHOD          wb97x-d
    BASIS          def2-TZVP
    VFB_CTA        forward/backward  # input "forward" or "backward" to choose
    the one-way CT state
    THRESH         14
    SCF_CONVERGENCE 9
    SYMMETRY       FALSE
    SCF_ALGORITHM  DIIS
    IDERIV         1
    SCFMI_MODE      1
$end

```

4.7. Characterization of the Fluxional Behavior of [Ni(cod)(1)]

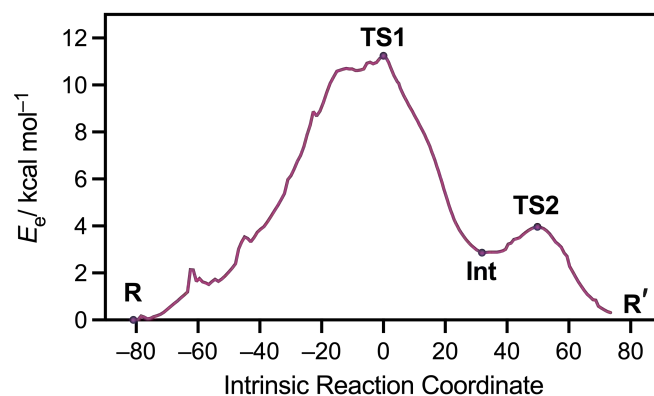


Figure S13. Reaction pathway connecting **R** and **R'** determined by IRC calculations.

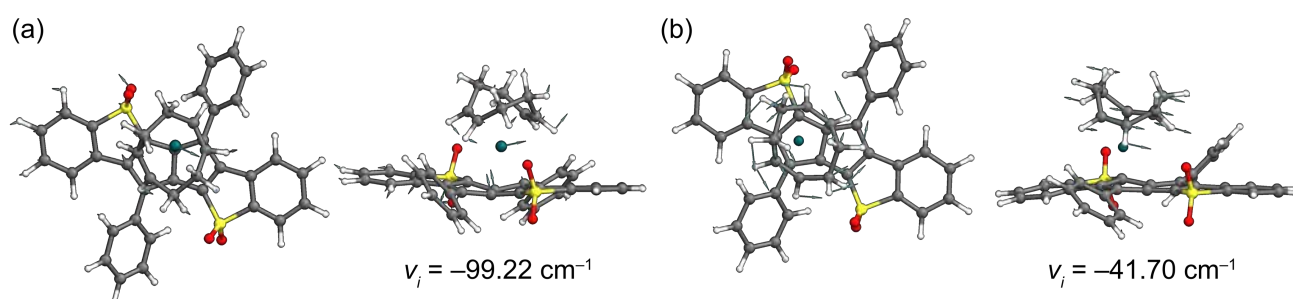


Figure S14. Top and front views of (a) **TS1** and (b) **TS2**. The light blue arrows indicate the vibrational mode corresponding to the imaginary frequency (ν_i).

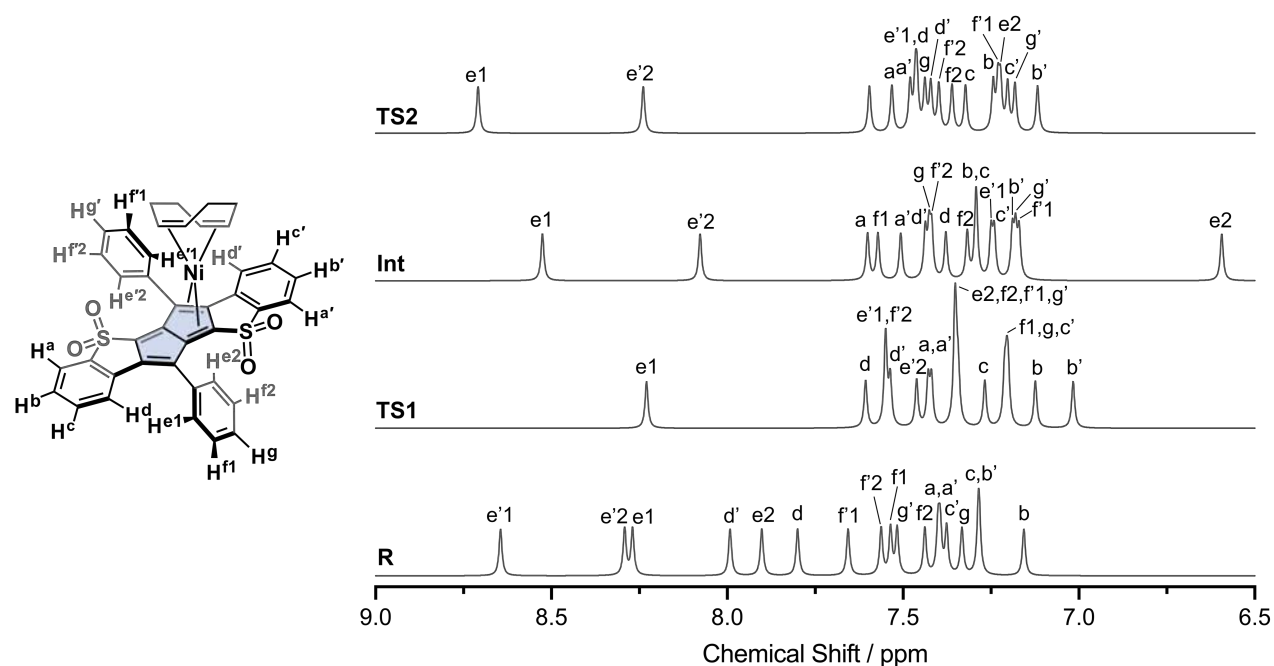
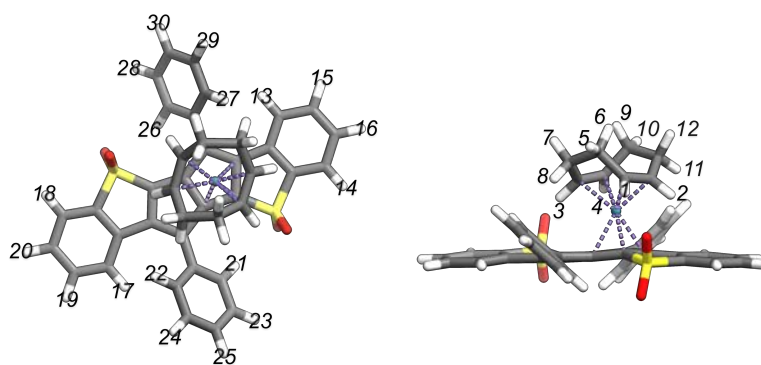


Figure S15. Simulated ^1H NMR spectra of [Ni(cod)(1)] in the aromatic region for **R**, **TS1**, **Int**, and **TS2**, generated by Lorentzian broadening of chemical shifts with a full width at half maximum (FWHM) of 0.01 ppm.

Table S9. Computed ^1H chemical shifts of $[\text{Ni}(\text{cod})(1)]: \text{R}, \text{TS1}, \text{Int}$ and TS2 .



H	R	TS1	Int	TS2	H	R	TS1	Int	TS2
1	3.5323	3.3457	5.2859	6.0382	16	7.2862	7.0164	7.1891	7.1175
2	5.2008	5.285	2.7128	4.0841	17	7.8001	7.6066	7.3784	7.4606
3	4.6844	5.3445	4.5389	4.9208	18	7.4006	7.4288	7.6009	7.5319
4	2.7017	5.6643	3.6493	3.9209	19	7.2833	7.2676	7.2915	7.3226
5	3.2356	3.2767	3.2313	2.8677	20	7.1565	7.1239	7.2932	7.2438
6	2.6700	1.9239	2.3300	2.4653	21	8.2694	8.2295	8.5254	8.7085
7	2.2851	2.1914	2.2621	2.1943	22	7.9019	7.3500	6.5937	7.2231
8	2.2368	1.9192	1.9230	1.9246	23	7.5357	7.2055	7.5714	7.5958
9	2.2183	2.6574	2.3538	2.3057	24	7.4382	7.3525	7.3175	7.3609
10	3.0160	3.6323	3.8791	3.8363	25	7.3327	7.2116	7.4262	7.4381
11	1.6775	2.0341	1.8613	1.8781	26	8.2918	7.4615	8.0771	8.2389
12	2.0686	2.0683	1.9952	2.1253	27	8.6447	7.5497	7.2498	7.4652
13	7.9919	7.5365	7.437	7.4213	28	7.5624	7.5496	7.4187	7.3984
14	7.3946	7.4193	7.5067	7.4801	29	7.6565	7.3431	7.1701	7.2304
15	7.3762	7.2010	7.2408	7.2027	30	7.5169	7.3527	7.1801	7.1817

5. X-Ray Absorption Spectroscopy

5.1. Ni K-edge X-ray Absorption Spectroscopy

Sample preparation. For [Ni(cod)(1)] and [Ni(cod)(dq)]: A mixture of the nickel complex and an appropriate amount of boron nitride were mixed in a tube using a mixer mill (Retsch MM400, 90 s at 30 Hz) with a zirconia ball. The resulting powder was pelletized using a tablet press (JASCO Corporation, $\phi = 10$ mm) and a hydraulic press (RIKEN KIKI Co., Ltd., CMD-5PA with the pressure of ca. 2–5 MPa). The thus prepared pellet was packed in a polypropylene bag and subjected to XAS measurements.

For [Ni(cod)(2)]: [Ni(cod)(2)] was dissolved in benzene (1.0 mL) in a glovebox filled with argon gas. An assembled in-house-build solution cell^{S30,S31} with the optical path length of 4.5 mm was filled with the solution and sealed with a screw cap. The thus prepared cell was taken out from the glovebox and placed on the optical axis of the X-ray beam and XAS measurements were performed.

Measurements and data processing. Ni K-edge XAS experiments were performed at the SPring-8 BL14B2 beamline using Si(111) double-crystal monochromatized synchrotron radiation under the ring-conditions of 8.0 GeV and 100 mA. All experiments were carried out using the transmission method with quick-scan XAS technique (QXAFS) at room temperature. Ionization chambers were used to measure the intensities of the incident (I_0) and transmitted (I) X-rays. XAS analysis was conducted using the Larch software package.^{S32} The pre-edge background of raw data was subtracted by a linear function with 1st or 2nd Victoreen order, and the normalization was performed using a quadratic polynomial function. E_0 was defined as the photon energy at the absorption edge where $\mu T = 0.5$ in the normalized XAS spectrum. E_0 values of Ni foil was set to 8333 eV for the photon-energy calibration. The pre-edge analysis was conducted using the pre-edge peak fitting program implemented in Larch software. The pre-edge baseline was approximated by fitting with the convolution of Lorentzian and linear functions with peak masking to negate the rising-edge background, and the thus subtracted pre-edge spectrum was fitted with pseudo-Voigt function to estimate the area of the unit-normalized peak.

5.2. Ni L_{2,3}-edge X-ray Absorption Spectroscopy

Sample preparation. In an argon-filled glovebox, the sample powder was gently pressed onto a piece of double-sided conductive carbon tape affixed to a copper sample holder using the back of a spatula (Figure S15). The sample holder was then mounted on a transfer rod in the glovebox and transferred into the measurement chamber. Subsequently, the chamber was evacuated using a dry pump followed by a turbomolecular pump to achieve a high vacuum environment ($\sim 10^{-5}$ Pa).

Measurements and data processing. Ni L_{2,3}-edge XAS experiments were performed at the Photon Factory (PF) BL-19B beamline equipped with an APPLE II type undulator^{S33} under the ring-conditions of 2.5 GeV and 450 mA. The synchrotron radiation was monochromatized by a variable-induced-angle Monk-Gillieson mounting monochromator with varied-line-spacing gratings (1200 lines/mm).^{S34} XAS experiments were carried out at room temperature under a pressure of less than 5×10^{-5} Pa using the total electron yield (TEY) method. The raw XAS data were obtained by dividing the drain current (I_1) by the mirror current (I_0) to negate the intensity variation in the

incident X-ray. The data were collected in the range from 830 eV to 920 eV with an energy step size of 0.05 eV. The background of raw data was subtracted by an exponential function obtained by fitting the pre-edge region between 830 and 845 eV using Igor Pro 9 software, and the subtracted data were normalized by dividing the average intensity in the post-edge region from 880 to 900 eV. The peak maxima of the Ni L₃-edge XAS spectrum of NiO were set to 853.2 eV for photon energy calibration. The area of the unit-normalized peak was obtained by fitting the L₃- and L₂-edge peaks with the pseudo-Voigt functions after edge jump subtraction by the arctangent function using Igor Pro 9.



Figure S16. Photo of XAS sample for [Ni(cod)(1)] and [Ni(cod)(2)].

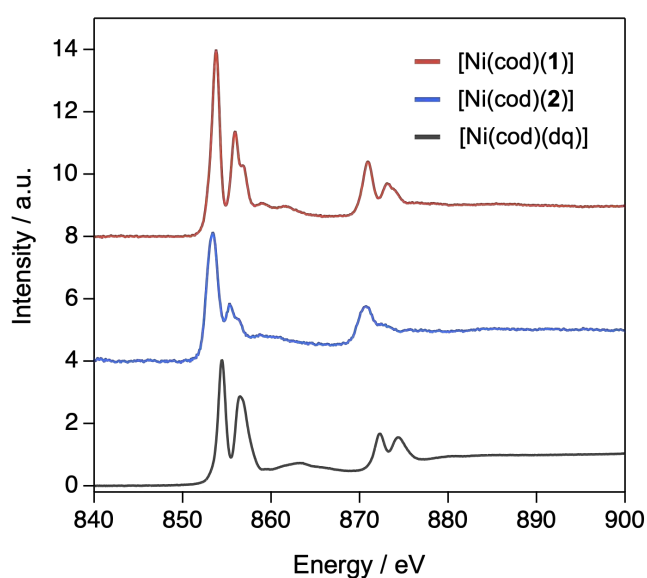


Figure S17. Offset view of Ni L_{2,3}-edge XAS data of [Ni(cod)(1)], [Ni(cod)(2)], and [Ni(cod)(dq)].

Table S10. Summary of Ni K- and Ni L_{2,3}-edge XAS data

	K-edge		L ₃ -edge			L ₂ -edge		
	Pre-edge	Amplitude	Peak maxima	Main peak	Second peak	Peak maxima	Main peak	Second peak
	/ eV	/ area	/ eV	/ area	/ area	/ eV	/ area	/ area
Ni(cod)(1)	8327.0	0.37	853.8	6.7	4.0	871.0	2.3	1.2
Ni(cod)(2)	8326.4	0.27	853.4	5.9	1.7	870.7	1.9	0.2
Ni(cod)(dq)	8326.8	0.20	854.5	5.6	3.5	872.3	1.4	1.3

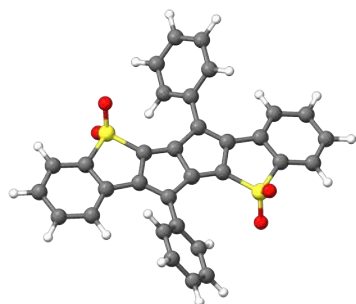
6. Appendix

List of Computed Geometries

Charge, spin multiplicity, point group, number of imaginary frequencies, energies (in Hartree), and Cartesian coordinates (in Å) of computed geometries are listed in this section.

Table S11. List of computed geometries.

Geometry	Page	Geometry	Page
1 @B3LYP	32	1 ^{CH2}	54
1' @B3LYP	33	[Ni(cod)(1 ^{Me})]	55
1 @ωB97X-D	34	[Ni(cod)(1 ^{CH2})]	57
1' @ωB97X-D	35	2 ^{Me}	59
2 @B3LYP	36	2 ^{CH2}	60
2' @ωB97X-D	37	[Ni(cod)(2 ^{Me})]	61
2 @ωB97X-D	38	[Ni(cod)(2 ^{CH2})]	63
COD	39	pentalene (<i>C</i> _{2h})	65
[Ni(cod) ₂]	40	pentalene (<i>D</i> _{2h})	66
[Ni(cod)(1): R	41	I	67
[Ni(cod)(1): TS1	43	II	68
[Ni(cod)(1): Int	45	III	69
[Ni(cod)(1): TS2	47	IV	70
[Ni(cod)(2)]	49	V	71
[Ni(cod)(2')]	51	VI	72
1 ^{Me}	53	VII	73



1

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2327.32467963 Hartree

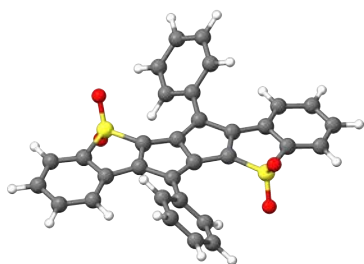
Sum of electronic and zero-point Energies = -2326.912540 Hartree

Sum of electronic and thermal Energies = -2326.883363 Hartree

Sum of electronic and thermal Enthalpies = -2326.882419 Hartree

Sum of electronic and thermal Free Energies = -2326.973477 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	2.177531	0.125411	-0.002214	C	1.312269	-1.088034	-0.011884
C	0.005746	-0.737827	-0.024136	C	-0.005685	0.737759	-0.024240
C	1.381774	1.248386	0.006716	C	-1.381733	-1.248424	0.007029
C	-2.177466	-0.125435	-0.001961	C	-1.312194	1.087986	-0.011917
C	-3.595015	0.210406	-0.075994	C	-3.824567	1.603802	-0.081717
S	-2.310093	2.572067	-0.013299	C	3.595081	-0.210418	-0.076366
C	3.824665	-1.603810	-0.081834	S	2.310214	-2.572084	-0.013109
C	-1.755460	-2.666026	0.090897	C	1.755414	2.666025	0.090337
C	-4.713932	-0.633133	-0.185246	C	-5.989037	-0.080530	-0.278030
C	-6.186147	1.306672	-0.263956	C	-5.090723	2.165933	-0.170255
C	4.713978	0.633113	-0.185917	C	5.989083	0.080513	-0.278733
C	6.186221	-1.306681	-0.264393	C	5.090822	-2.165938	-0.170397
C	1.103363	3.625649	-0.704476	C	1.452387	4.971935	-0.615781
C	2.447324	5.385817	0.271332	C	3.089197	4.444184	1.080130
C	2.746250	3.097338	0.992760	C	-1.103679	-3.625810	-0.703946
C	-1.452846	-4.972045	-0.615043	C	-2.447681	-5.385713	0.272284
C	-3.089330	-4.443912	1.081064	C	-2.746244	-3.097116	0.993484
O	-2.194759	3.284776	1.266330	O	-2.141559	3.318459	-1.271475
O	2.195055	-3.284603	1.266642	O	2.141653	-3.318699	-1.271150
H	-4.584466	-1.708752	-0.204653	H	-6.846776	-0.741592	-0.364472
H	-7.189946	1.714787	-0.332788	H	-5.219232	3.244047	-0.173396
H	4.584496	1.708724	-0.205531	H	6.846801	0.741572	-0.365410
H	7.190020	-1.714793	-0.333257	H	5.219348	-3.244051	-0.173342
H	0.323322	3.321229	-1.394287	H	0.937356	5.697644	-1.238423
H	2.712384	6.437052	0.343137	H	3.847097	4.761398	1.791341
H	3.219691	2.373795	1.649791	H	-0.323759	-3.321559	-1.393962
H	-0.938017	-5.697879	-1.237704	H	-2.712861	-6.436907	0.344240
H	-3.847172	-4.760954	1.792415	H	-3.219530	-2.373436	1.650477



1'

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2327.32288708 Hartree

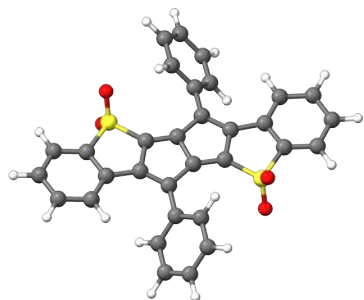
Sum of electronic and zero-point Energies = -2326.910632 Hartree

Sum of electronic and thermal Energies = -2326.881453 Hartree

Sum of electronic and thermal Enthalpies = -2326.880509 Hartree

Sum of electronic and thermal Free Energies = -2326.971259 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	2.174802	0.128518	-0.011635	C	1.454786	-1.035144	-0.033942
C	0.039297	-0.727286	-0.086471	C	-0.039357	0.727387	-0.086410
C	1.218162	1.284978	0.007771	C	-1.218191	-1.284914	0.007858
C	-2.174846	-0.128478	-0.011372	C	-1.454858	1.035206	-0.033580
C	-3.630306	0.042271	-0.125346	C	-3.983873	1.407328	-0.165345
C	3.630273	-0.042280	-0.125433	C	3.983794	-1.407352	-0.165287
C	-4.656782	-0.900593	-0.239108	C	-5.983108	-0.465013	-0.351577
C	-6.304018	0.892116	-0.359215	C	-5.286322	1.853617	-0.271889
C	4.656785	0.900541	-0.239222	C	5.983106	0.464908	-0.351528
C	6.303971	-0.892232	-0.359010	C	5.286235	-1.853695	-0.271696
C	1.557243	2.697844	0.169707	C	-1.557189	-2.697806	0.169761
C	0.852854	3.687949	-0.542111	C	1.144646	5.035750	-0.349729
C	2.133045	5.420602	0.558599	C	2.831094	4.448124	1.279969
C	2.552118	3.098448	1.084672	C	-0.852851	-3.687827	-0.542221
C	-1.144532	-5.035658	-0.349874	C	-2.132777	-5.420616	0.558576
C	-2.830772	-4.448221	1.280110	C	-2.551900	-3.098518	1.084854
H	-4.433513	-1.960601	-0.252005	H	-6.774539	-1.204045	-0.438975
H	-7.339086	1.208447	-0.445698	H	-5.512418	2.915292	-0.298321
H	4.433545	1.960555	-0.252289	H	6.774569	1.203906	-0.438941
H	7.339039	-1.208601	-0.445347	H	5.512295	-2.915380	-0.298021
H	0.085024	3.401058	-1.253195	H	0.593822	5.785353	-0.910019
H	2.355383	6.473403	0.708905	H	3.588002	4.742161	2.001818
H	3.075840	2.348491	1.669464	H	-0.085125	-3.400850	-1.253383
H	-0.593744	-5.785198	-0.910281	H	-2.355033	-6.473439	0.708854
H	-3.587553	-4.742346	2.002056	H	-3.075568	-2.348620	1.669774
S	-2.516966	2.461379	-0.118112	S	2.516854	-2.461363	-0.118188
O	-2.467141	3.247449	1.122980	O	-2.357975	3.156869	-1.408149
O	2.466679	-3.247282	1.122994	O	2.358149	-3.156985	-1.408180



1

#p opt freq wb97xd/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2326.79271841 Hartree

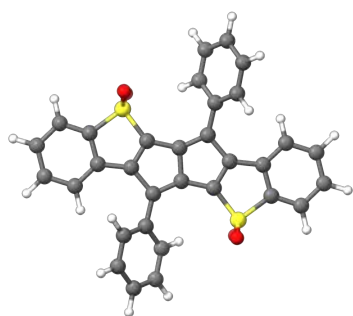
Sum of electronic and zero-point Energies = -2326.373944 Hartree

Sum of electronic and thermal Energies = -2326.345224 Hartree

Sum of electronic and thermal Enthalpies = -2326.344279 Hartree

Sum of electronic and thermal Free Energies = -2326.434460 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.006999	0.737021	-0.021681	C	-0.006983	-0.737005	-0.021681
C	-1.301604	-1.086377	-0.018160	C	-2.163472	0.129452	-0.011946
C	-1.381540	1.243235	0.002779	C	1.381548	-1.243225	0.002789
C	2.163486	-0.129446	-0.011937	C	1.301624	1.086387	-0.018141
C	1.751005	-2.660881	0.087024	C	2.421660	-5.368876	0.275047
C	2.725075	-3.089278	0.997681	C	1.102436	-3.607612	-0.713319
C	1.439095	-4.952987	-0.618700	C	3.061555	-4.434302	1.086357
H	3.195269	-2.364484	1.656294	H	0.331033	-3.295503	-1.410490
H	0.924096	-5.676421	-1.242533	H	3.812262	-4.755772	1.801972
H	2.679424	-6.420908	0.350182	C	-1.751022	2.660884	0.087022
C	-2.421763	5.368857	0.275049	C	-1.102495	3.607634	-0.713333
C	-2.725110	3.089248	0.997674	C	-3.061631	4.434261	1.086355
C	-1.439195	4.952998	-0.618709	H	-0.331096	3.295550	-1.410518
H	-3.195287	2.364436	1.656280	H	-3.812347	4.755706	1.801971
H	-0.924225	5.676448	-1.242547	H	-2.679559	6.420881	0.350185
C	3.582836	0.210123	-0.078883	C	6.158195	1.301592	-0.244941
C	4.692114	-0.633915	-0.185216	C	3.802986	1.596094	-0.078070
C	5.066151	2.158882	-0.156198	C	5.964868	-0.081388	-0.266293
H	4.558542	-1.709293	-0.212962	H	5.193861	3.236570	-0.154814
H	6.823700	-0.739980	-0.350185	H	7.161464	1.710270	-0.306109
C	-3.582819	-0.210123	-0.078878	C	-6.158170	-1.301597	-0.244934
C	-4.692097	0.633915	-0.185197	C	-3.802959	-1.596094	-0.078080
C	-5.066122	-2.158886	-0.156208	C	-5.964849	0.081385	-0.266274
H	-4.558526	1.709294	-0.212929	H	-5.193830	-3.236574	-0.154832
H	-6.823684	0.739974	-0.350153	H	-7.161437	-1.710279	-0.306102
S	-2.296091	-2.558922	-0.018687	S	2.296123	2.558925	-0.018670
O	2.135637	3.292570	-1.270256	O	2.178116	3.264981	1.249581
O	-2.135577	-3.292565	-1.270269	O	-2.178049	-3.264964	1.249570



1'

#p opt freq rwb97xd/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2326.79133313 Hartree

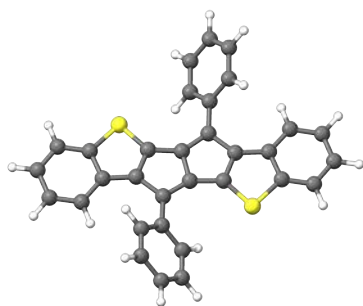
Sum of electronic and zero-point Energies = -2326.372580 Hartree

Sum of electronic and thermal Energies = -2326.343817 Hartree

Sum of electronic and thermal Enthalpies = -2326.342873 Hartree

Sum of electronic and thermal Free Energies = -2326.432995 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	2.158468	0.136256	-0.024731	C	1.462529	-1.026642	-0.048436
C	0.042203	-0.727115	-0.101474	C	-0.042235	0.727101	-0.101442
C	1.195663	1.284780	-0.005844	C	-1.195692	-1.284796	-0.005831
C	-2.158502	-0.136267	-0.024703	C	-1.462556	1.026630	-0.048357
C	-3.618828	0.016297	-0.119272	C	-3.979996	1.371453	-0.156082
C	3.618796	-0.016290	-0.119230	C	3.979979	-1.371439	-0.156097
C	-4.624277	-0.939776	-0.217402	C	-5.953227	-0.520695	-0.313967
C	-6.287299	0.829590	-0.320555	C	-5.285686	1.802039	-0.248187
C	4.624234	0.939804	-0.217235	C	5.953194	0.520746	-0.313763
C	6.287282	-0.829536	-0.320430	C	5.285678	-1.802004	-0.248170
C	1.521812	2.701854	0.158479	C	-1.521819	-2.701871	0.158519
C	0.825707	3.672541	-0.571842	C	1.092190	5.021556	-0.371435
C	2.051447	5.415813	0.556880	C	2.743181	4.455991	1.293353
C	2.482236	3.106330	1.095995	C	-0.825816	-3.672547	-0.571915
C	-1.092257	-5.021566	-0.371482	C	-2.051361	-5.415842	0.556982
C	-2.742978	-4.456035	1.293584	C	-2.482076	-3.106369	1.096198
H	-4.383880	-1.996510	-0.230636	H	-6.737739	-1.267083	-0.389820
H	-7.326607	1.132227	-0.394969	H	-5.524488	2.860317	-0.273544
H	4.383820	1.996537	-0.230392	H	6.737702	1.267148	-0.389520
H	7.326598	-1.132155	-0.394810	H	5.524495	-2.860277	-0.273576
H	0.078797	3.371877	-1.300394	H	0.543308	5.763881	-0.941597
H	2.256220	6.470598	0.712686	H	3.479599	4.760041	2.030836
H	3.000243	2.360197	1.691965	H	-0.079026	-3.371871	-1.300585
H	-0.543460	-5.763880	-0.941742	H	-2.256099	-6.470630	0.712810
H	-3.479264	-4.760099	2.031192	H	-2.999970	-2.360252	1.692284
S	-2.530934	2.432508	-0.127778	S	2.530923	-2.432508	-0.127894
O	-2.477860	3.214032	1.100581	O	-2.391291	3.113914	-1.412433
O	2.477810	-3.214092	1.100426	O	2.391335	-3.113855	-1.412586



2

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_2

Number of imaginary frequencies = 0

Electronic Energy = -2026.61321785 Hartree

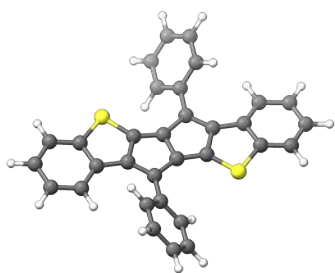
Sum of electronic and zero-point Energies = -2026.218278 Hartree

Sum of electronic and thermal Energies = -2026.193044 Hartree

Sum of electronic and thermal Enthalpies = -2026.192100 Hartree

Sum of electronic and thermal Free Energies = -2026.273556 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.012033	-2.184650	-0.066490	C	-1.126178	-1.386728	-0.110614
C	-0.728414	0.004841	-0.185708	C	0.728414	-0.004841	-0.185708
C	1.201953	-1.297936	-0.078103	C	-1.201953	1.297936	-0.078103
C	-0.012033	2.184650	-0.066490	C	1.126178	1.386728	-0.110614
C	0.286898	3.593263	-0.098263	C	1.696547	3.808102	-0.113656
S	2.605198	2.288870	-0.140634	C	-0.286898	-3.593263	-0.098263
C	-1.696547	-3.808102	-0.113656	S	-2.605198	-2.288870	-0.140634
C	-0.546598	4.730241	-0.139849	C	0.012033	6.001893	-0.165419
C	1.404313	6.186068	-0.153120	C	2.256268	5.085543	-0.131185
C	0.546598	-4.730241	-0.139849	C	-0.012033	-6.001893	-0.165419
C	-1.404313	-6.186068	-0.153120	C	-2.256268	-5.085543	-0.131185
C	2.604687	-1.699251	0.072960	C	-2.604687	1.699251	0.072960
C	3.609234	-1.075607	-0.690174	C	4.948427	-1.421668	-0.522746
C	5.311645	-2.389788	0.415964	C	4.325225	-3.012127	1.185034
C	2.984131	-2.676059	1.012747	C	-3.609234	1.075607	-0.690174
C	-4.948427	1.421668	-0.522746	C	-5.311645	2.389788	0.415964
C	-4.325225	3.012127	1.185034	C	-2.984131	2.676059	1.012747
H	-1.624023	4.611412	-0.163775	H	-0.641968	6.869128	-0.199845
H	1.820339	7.189266	-0.171258	H	3.334722	5.217621	-0.137303
H	1.624023	-4.611412	-0.163775	H	0.641968	-6.869128	-0.199845
H	-1.820339	-7.189266	-0.171258	H	-3.334722	-5.217621	-0.137303
H	3.328182	-0.337825	-1.435927	H	5.708714	-0.936979	-1.129100
H	6.356366	-2.657602	0.547894	H	4.601315	-3.758891	1.924628
H	2.223327	-3.148138	1.626567	H	-3.328182	0.337825	-1.435927
H	-5.708714	0.936979	-1.129100	H	-6.356366	2.657602	0.547894
H	-4.601315	3.758891	1.924628	H	-2.223327	3.148138	1.626567



2'

#p opt freq rwb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2026.15318155 Hartree

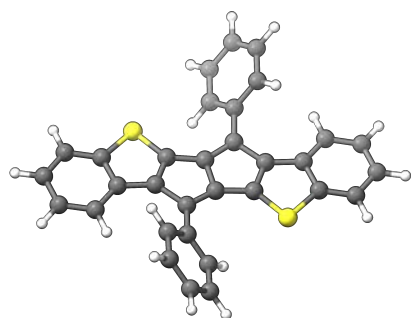
Sum of electronic and zero-point Energies = -2025.754515 Hartree

Sum of electronic and thermal Energies = -2025.729232 Hartree

Sum of electronic and thermal Enthalpies = -2025.728287 Hartree

Sum of electronic and thermal Free Energies = -2025.811209 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.136285	0.702158	-0.181875	C	-0.136346	-0.702152	-0.181884
C	1.087566	-1.331108	-0.079482	C	2.150184	-0.330045	-0.046805
C	1.567531	0.925762	-0.090305	C	-1.567575	-0.925761	-0.090065
C	-2.150229	0.330048	-0.046499	C	-1.087607	1.331107	-0.079220
C	-2.198519	-2.246230	-0.023176	C	-3.344675	-4.797092	0.124606
C	-1.738474	-3.282801	-0.845530	C	-3.230894	-2.510839	0.885577
C	-3.799595	-3.776781	0.957083	C	-2.311835	-4.547016	-0.775279
H	-0.941694	-3.083979	-1.557030	H	-3.565328	-1.722723	1.553724
H	-4.593777	-3.970403	1.671931	H	-1.950988	-5.338275	-1.425332
H	-3.789866	-5.785795	0.181977	C	2.198509	2.246216	-0.023471
C	3.344788	4.797027	0.124211	C	3.231233	2.510678	0.884927
C	1.738200	3.282897	-0.845538	C	2.311616	4.547090	-0.775332
C	3.799992	3.776596	0.956385	H	3.565917	1.722454	1.552825
H	0.941158	3.084183	-1.556776	H	1.950553	5.338440	-1.425156
H	4.594451	3.970103	1.670957	H	3.790026	5.785711	0.181543
C	-3.467173	0.941147	-0.024189	C	-5.753811	2.564619	0.030632
C	-4.740337	0.366682	-0.077456	C	-3.374205	2.347691	0.032348
C	-4.495986	3.161967	0.066304	C	-5.870044	1.177442	-0.048993
H	-4.845865	-0.710155	-0.152659	H	-4.397775	4.242315	0.112653
H	-6.854236	0.720946	-0.091877	H	-6.643776	3.185511	0.056587
C	3.467132	-0.941127	-0.024592	C	5.753798	-2.564556	0.029994
C	4.740279	-0.366627	-0.077843	C	3.374187	-2.347675	0.031829
C	4.495985	-3.161933	0.065665	C	5.870003	-1.177368	-0.049500
H	4.845776	0.710222	-0.152931	H	4.397798	-4.242287	0.111925
H	6.854184	-0.720849	-0.092372	H	6.643775	-3.185434	0.055859
S	-1.693126	2.945507	0.015516	S	1.693113	-2.945511	0.015045



2

#p opt freq wb97xd/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_2

Number of imaginary frequencies = 0

Electronic Energy = -2026.16319664 Hartree

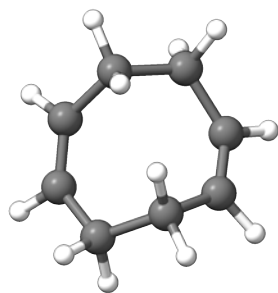
Sum of electronic and zero-point Energies = -2025.762315 Hartree

Sum of electronic and thermal Energies = -2025.737464 Hartree

Sum of electronic and thermal Enthalpies = -2025.736520 Hartree

Sum of electronic and thermal Free Energies = -2025.817172 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.329224	2.145177	-0.085040	C	1.328366	1.200141	-0.128262
C	0.719569	-0.117980	-0.202440	C	-0.719569	0.117980	-0.202440
C	-0.984203	1.450834	-0.094425	C	0.984203	-1.450834	-0.094425
C	-0.329224	-2.145177	-0.085040	C	-1.328366	-1.200141	-0.128262
C	-0.831772	-3.493003	-0.102466	C	-2.248317	-3.492090	-0.112573
S	-2.917919	-1.865301	-0.145458	C	0.831772	3.493003	-0.102466
C	2.248317	3.492090	-0.112573	S	2.917919	1.865301	-0.145458
C	-0.168776	-4.731651	-0.136699	C	-0.903498	-5.904281	-0.150066
C	-2.305359	-5.880782	-0.133418	C	-2.987172	-4.674401	-0.119128
C	0.168776	4.731651	-0.136699	C	0.903498	5.904281	-0.150066
C	2.305359	5.880782	-0.133418	C	2.987172	4.674401	-0.119128
C	-2.305359	2.067838	0.065141	C	2.305359	-2.067838	0.065141
C	-3.384979	1.632863	-0.712516	C	-4.650003	2.180517	-0.531735
C	-4.853601	3.166365	0.429980	C	-3.785970	3.602849	1.211470
C	-2.519592	3.060300	1.029870	C	3.384979	-1.632863	-0.712516
C	4.650003	-2.180517	-0.531735	C	4.853601	-3.166365	0.429980
C	3.785970	-3.602849	1.211470	C	2.519592	-3.060300	1.029870
H	0.914866	-4.765910	-0.165591	H	-0.385166	-6.858016	-0.178683
H	-2.862209	-6.812551	-0.142085	H	-4.072698	-4.649602	-0.121130
H	-0.914866	4.765910	-0.165591	H	0.385166	6.858016	-0.178683
H	2.862209	6.812551	-0.142085	H	4.072698	4.649602	-0.121130
H	-3.220632	0.878436	-1.476862	H	-5.477013	1.838792	-1.146588
H	-5.841725	3.593738	0.571081	H	-3.941212	4.365001	1.969134
H	-1.691516	3.387581	1.651887	H	3.220632	-0.878436	-1.476862
H	5.477013	-1.838792	-1.146588	H	5.841725	-3.593738	0.571081
H	3.941212	-4.365001	1.969134	H	1.691516	-3.387581	1.651887



COD

#p opt freq wb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -311.933224524 Hartree

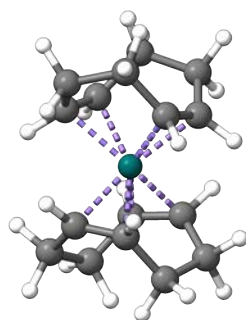
Sum of electronic and zero-point Energies = -311.750175 Hartree

Sum of electronic and thermal Energies = -311.742790 Hartree

Sum of electronic and thermal Enthalpies = -311.741846 Hartree

Sum of electronic and thermal Free Energies = -311.781695 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	-1.183095	1.244669	-0.503978	H	-1.772371	1.847575	-1.195507
C	0.045327	1.689430	-0.231748	H	0.354490	2.600040	-0.744330
C	-0.045316	-1.689434	-0.231752	H	-0.354447	-2.600062	-0.744320
C	1.183106	-1.244663	-0.503966	H	1.772400	-1.847581	-1.195471
C	1.095919	1.084747	0.669143	H	0.661090	0.719155	1.601500
H	1.795281	1.876188	0.962744	C	1.917212	-0.026210	-0.007119
H	2.709445	-0.351180	0.683683	C	-1.095938	-1.084762	0.669115
H	-0.661135	-0.719216	1.601505	H	-1.795319	-1.876207	0.962663
C	-1.917213	0.026230	-0.007113	H	-2.709413	0.351209	0.683722
H	2.445051	0.411233	-0.865847	H	-2.445093	-0.411190	-0.865827



[Ni(cod)₂]

#p opt freq wb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -793.22111475 Hartree

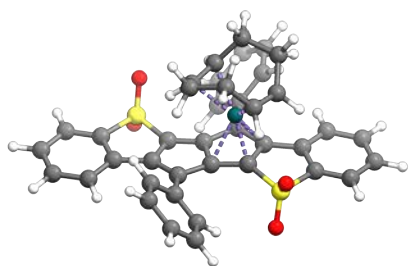
Sum of electronic and zero-point Energies = -792.851226 Hartree

Sum of electronic and thermal Energies = -792.835067 Hartree

Sum of electronic and thermal Enthalpies = -792.834123 Hartree

Sum of electronic and thermal Free Energies = -792.892511 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	1.350939	1.533522	-0.602593	H	0.742982	2.369075	-0.945095
C	1.493949	0.471828	-1.470420	H	0.964198	0.544422	-2.416877
C	1.494117	-0.471628	1.470246	H	0.964513	-0.543709	2.416834
C	1.350691	-1.533544	0.602670	H	0.742938	-2.369129	0.945465
C	-1.494330	0.471568	1.470467	H	-0.964930	0.543995	2.417141
C	-1.350736	1.533364	0.602751	H	-0.742645	2.368655	0.945671
C	-1.494094	-0.471313	-1.470545	H	-0.963848	-0.543282	-2.416765
C	-1.351179	-1.533331	-0.603162	H	-0.743148	-2.368776	-0.945781
C	2.595070	-0.573696	-1.417403	H	3.508937	-0.127841	-1.013102
H	2.844353	-0.877702	-2.440205	C	2.204895	-1.826519	-0.610539
H	3.109213	-2.383834	-0.321406	C	2.595792	0.573271	1.416673
H	3.508952	0.127128	1.011131	H	2.846295	0.876741	2.439332
C	2.205172	1.826398	0.610572	H	3.109212	2.384167	0.321429
C	-2.204575	1.826830	-0.610554	H	-1.630848	2.496844	-1.262447
H	-3.108580	2.384697	-0.321511	C	-2.595297	0.574040	-1.417135
H	-2.845281	0.877813	-2.439833	H	-3.508789	0.128153	-1.012068
C	-2.205510	-1.826380	0.609918	H	-1.632310	-2.496897	1.261755
H	-3.109790	-2.383656	0.320587	C	-2.595699	-0.573647	1.416907
H	-2.845404	-0.877835	2.439550	H	-3.509323	-0.127512	1.012380
Ni	0.000167	-0.000195	0.000581	H	1.631624	-2.496968	-1.262392
H	1.631773	2.496358	1.262814				



#p opt=(maxstep=5,notrust) freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2808.10164705 Hartree

Sum of electronic and zero-point Energies = -2807.495737 Hartree

Sum of electronic and thermal Energies = -2807.457665 Hartree

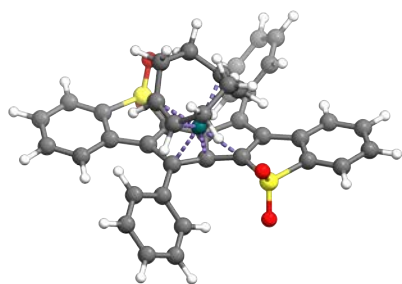
Sum of electronic and thermal Enthalpies = -2807.456721 Hartree

Sum of electronic and thermal Free Energies = -2807.565480 Hartree

[Ni(cod)(1)]: **R**

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-2.407777	-0.795413	2.224476	H	-3.204373	-1.039725	1.522278
C	-1.404308	-1.728189	2.370482	H	-1.457580	-2.610189	1.737060
C	-0.309640	1.182696	2.520899	H	-0.184280	2.114783	1.971802
C	0.695162	0.257341	2.369832	H	1.519886	0.545750	1.721541
C	-2.673163	0.395800	3.117769	H	-3.318307	1.083271	2.559799
H	-3.254602	0.083352	3.996666	C	-1.403792	1.151348	3.571471
H	-1.010658	0.717009	4.494876	H	-1.679129	2.182012	3.815376
C	0.902947	-1.022358	3.142738	H	1.488782	-0.817116	4.049591
H	1.514827	-1.681664	2.518243	C	-0.399574	-1.757550	3.503124
H	-0.161650	-2.801215	3.730217	H	-0.846658	-1.339246	4.409209
Ni	-0.827320	-0.203913	0.987852	C	-1.120807	-0.979700	-0.796654
C	0.259902	-0.669946	-0.823917	C	0.340491	0.784165	-0.809564
C	-1.016786	1.350179	-0.755973	C	-1.891357	0.281394	-0.884277
C	1.594361	-1.221031	-0.740591	C	2.438493	-0.121553	-0.729626
C	1.675502	1.100717	-0.777638	C	-3.323669	0.060206	-1.085025
C	-5.945181	-0.838764	-1.495327	C	-4.370553	0.981157	-1.150357
C	-3.622665	-1.299266	-1.255166	C	-4.910697	-1.769928	-1.446705
C	-5.668431	0.522504	-1.357961	H	-4.173473	2.042380	-1.049957
H	-5.100468	-2.832324	-1.561758	H	-6.480425	1.241118	-1.412147
H	-6.966016	-1.172822	-1.649033	C	3.883521	0.131586	-0.647951
C	6.509811	1.099954	-0.501966	C	4.956861	-0.753976	-0.744014
C	4.163517	1.498078	-0.496391	C	5.449109	2.001703	-0.424676
C	6.258424	-0.262039	-0.665877	H	4.779417	-1.813658	-0.891716
H	5.621178	3.068286	-0.320117	H	7.091502	-0.954327	-0.743076
H	7.531658	1.461354	-0.446439	S	-2.171545	-2.346847	-1.231983
S	2.674168	2.497557	-0.455961	O	2.699974	3.471488	-1.547173
O	2.424852	2.995665	0.901162	O	-2.296192	-3.347336	-0.174740
O	-1.854505	-2.793099	-2.584702	C	1.894018	-2.648959	-0.564774

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
C	1.205869	-3.619516	-1.302115	C	2.817584	-3.070113	0.402355
C	1.438447	-4.972922	-1.079880	H	0.489092	-3.314742	-2.059341
C	3.057246	-4.423312	0.615120	H	3.340815	-2.325974	0.997437
C	2.366048	-5.380474	-0.124811	H	0.889836	-5.709628	-1.658632
H	3.776351	-4.730681	1.369135	H	2.546480	-6.437584	0.046005
C	-1.308918	2.786027	-0.630198	C	-0.560467	3.715919	-1.357517
C	-2.293404	3.244495	0.253877	C	-0.809410	5.077310	-1.217513
H	0.228113	3.380250	-2.023532	C	-2.545457	4.604317	0.386439
H	-2.841297	2.524389	0.855941	C	-1.804181	5.524552	-0.353072
H	-0.211532	5.786900	-1.779996	H	-3.308200	4.947686	1.079130
H	-1.991785	6.588186	-0.241986				



[Ni(cod)(1)]: TS1

#p opt=(calcfc,ts,noeigen) freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 1, $\nu_i = -99.22$

Electronic Energy = -2808.08372589 Hartree

Sum of electronic and zero-point Energies = -2807.478547 Hartree

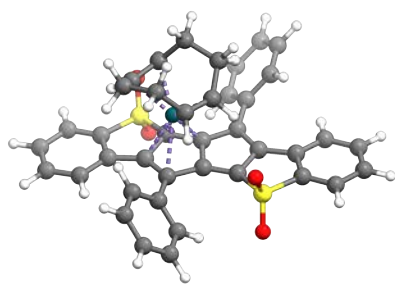
Sum of electronic and thermal Energies = -2807.440888 Hartree

Sum of electronic and thermal Enthalpies = -2807.439944 Hartree

Sum of electronic and thermal Free Energies = -2807.547416 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-0.800507	0.414165	2.370834	H	-1.235784	1.111464	1.658327
C	-1.471852	-0.755673	2.568098	H	-2.381287	-0.914488	1.995105
C	1.647561	-1.192235	2.402288	H	2.538545	-1.066529	1.788594
C	0.948739	-2.365921	2.260517	H	1.336549	-3.088542	1.545925
C	0.412570	0.930261	3.106533	H	0.893852	1.677966	2.467405
H	0.091386	1.462346	4.012480	C	1.447636	-0.151442	3.485597
H	1.171968	-0.645177	4.421176	H	2.404442	0.346006	3.669299
C	-0.202334	-2.879398	3.091427	H	0.190994	-3.473437	3.928395
H	-0.772790	-3.564184	2.456707	C	-1.159956	-1.794334	3.619172
H	-2.093294	-2.279467	3.919547	H	-0.756107	-1.314853	4.514495
Ni	0.087132	-0.955939	0.955320	C	-1.454168	-0.716588	-0.733712
C	-0.091875	-0.402731	-1.021054	C	-0.016291	1.021762	-0.943918
C	-1.300657	1.580554	-0.788792	C	-2.199630	0.486924	-0.699872
C	1.186728	-1.021296	-0.778785	C	2.117135	0.145306	-0.715667
C	1.381487	1.300298	-0.763534	C	-3.648815	0.262959	-0.662924
C	-6.305073	-0.659019	-0.564497	C	-4.705836	1.173566	-0.646471
C	-3.969984	-1.106090	-0.662931	C	-5.263891	-1.586862	-0.605299
C	-6.018307	0.704278	-0.597102	H	-4.509920	2.239233	-0.681868
H	-5.459533	-2.654656	-0.606762	H	-6.833346	1.421917	-0.585510
H	-7.334208	-1.000332	-0.523567	C	3.569197	0.366291	-0.599625
C	6.202836	1.312979	-0.323963	C	4.643047	-0.515111	-0.703522
C	3.861569	1.725459	-0.392347	C	5.140832	2.218886	-0.253179
C	5.946998	-0.033750	-0.557914	H	4.480746	-1.560654	-0.931503
H	5.313287	3.280309	-0.105603	H	6.776152	-0.729752	-0.642772
H	7.223479	1.665350	-0.217076	S	-2.510242	-2.129997	-0.820168
S	2.370557	2.716709	-0.409491	O	2.419090	3.674041	-1.508994
O	2.059737	3.198284	0.937536	O	-2.337169	-2.979878	0.363614
O	-2.474910	-2.765666	-2.135298	C	1.533322	-2.409175	-1.174924

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	0.788972	-3.051581	-2.169979	C	2.567182	-3.122877	-0.553097
C	1.066978	-4.367758	-2.526669	H	-0.019060	-2.522803	-2.663578
C	2.853828	-4.432592	-0.918446	H	3.139334	-2.657189	0.244016
C	2.101960	-5.061837	-1.908814	H	0.466075	-4.846959	-3.293015
H	3.658436	-4.965878	-0.420511	H	2.320203	-6.087096	-2.191532
C	-1.571251	3.014267	-0.606785	C	-0.878051	3.959523	-1.371450
C	-2.463834	3.464979	0.374855	C	-1.077621	5.319726	-1.163653
H	-0.174242	3.626041	-2.127608	C	-2.670063	4.825104	0.574325
H	-2.981758	2.742311	1.000287	C	-1.977997	5.757083	-0.196189
H	-0.521980	6.037291	-1.759001	H	-3.361311	5.158118	1.343089
H	-2.133059	6.819715	-0.035382				



[Ni(cod)(1)]: **Int**

#p opt=(maxstep=5,notrust) freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2808.09725768 Hartree

Sum of electronic and zero-point Energies = -2807.491324 Hartree

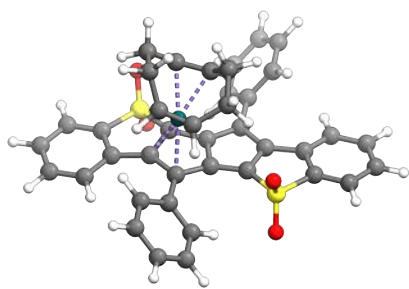
Sum of electronic and thermal Energies = -2807.453287 Hartree

Sum of electronic and thermal Enthalpies = -2807.452343 Hartree

Sum of electronic and thermal Free Energies = -2807.560317 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	0.505329	1.583008	2.256771	H	0.399507	2.392106	1.537422
C	-0.544674	0.708260	2.395177	H	-1.386155	0.831414	1.714979
C	2.128494	-0.802460	2.557661	H	3.043376	-1.067673	2.027723
C	1.053821	-1.661941	2.429117	H	1.203123	-2.555497	1.829555
C	1.730784	1.663639	3.134949	H	2.499659	2.175828	2.549184
H	1.527523	2.302491	4.005710	C	2.266236	0.292956	3.601943
H	1.767824	-0.025305	4.521745	H	3.325692	0.401884	3.854261
C	-0.237052	-1.659199	3.211528	H	-0.117458	-2.252212	4.129339
H	-0.985222	-2.173689	2.599672	C	-0.752677	-0.250447	3.545851
H	-1.824607	-0.310283	3.757558	H	-0.279148	0.143405	4.449102
Ni	0.891670	-0.109685	1.021764	C	-1.713884	-1.014304	-0.686762
C	-0.324255	-0.723852	-0.778545	C	-0.245426	0.728445	-0.852466
C	-1.510426	1.298717	-0.789674	C	-2.415439	0.183094	-0.704594
C	0.975707	-1.293522	-0.733649	C	1.917954	-0.171270	-0.807370
C	1.177788	1.021768	-0.719334	C	-3.875058	0.005741	-0.623848
C	-6.539361	-0.835492	-0.443588	C	-4.902489	0.937215	-0.742939
C	-4.216228	-1.341131	-0.434190	C	-5.519361	-1.785231	-0.344153
C	-6.227231	0.506291	-0.646956	H	-4.677338	1.983475	-0.918025
H	-5.740550	-2.839141	-0.208576	H	-7.027838	1.233930	-0.740039
H	-7.576221	-1.148304	-0.373781	C	3.366793	0.027272	-0.995215
C	6.005165	0.925311	-1.306387	C	4.395635	-0.893226	-1.174781
C	3.689908	1.391429	-1.026268	C	4.979963	1.862314	-1.162182
C	5.705558	-0.433584	-1.326580	H	4.188092	-1.954924	-1.220380
H	5.183230	2.928397	-1.172808	H	6.503240	-1.156270	-1.468401
H	7.030663	1.259645	-1.423941	S	-2.756616	-2.385714	-0.395859
S	2.234255	2.429504	-0.940332	O	1.989840	3.042398	-2.242090
O	2.290955	3.287111	0.241461	O	-2.538594	-2.924353	0.950756
O	-2.800383	-3.335812	-1.507699	C	1.277883	-2.734270	-0.819614

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
C	0.435730	-3.556515	-1.574786	C	2.370884	-3.313172	-0.161360
C	0.680370	-4.922845	-1.666498	H	-0.419409	-3.133793	-2.090895
C	2.619125	-4.676356	-0.260734	H	3.018694	-2.695804	0.453762
C	1.772845	-5.486451	-1.015722	H	0.004872	-5.543099	-2.246370
H	3.466872	-5.108288	0.262634	H	1.961583	-6.553168	-1.087706
C	-1.802834	2.735593	-0.657427	C	-1.144962	3.671938	-1.463413
C	-2.685843	3.202516	0.326880	C	-1.360210	5.035161	-1.286554
H	-0.456196	3.329976	-2.229428	C	-2.913548	4.564304	0.490302
H	-3.186072	2.490095	0.977961	C	-2.248660	5.486869	-0.315042
H	-0.830053	5.743884	-1.915350	H	-3.600407	4.906303	1.259293
H	-2.418925	6.550977	-0.181267				



[Ni(cod)(1)]: TS2

#p opt=(calcf,ts,noeigen) freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 1, $\nu_i = -41.70$

Electronic Energy = -2808.09550112 Hartree

Sum of electronic and zero-point Energies = -2807.489648 Hartree

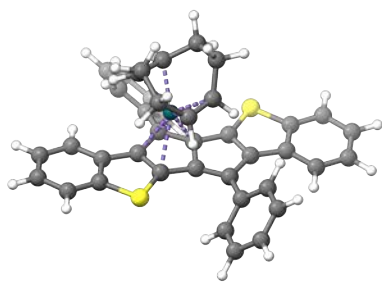
Sum of electronic and thermal Energies = -2807.452430 Hartree

Sum of electronic and thermal Enthalpies = -2807.451486 Hartree

Sum of electronic and thermal Free Energies = -2807.557041 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-1.056303	-1.593417	2.420378	H	-1.258222	-2.440155	1.769050
C	0.228326	-1.110020	2.446804	H	0.932192	-1.573597	1.759673
C	-1.794673	1.206348	2.656050	H	-2.619897	1.717752	2.168397
C	-0.528976	1.674145	2.387752	H	-0.444354	2.529747	1.720721
C	-2.188806	-1.216538	3.343724	H	-3.121569	-1.434442	2.812099
H	-2.181012	-1.869058	4.227849	C	-2.185192	0.261725	3.776491
H	-1.523095	0.414378	4.632205	H	-3.189671	0.528791	4.119853
C	0.759459	1.288813	3.069029	H	0.921662	1.941078	3.938440
H	1.568924	1.515537	2.368741	C	0.830376	-0.189664	3.486144
H	1.881469	-0.463739	3.619225	H	0.347471	-0.353196	4.453025
Ni	-0.950359	0.058135	1.062369	C	1.692772	1.008872	-0.682637
C	0.316021	0.707929	-0.779483	C	0.238285	-0.735635	-0.822523
C	1.515788	-1.301251	-0.740747	C	2.409798	-0.190403	-0.679464
C	-1.002002	1.269946	-0.699843	C	-1.938230	0.166830	-0.789027
C	-1.182803	-1.027444	-0.654629	C	3.867910	0.004555	-0.640487
C	6.530068	0.880339	-0.540561	C	4.905926	-0.918716	-0.741710
C	4.201351	1.361075	-0.512072	C	5.501341	1.821651	-0.460956
C	6.227092	-0.471174	-0.686882	H	4.690974	-1.973464	-0.872403
H	5.712522	2.882566	-0.371201	H	7.033183	-1.194511	-0.766686
H	7.564628	1.205811	-0.501320	C	-3.372025	-0.039558	-1.046616
C	-5.987974	-0.953071	-1.497231	C	-4.400295	0.875569	-1.263918
C	-3.684377	-1.404685	-1.115992	C	-4.964071	-1.883534	-1.316496
C	-5.696429	0.408668	-1.485761	H	-4.200305	1.939697	-1.282278
H	-5.157452	-2.951006	-1.349883	H	-6.491694	1.127815	-1.656714
H	-7.004282	-1.292380	-1.667837	S	2.738086	2.397636	-0.500352
S	-2.230366	-2.434681	-0.986301	O	-1.893861	-2.977779	-2.297911
O	-2.356263	-3.349418	0.146033	O	2.545961	3.025284	0.812686
O	2.752586	3.277889	-1.670673	C	-1.305513	2.712370	-0.752422

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
C	-0.449912	3.558493	-1.465152	C	-2.414367	3.268861	-0.102298
C	-0.693876	4.926823	-1.517899	H	0.413065	3.155187	-1.983210
C	-2.659826	4.634676	-0.159055	H	-3.085048	2.628543	0.460190
C	-1.797147	5.469362	-0.867433	H	-0.008193	5.564132	-2.066381
H	-3.520361	5.048765	0.357741	H	-1.983916	6.538155	-0.906460
C	1.822802	-2.733524	-0.594463	C	1.173441	-3.690872	-1.382476
C	2.722255	-3.174224	0.387655	C	1.413700	-5.047716	-1.190046
H	0.474158	-3.369792	-2.147785	C	2.972877	-4.529985	0.568349
H	3.219189	-2.443432	1.021328	C	2.316473	-5.473174	-0.219441
H	0.891872	-5.773395	-1.806498	H	3.672420	-4.850649	1.335143
H	2.504936	-6.532689	-0.073927				



[Ni(cod)(2)]

#p opt freq wb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2507.44777351 Hartree

Sum of electronic and zero-point Energies = -2506.860494 Hartree

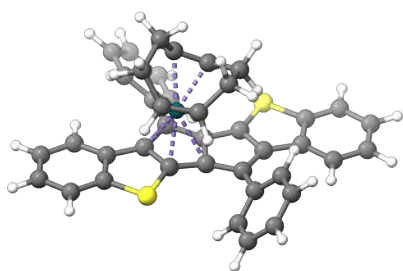
Sum of electronic and thermal Energies = -2506.826051 Hartree

Sum of electronic and thermal Enthalpies = -2506.825107 Hartree

Sum of electronic and thermal Free Energies = -2506.926090 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
H	-5.075091	0.832486	-0.790026	C	-5.028331	-0.240037	-0.632770
C	-4.931005	-3.030967	-0.275468	C	-3.782017	-0.882249	-0.571310
C	-6.193763	-0.979552	-0.506262	C	-6.149123	-2.367364	-0.321706
C	-3.759415	-2.287677	-0.405164	H	-7.154605	-0.475786	-0.556160
H	-7.071885	-2.931053	-0.223258	H	-4.891541	-4.108693	-0.147088
C	-2.442512	-0.350782	-0.675056	C	-1.800108	0.949181	-0.837367
C	-1.477782	-1.349145	-0.624937	S	-2.124592	-2.947584	-0.416187
C	0.865476	1.306973	-0.903778	C	-0.451073	0.684493	-0.910922
C	1.219063	-1.003138	-0.756777	C	1.861224	0.328476	-0.926700
C	-0.184486	-0.742928	-0.759622	S	1.476342	2.926735	-1.003716
C	3.126001	2.309662	-1.054626	C	5.588435	1.067377	-1.220479
C	4.274072	3.095208	-1.128653	C	3.188290	0.896159	-1.031486
C	4.446488	0.285482	-1.139529	C	5.508011	2.464890	-1.204683
H	4.203054	4.178702	-1.137040	H	4.524678	-0.794136	-1.184843
H	6.413084	3.060952	-1.267457	H	6.558345	0.586754	-1.305612
Ni	0.838779	-0.110053	0.985130	C	-0.693400	0.400947	2.396403
H	-1.575641	0.303708	1.764161	C	0.105092	1.496404	2.188917
H	-0.237886	2.219408	1.451470	C	2.298719	-0.377330	2.450999
H	3.230216	-0.407131	1.885890	C	1.482359	-1.493292	2.388082
H	1.836754	-2.341907	1.808564	C	2.189008	0.740990	3.474162
H	1.808599	0.339999	4.417874	H	3.192841	1.120024	3.692190
C	1.315083	1.916843	2.990742	H	1.919222	2.559295	2.339208
H	1.013964	2.540278	3.844769	C	-0.593809	-0.556227	3.565243
H	-1.601780	-0.892794	3.827916	H	-0.208958	-0.031890	4.444002
C	0.270207	-1.790678	3.241986	H	0.572383	-2.291867	4.172951
H	-0.342274	-2.509972	2.687113	C	1.840940	-2.328838	-0.959935
C	2.955248	-4.870670	-1.394316	C	1.210106	-3.253278	-1.800063
C	3.040765	-2.701711	-0.341980	C	3.595007	-3.958111	-0.558524

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
C	1.760282	-4.513800	-2.011756	H	0.287374	-2.973617	-2.299516
H	3.536462	-2.006875	0.328950	H	4.524631	-4.227326	-0.065735
H	1.253139	-5.215597	-2.666900	H	3.385731	-5.853348	-1.561274
C	-2.405078	2.286081	-0.764583	C	-3.452878	4.886617	-0.551537
C	-1.993918	3.300872	-1.639260	C	-3.354532	2.600277	0.218481
C	-3.876904	3.884928	0.319645	C	-2.506973	4.589546	-1.529072
H	-1.278944	3.065016	-2.422964	H	-3.669992	1.829575	0.916667
H	-4.609892	4.107704	1.089884	H	-2.173951	5.360775	-2.217694
H	-3.858465	5.890607	-0.469268				



#p opt freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2507.44473069 Hartree

Sum of electronic and zero-point Energies = -2506.857360 Hartree

Sum of electronic and thermal Energies = -2506.823063 Hartree

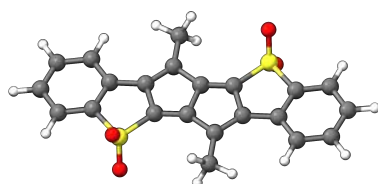
Sum of electronic and thermal Enthalpies = -2506.822119 Hartree

Sum of electronic and thermal Free Energies = -2506.922317 Hartree

[Ni(cod)(2')]

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-1.423515	-1.740338	2.171795	H	-1.936164	-2.425845	1.499470
C	-0.052967	-1.705134	2.086623	H	0.407500	-2.337865	1.329848
C	-1.277303	1.120775	2.639505	H	-1.949470	1.894851	2.278817
C	0.029403	1.213844	2.232023	H	0.281896	2.059389	1.595060
C	-2.303651	-1.125204	3.234142	H	-3.299568	-0.996719	2.795084
H	-2.423880	-1.826214	4.072462	C	-1.808411	0.236291	3.749266
H	-1.042316	0.103490	4.517215	H	-2.641032	0.751105	4.239941
C	1.220556	0.407450	2.691010	H	1.714559	0.915546	3.531220
H	1.938668	0.415399	1.861703	C	0.905418	-1.055153	3.062837
H	1.841530	-1.622577	3.051464	H	0.515522	-1.125928	4.081870
Ni	-0.899175	-0.135713	0.886695	C	-0.907395	-1.166721	-0.944997
C	0.402970	-0.578510	-1.023487	C	0.188382	0.849290	-0.929869
C	-1.239107	1.102110	-0.739792	C	-1.919516	-0.146382	-0.946554
C	1.775531	-0.878555	-0.909494	C	2.423901	0.379450	-0.790488
C	1.458926	1.418300	-0.786794	C	-3.253786	-0.692822	-1.124596
C	-5.588210	-2.207452	-1.466067	C	-4.494898	-0.056518	-1.234559
C	-3.208766	-2.100379	-1.238862	C	-4.362546	-2.860705	-1.389699
C	-5.647771	-0.813682	-1.401117	H	-4.555042	1.025481	-1.203403
H	-4.306585	-3.942714	-1.462198	H	-6.606764	-0.311568	-1.485468
H	-6.498263	-2.785618	-1.591641	C	3.773707	0.895081	-0.683495
C	6.168970	2.333870	-0.423586	C	5.006100	0.228398	-0.716527
C	3.777939	2.301687	-0.539894	C	4.961394	3.021739	-0.405077
C	6.187619	0.944821	-0.585245	H	5.032242	-0.847687	-0.854624
H	4.941924	4.102056	-0.294628	H	7.137678	0.419240	-0.613418
H	7.100882	2.881666	-0.321029	S	-1.562477	-2.762791	-1.199455
S	2.157402	3.008858	-0.580591	C	2.326359	-2.226485	-0.743483
C	1.880418	-3.292891	-1.537030	C	3.246736	-2.503351	0.279949
C	2.330928	-4.589837	-1.310923	H	1.189029	-3.091363	-2.351057

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
C	3.706896	-3.797446	0.496788	H	3.588798	-1.690143	0.915101
C	3.247397	-4.848609	-0.294752	H	1.971587	-5.399911	-1.939216
H	4.419102	-3.988210	1.294893	H	3.602611	-5.860105	-0.121794
C	-1.849514	2.442063	-0.653931	C	-1.255390	3.514267	-1.327259
C	-3.016552	2.674790	0.084519	C	-1.813185	4.787531	-1.261362
H	-0.362011	3.343018	-1.920416	C	-3.574931	3.945680	0.148105
H	-3.478807	1.850148	0.619962	C	-2.972585	5.007937	-0.523917
H	-1.338549	5.607721	-1.791104	H	-4.478485	4.108904	0.728148
H	-3.405753	6.002163	-0.471203				



1^{Me}

#p opt freq wb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -1943.99311577 Hartree

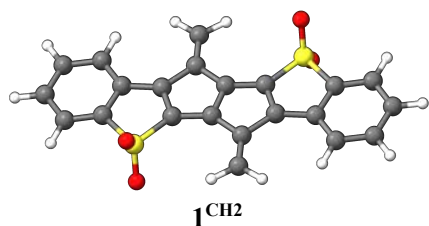
Sum of electronic and zero-point Energies = -1943.132886 Hartree

Sum of electronic and thermal Energies = -1943.109642 Hartree

Sum of electronic and thermal Enthalpies = -1943.108698 Hartree

Sum of electronic and thermal Free Energies = -1943.187222 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-0.237681	-0.696346	-0.000121	C	0.237675	0.696320	-0.000155
C	1.573052	0.614430	-0.000164	C	2.005877	-0.814832	-0.000099
C	0.912405	-1.619557	-0.000069	C	-0.912407	1.619539	-0.000148
C	-2.005877	0.814816	-0.000090	C	-1.573057	-0.614444	-0.000076
C	3.463046	-0.927512	-0.000012	C	6.256581	-0.689873	0.000182
C	4.261930	-2.073429	-0.000005	C	4.102504	0.323927	0.000065
C	5.480183	0.464814	0.000156	C	5.646246	-1.945461	0.000101
H	3.810241	-3.058721	-0.000096	H	5.936200	1.449641	0.000200
H	6.261626	-2.839679	0.000114	H	7.338769	-0.612402	0.000259
C	-3.463035	0.927515	-0.000008	C	-6.256582	0.689912	0.000228
C	-4.261909	2.073440	0.000070	C	-4.102518	-0.323918	0.000047
C	-5.480194	-0.464783	0.000159	C	-5.646222	1.945495	0.000183
H	-3.810200	3.058724	0.000048	H	-5.936239	-1.449597	0.000209
H	-6.261604	2.839712	0.000245	H	-7.338769	0.612468	0.000317
S	-2.963480	-1.707347	-0.000043	S	2.963463	1.707339	0.000018
C	-0.757496	3.102018	-0.000212	H	-1.714891	3.624480	-0.000346
H	-0.186168	3.421475	-0.879725	H	-0.186333	3.421592	0.879365
C	0.757539	-3.102044	-0.000011	H	0.186547	-3.421639	-0.879692
H	0.186038	-3.421511	0.879386	H	1.714952	-3.624472	0.000313
O	-3.040582	-2.439582	1.258666	O	-3.040678	-2.439461	-1.258817
O	3.040460	2.439454	1.258795	O	3.040769	2.439578	-1.258685



#p opt freq wb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1942.75755652 Hartree

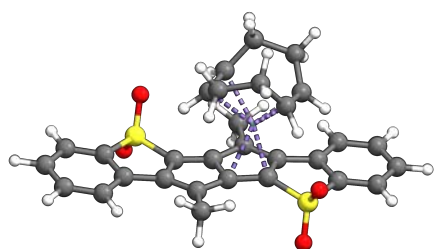
Sum of electronic and zero-point Energies = -1941.917571 Hartree

Sum of electronic and thermal Energies = -1941.895563 Hartree

Sum of electronic and thermal Enthalpies = -1941.894619 Hartree

Sum of electronic and thermal Free Energies = -1941.969522 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-0.237028	0.641842	0.000016	C	0.237015	-0.641815	-0.000000
C	-0.878335	-1.595714	0.000066	C	-2.083131	-0.731165	0.000151
C	-1.674550	0.565418	0.000143	C	1.674545	-0.565401	0.000131
C	2.083126	0.731170	0.000182	C	0.878335	1.595746	0.000195
C	0.791053	2.930567	0.000512	H	1.668586	3.567385	0.000769
H	-0.175894	3.425443	0.000554	C	-0.791012	-2.930537	-0.000099
H	0.175954	-3.425376	-0.000272	H	-1.668519	-3.567391	-0.000107
C	-3.539753	-0.913330	0.000050	C	-6.332562	-0.740572	-0.000109
C	-4.206487	0.322392	-0.000111	C	-4.302930	-2.075228	0.000148
C	-5.695874	-1.977547	0.000056	C	-5.579889	0.437177	-0.000180
H	-3.832374	-3.052013	0.000319	H	-6.290005	-2.886013	0.000132
H	-6.058677	1.411082	-0.000297	H	-7.416126	-0.686829	-0.000177
C	3.539756	0.913332	0.000056	C	6.332570	0.740529	-0.000252
C	4.206479	-0.322393	-0.000048	C	4.302959	2.075216	-0.000038
C	5.695896	1.977518	-0.000184	C	5.579881	-0.437206	-0.000184
H	3.832420	3.052010	-0.000008	H	6.290051	2.885969	-0.000258
H	6.058660	-1.411115	-0.000260	H	7.416134	0.686781	-0.000375
S	-3.039124	1.689829	-0.000103	S	3.039112	-1.689826	0.000045
O	3.100185	-2.426512	1.257297	O	3.099964	-2.426546	-1.257209
O	-3.100278	2.426638	1.257083	O	-3.099923	2.426424	-1.257431



[Ni(cod)(1^{Me})]

#p opt freq wb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -2424.75012644 Hartree

Sum of electronic and zero-point Energies = -2424.252705 Hartree

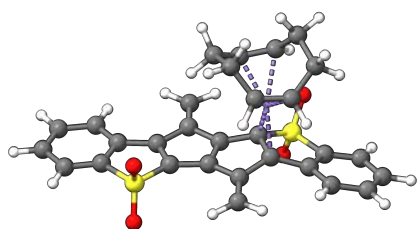
Sum of electronic and thermal Energies = -2424.220331 Hartree

Sum of electronic and thermal Enthalpies = -2424.219386 Hartree

Sum of electronic and thermal Free Energies = -2424.315234 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-2.423097	1.341770	1.682042	H	-3.270964	0.934818	1.132228
C	-1.795421	0.488411	2.562360	H	-2.167316	-0.532604	2.600789
C	0.229184	2.388713	1.145446	H	0.660480	2.686467	0.191249
C	0.857507	1.345056	1.779853	H	1.716616	0.916765	1.267886
C	-2.244401	2.838863	1.563755	H	-2.578652	3.123827	0.559378
H	-2.912130	3.355945	2.267293	C	-0.793002	3.324362	1.760231
H	-0.574135	3.465494	2.821952	H	-0.688117	4.310420	1.297211
C	0.638666	0.852957	3.190209	H	1.275036	1.419508	3.884215
H	0.988176	-0.184506	3.226613	C	-0.828773	0.902771	3.653797
H	-0.951845	0.224050	4.503495	H	-1.086197	1.899866	4.020709
Ni	-0.756361	0.539803	0.693774	C	-1.280428	-1.081047	-0.292718
C	0.129366	-1.148439	-0.288403	C	0.576540	-0.057727	-1.151071
C	-0.584225	0.670724	-1.676639	C	-1.698585	-0.022533	-1.246081
C	1.278022	-1.810445	0.284785	C	2.368146	-1.142381	-0.236204
C	1.943936	-0.082881	-1.124612	C	-3.149155	0.020420	-1.424321
C	-5.935043	-0.260575	-1.446859	C	-3.933751	0.943820	-2.115497
C	-3.790530	-1.040725	-0.766127	C	-5.166192	-1.195141	-0.756300
C	-5.318179	0.793770	-2.121901	H	-3.473391	1.768895	-2.647662
H	-5.626264	-2.022479	-0.225399	H	-5.927185	1.512650	-2.661279
H	-7.015741	-0.356030	-1.462219	C	3.819732	-1.140515	-0.033447
C	6.587569	-0.729910	0.093109	C	4.623120	-2.038343	0.668053
C	4.436015	-0.047741	-0.665454	C	5.800236	0.171700	-0.623386
C	5.998944	-1.822345	0.729182	H	4.184784	-2.907101	1.148057
H	6.241280	1.020878	-1.135837	H	6.623157	-2.523543	1.275016
H	7.661546	-0.583751	0.147968	S	-2.643697	-2.160475	0.040396
S	3.254350	0.986976	-1.539942	O	3.567446	1.022923	-2.966158
O	3.081072	2.273056	-0.850309	O	-2.957247	-2.262498	1.465053
O	-2.504001	-3.388776	-0.731431	C	-0.474971	1.860734	-2.573491

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
H	0.389336	2.469601	-2.289285	H	-0.317172	1.543405	-3.611050
H	-1.367327	2.490152	-2.535014	C	1.218326	-2.951427	1.250919
H	1.323800	-3.919739	0.746999	H	2.008958	-2.879992	2.005888
H	0.255429	-2.960513	1.773217				



[Ni(cod)(1^{CH2})]

#p opt freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -2423.49479311 Hartree

Sum of electronic and zero-point Energies = -2423.021041 Hartree

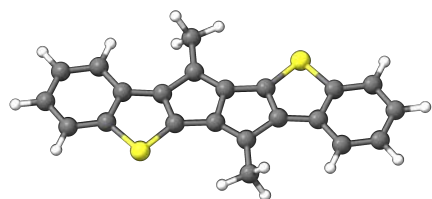
Sum of electronic and thermal Energies = -2422.989453 Hartree

Sum of electronic and thermal Enthalpies = -2422.988509 Hartree

Sum of electronic and thermal Free Energies = -2423.084430 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-1.554891	-0.704712	-1.564064	C	-1.461114	-0.688020	-2.898876
H	-0.495268	-0.797178	-3.385387	H	-2.327330	-0.572921	-3.541561
C	-2.328846	-0.597887	0.595377	C	-2.748970	-0.567778	-0.696438
C	0.203047	-0.825223	1.638222	C	0.091736	-0.889180	2.970633
H	-0.883285	-0.916483	3.447964	H	0.959897	-0.940847	3.618800
C	-0.897267	-0.775136	0.667087	C	-0.437307	-0.830081	-0.617615
C	1.424975	-0.772112	0.778452	C	1.020389	-0.781423	-0.612617
C	-4.196168	-0.403722	-0.870022	C	-6.967963	-0.064812	-0.677811
C	-4.841394	-0.257949	0.368793	C	-4.970013	-0.379433	-2.024995
C	-6.352314	-0.208843	-1.917462	C	-6.203965	-0.090213	0.493010
H	-4.516007	-0.500343	-3.002495	H	-6.955389	-0.192352	-2.820048
H	-6.666401	0.014799	1.469221	H	-8.043562	0.064019	-0.616735
C	2.771635	-1.352328	0.966916	C	5.372878	-2.396664	0.851378
C	3.329150	-1.826320	-0.229753	C	3.547476	-1.418670	2.123238
C	4.836814	-1.946127	2.057023	C	4.612577	-2.333592	-0.316299
H	3.157384	-1.056022	3.068855	H	5.434720	-2.000849	2.961939
H	5.007951	-2.679972	-1.266077	H	6.380104	-2.799231	0.820566
Ni	1.567024	0.987870	-0.068634	C	2.646692	2.241603	1.246634
H	3.052680	1.525502	1.960077	C	1.318761	2.556525	1.365390
H	0.754789	2.033809	2.137404	C	0.810960	2.341596	-1.524444
H	0.256271	1.670461	-2.178476	C	2.172454	2.342811	-1.649297
H	2.602075	1.628251	-2.350181	C	-0.042913	3.259287	-0.678802
H	-0.322901	4.154119	-1.251817	H	-0.978000	2.727573	-0.466904
C	3.126049	3.343751	-1.031496	H	2.645765	4.321828	-0.950887
H	3.976153	3.477722	-1.708033	C	3.670310	2.889494	0.342021
H	4.466308	2.154214	0.177704	H	4.140576	3.741922	0.851953
C	0.594550	3.687429	0.664106	H	-0.204314	4.047381	1.319798
H	1.272306	4.531675	0.516643	S	-3.660321	-0.317602	1.722658

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
S	2.179728	-1.683566	-1.599677	O	-3.900913	-1.475312	2.574988
O	-3.520110	0.999968	2.335703	O	1.637431	-2.989911	-1.968152
O	2.760384	-0.858793	-2.663540				



2^{Me}

#p opt freq wb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1643.15743196 Hartree

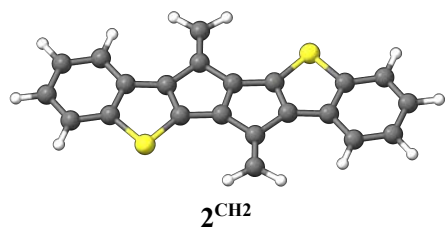
Sum of electronic and zero-point Energies = -1642.523387 Hartree

Sum of electronic and thermal Energies = -1642.504339 Hartree

Sum of electronic and thermal Enthalpies = -1642.503395 Hartree

Sum of electronic and thermal Free Energies = -1642.570710 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	-0.145124	-0.717952	-0.000028	C	0.145120	0.717961	-0.000055
C	1.597578	0.814850	-0.000087	C	2.115366	-0.457994	-0.000023
C	1.003365	-1.439973	0.000092	C	-1.003368	1.439981	0.000096
C	-2.115368	0.457999	0.000107	C	-1.597580	-0.814847	0.000030
C	-1.176615	2.920316	0.000185	H	-1.742627	3.242654	-0.882103
H	-0.213834	3.437667	0.000208	H	-1.742619	3.242552	0.882513
C	1.176608	-2.920312	0.000242	H	0.213823	-3.437654	0.000102
H	1.742458	-3.242515	0.882682	H	1.742787	-3.242667	-0.881932
C	3.549902	-0.483010	-0.000034	C	6.309849	-0.001549	-0.000010
C	4.451766	-1.559291	0.000024	C	4.071253	0.833566	-0.000061
C	5.443154	1.080493	-0.000043	C	5.813562	-1.313400	0.000024
H	4.079801	-2.579261	0.000060	H	5.824608	2.097057	-0.000078
H	6.508596	-2.147623	0.000057	H	7.381671	0.169897	-0.000009
C	-3.549908	0.483008	0.000056	C	-6.309843	0.001539	-0.000094
C	-4.451771	1.559290	0.000122	C	-4.071247	-0.833568	-0.000080
C	-5.443150	-1.080503	-0.000129	C	-5.813563	1.313395	0.000031
H	-4.079808	2.579261	0.000209	H	-5.824590	-2.097071	-0.000220
H	-6.508613	2.147605	0.000058	H	-7.381665	-0.169911	-0.000153
S	-2.798957	-2.051881	-0.000102	S	2.798963	2.051881	-0.000121



#p opt freq wb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1641.91943557 Hartree

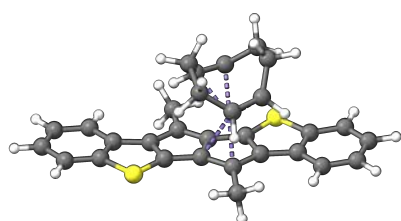
Sum of electronic and zero-point Energies = -1641.306866 Hartree

Sum of electronic and thermal Energies = -1641.288922 Hartree

Sum of electronic and thermal Enthalpies = -1641.287978 Hartree

Sum of electronic and thermal Free Energies = -1641.353029 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.169535	0.661575	0.000260	C	-0.169549	-0.661601	0.000304
C	1.040857	-1.499139	0.000009	C	2.150969	-0.519098	-0.000118
C	1.615104	0.746777	-0.000010	C	-1.615116	-0.746801	0.000150
C	-2.150983	0.519081	0.000031	C	-1.040878	1.499121	0.000036
C	-1.088278	2.834992	0.000040	H	-2.026396	3.379114	-0.000085
H	-0.178750	3.428542	0.000161	C	1.088352	-2.835008	-0.000057
H	0.178891	-3.428657	0.000189	H	2.026543	-3.379003	-0.000272
C	3.586469	-0.519506	-0.000067	C	6.339607	0.010363	-0.000075
C	4.086167	0.805628	-0.000025	C	4.508448	-1.578073	-0.000168
C	5.865941	-1.309261	-0.000142	C	5.453259	1.076190	-0.000077
H	4.159548	-2.605483	-0.000260	H	6.574679	-2.131772	-0.000170
H	5.816131	2.099548	-0.000006	H	7.408153	0.200792	-0.000022
C	-3.586476	0.519479	-0.000026	C	-6.339609	-0.010343	-0.000158
C	-4.086183	-0.805630	0.000032	C	-4.508433	1.578086	-0.000198
C	-5.865922	1.309285	-0.000216	C	-5.453298	-1.076182	-0.000033
H	-4.159480	2.605474	-0.000303	H	-6.574654	2.131800	-0.000304
H	-5.816173	-2.099539	0.000022	H	-7.408168	-0.200716	-0.000205
S	-2.798491	-2.007086	0.000139	S	2.798477	2.007104	0.000130



[Ni(cod)(2^{Me})]

#p opt freq wb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -2124.10107277 Hartree

Sum of electronic and zero-point Energies = -2123.622080 Hartree

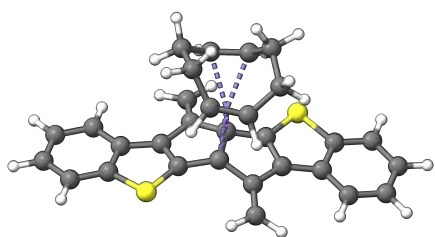
Sum of electronic and thermal Energies = -2123.593667 Hartree

Sum of electronic and thermal Enthalpies = -2123.592722 Hartree

Sum of electronic and thermal Free Energies = -2123.679733 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
H	-4.494215	-2.286263	-1.677925	C	-4.848637	-1.577165	-0.935807
C	-5.786834	0.253083	0.987743	C	-3.934736	-0.958386	-0.070304
C	-6.199837	-1.283149	-0.838561	C	-6.669215	-0.373074	0.117554
C	-4.428689	-0.040668	0.887460	H	-6.903729	-1.765185	-1.510616
H	-7.730877	-0.154617	0.181095	H	-6.148643	0.957870	1.730600
C	-2.502942	-1.090701	0.026341	C	-1.444791	-1.816328	-0.665496
C	-1.964329	-0.289786	1.022537	S	-3.146308	0.641294	1.889220
C	1.170544	-1.537371	-0.100869	C	-0.281805	-1.431669	-0.052727
C	0.669971	-0.058331	1.649630	C	1.735455	-0.810377	0.944818
C	-0.538315	-0.449750	1.008487	S	2.340674	-2.380205	-1.064182
C	3.648115	-1.712981	-0.085441	C	5.463788	-0.504374	1.617682
C	5.009213	-1.929406	-0.287504	C	3.175554	-0.889378	0.962369
C	4.109072	-0.289172	1.818817	C	5.913929	-1.318686	0.570610
H	5.356004	-2.562851	-1.098341	H	3.770257	0.336814	2.638250
H	6.978316	-1.477459	0.428672	H	6.184874	-0.038071	2.282134
Ni	0.700084	0.841861	-0.103876	C	-0.473091	1.505155	-1.745591
H	-1.367575	0.895451	-1.617937	C	0.650956	0.876905	-2.222515
H	0.550776	-0.175687	-2.480787	C	2.071663	2.407539	-0.219420
H	2.901507	2.085012	0.409175	C	0.918945	2.825018	0.424355
H	0.937631	2.852428	1.511826	C	2.406039	2.663452	-1.679318
H	1.957029	3.607793	-2.000337	H	3.488445	2.797744	-1.776374
C	1.969108	1.510718	-2.606933	H	2.726610	0.719290	-2.561309
H	1.944508	1.854822	-3.651053	C	-0.648662	3.003787	-1.603692
H	-1.696174	3.254730	-1.798733	H	-0.064826	3.519165	-2.371291
C	-0.273867	3.519141	-0.197637	H	-0.111377	4.606397	-0.231615
H	-1.126086	3.354073	0.471154	C	0.808993	0.613370	2.980492
H	-0.068472	1.231473	3.193240	H	0.898162	-0.132516	3.779146
H	1.695213	1.254819	3.022022	C	-1.621869	-2.738464	-1.830318

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
H	-2.099205	-2.231571	-2.678931	H	-0.658893	-3.129864	-2.174381
H	-2.253813	-3.597029	-1.570365				



[Ni(cod)(2^{CH2})]

#p opt freq rwb97xd/genecp

Charge = 0, Multiplicity = 1, Point group = C₁

Number of imaginary frequencies = 0

Electronic Energy = -2122.85566777 Hartree

Sum of electronic and zero-point Energies = -2122.400488 Hartree

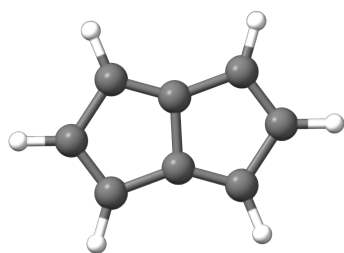
Sum of electronic and thermal Energies = -2122.372681 Hartree

Sum of electronic and thermal Enthalpies = -2122.371737 Hartree

Sum of electronic and thermal Free Energies = -2122.460089 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-1.026978	-1.058982	-1.503381	C	-1.078066	-1.283552	-2.822242
H	-0.187668	-1.198640	-3.438714	H	-1.992182	-1.584228	-3.322179
C	-1.589772	-0.975518	0.758425	C	-2.112199	-1.145811	-0.500511
C	1.029029	-1.058125	1.503518	C	1.080303	-1.282335	2.822435
H	0.189856	-1.197889	3.438901	H	1.994647	-1.582183	3.322446
C	-0.160863	-0.668630	0.703976	C	0.162630	-0.668506	-0.703882
C	2.114283	-1.144420	0.500614	C	1.591751	-0.974451	-0.758327
S	-2.779501	-1.011139	2.015981	S	2.781477	-1.009415	-2.015893
C	-3.542862	-1.296425	-0.498168	C	-6.283593	-1.575221	0.028494
C	-4.051473	-1.256669	0.822263	C	-4.450148	-1.468579	-1.555053
C	-5.802638	-1.603871	-1.287648	C	-5.410550	-1.403123	1.093051
H	-4.093787	-1.492847	-2.579436	H	-6.500007	-1.736425	-2.109457
H	-5.779057	-1.377595	2.114183	H	-7.346687	-1.686327	0.218324
C	3.545007	-1.294463	0.498238	C	6.285799	-1.572429	-0.028529
C	4.053558	-1.254506	-0.822211	C	4.452379	-1.466397	1.555085
C	5.804897	-1.601284	1.287630	C	5.412674	-1.400539	-1.093048
H	4.096061	-1.490861	2.579476	H	6.502331	-1.733692	2.109408
H	5.781136	-1.374883	-2.114193	H	7.348919	-1.683236	-0.218398
Ni	0.000499	1.160641	0.000172	C	0.591689	2.457528	1.547123
H	0.915200	1.749508	2.309277	C	-0.758422	2.617436	1.372582
H	-1.414667	2.007117	1.992058	C	-0.595296	2.454494	-1.547570
H	-0.915483	1.744882	-2.309624	C	0.753971	2.620794	-1.372735
H	1.413325	2.013237	-1.991652	C	-1.705460	3.260163	-0.909810
H	-1.931708	4.146433	-1.519394	H	-2.606950	2.637292	-0.927799
C	1.424385	3.702338	-0.550650	H	0.824958	4.615914	-0.574137
H	2.381850	3.956922	-1.016705	C	1.697976	3.268411	0.909271
H	2.602399	2.649795	0.926951	H	1.920216	4.155669	1.518896
C	-1.434452	3.695356	0.550289	H	-2.393440	3.944573	1.016099

Cartesian Coordinates				Cartesian Coordinates			
Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>	Atoms	<i>X</i>	<i>Y</i>	<i>Z</i>
H	-0.840188	4.612274	0.574071				



pentalene (C_{2h})

#p opt freq rwb97xd/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_{2h}

Number of imaginary frequencies = 0

Electronic Energy = -308.258088326 Hartree

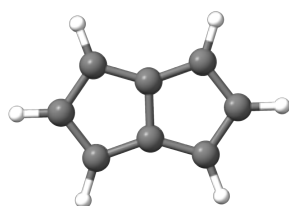
Sum of electronic and zero-point Energies = -308.146150 Hartree

Sum of electronic and thermal Energies = -308.140534 Hartree

Sum of electronic and thermal Enthalpies = -308.139590 Hartree

Sum of electronic and thermal Free Energies = -308.174968 Hartree

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	-0.040320	0.728824	0.000000	C	0.040320	-0.728824	0.000000
C	1.340468	-1.081697	0.000000	C	2.157003	0.171543	0.000000
C	1.340468	1.245427	0.000000	H	1.761211	-2.081240	0.000000
H	3.240165	0.187426	0.000000	H	1.636854	2.286066	0.000000
C	-1.340468	1.081697	0.000000	H	-1.761211	2.081240	0.000000
C	-2.157003	-0.171543	0.000000	H	-3.240165	-0.187426	0.000000
C	-1.340468	-1.245427	0.000000	H	-1.636854	-2.286066	0.000000



pentalene (D_{2h})

#p opt freq rwb97xd/def2svp

Charge = 0, Multiplicity = 1, Point group = D_{2h}

Number of imaginary frequencies = 1, $\nu_i = -2646.15$

Electronic Energy = -308.025015262 Hartree

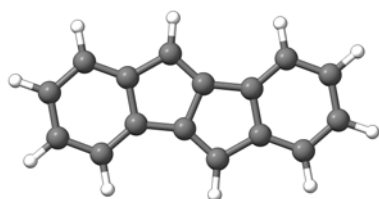
Sum of electronic and zero-point Energies = -307.916333 Hartree

Sum of electronic and thermal Energies = -307.910744 Hartree

Sum of electronic and thermal Enthalpies = -307.909800 Hartree

Sum of electronic and thermal Free Energies = -307.944661 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	-0.000000	0.705281	0.000000	C	-0.000000	-0.705281	0.000000
C	0.000000	-1.140548	1.350653	C	-0.000000	0.000000	2.182670
C	0.000000	1.140548	1.350653	H	0.000000	-2.173251	1.705144
H	-0.000000	0.000000	3.270714	H	0.000000	2.173251	1.705144
C	0.000000	1.140548	-1.350653	H	0.000000	2.173251	-1.705144
C	0.000000	0.000000	-2.182670	H	0.000000	0.000000	-3.270714
C	0.000000	-1.140548	-1.350653	H	0.000000	-2.173251	-1.705144



I

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -615.706279484 Hartree

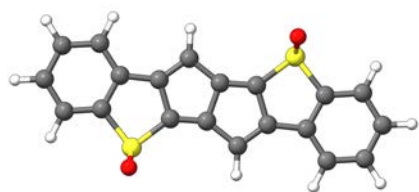
Sum of electronic and zero-point Energies = -615.500421 Hartree

Sum of electronic and thermal Energies = -615.489826 Hartree

Sum of electronic and thermal Enthalpies = -615.488882 Hartree

Sum of electronic and thermal Free Energies = -615.536883 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	-0.715586	1.593854	-0.000064	H	-0.627973	2.675436	-0.000010
C	0.286502	0.678444	-0.000146	C	-0.286509	-0.678423	-0.000146
C	0.715578	-1.593837	-0.000060	H	0.627973	-2.675420	-0.000000
C	1.744452	0.535201	-0.000063	C	2.003263	-0.870193	-0.000003
C	-1.744431	-0.535199	-0.000063	C	-2.003272	0.870195	0.000005
C	-3.308794	1.343078	0.000100	H	-3.515178	2.410481	0.000162
C	-2.793517	-1.441763	-0.000046	H	-2.606630	-2.512476	-0.000101
C	-4.366139	0.416548	0.000119	H	-5.392107	0.774495	0.000170
C	-4.112742	-0.954062	0.000048	H	-4.942440	-1.655650	0.000059
C	2.793535	1.441762	-0.000043	H	2.606665	2.512477	-0.000096
C	4.112745	0.954053	0.000045	H	4.942456	1.655625	0.000063
C	3.308780	-1.343086	0.000100	H	3.515149	-2.410494	0.000163
C	4.366135	-0.416562	0.000119	H	5.392094	-0.774529	0.000184



II

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1865.20469757 Hartree

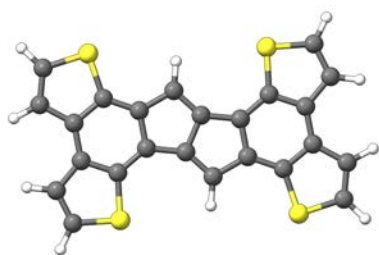
Sum of electronic and zero-point Energies = -1864.954796 Hartree

Sum of electronic and thermal Energies = -1864.935238 Hartree

Sum of electronic and thermal Enthalpies = -1864.934294 Hartree

Sum of electronic and thermal Free Energies = -1865.003480 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-0.247940	-0.703070	0.000375	C	0.247934	0.703085	0.000393
C	-0.897315	1.615233	0.000335	C	-2.006746	0.828564	0.000212
C	-1.593235	-0.614126	0.000226	H	-0.846231	2.695710	0.000358
C	1.593230	0.614146	0.000252	C	2.006736	-0.828554	0.000192
C	0.897304	-1.615218	0.000288	H	0.846218	-2.695695	0.000287
C	-3.450023	0.983671	0.000016	C	-6.250880	0.850834	-0.000393
C	-4.132868	-0.251319	-0.000096	C	-4.204632	2.166430	-0.000060
C	-5.594681	2.090332	-0.000255	C	-5.518506	-0.338043	-0.000316
H	-3.704678	3.130483	0.000009	H	-6.179628	3.005468	-0.000286
H	-6.015761	-1.303156	-0.000395	H	-7.336071	0.813591	-0.000569
C	3.450015	-0.983672	0.000008	C	6.250866	-0.850857	-0.000359
C	4.132864	0.251304	-0.000092	C	4.204612	-2.166444	-0.000075
C	5.594659	-2.090350	-0.000256	C	5.518501	0.338031	-0.000279
H	3.704633	-3.130484	0.000008	H	6.179606	-3.005487	-0.000317
H	6.015776	1.303132	-0.000354	H	7.336057	-0.813626	-0.000496
S	-3.022304	-1.678569	0.000036	S	3.022319	1.678577	0.000021
O	3.129632	2.413366	1.268381	O	3.129266	2.413272	-1.268420
O	-3.129566	-2.413309	1.268424	O	-3.129275	-2.413318	-1.268369



III

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -2513.32490141 Hartree

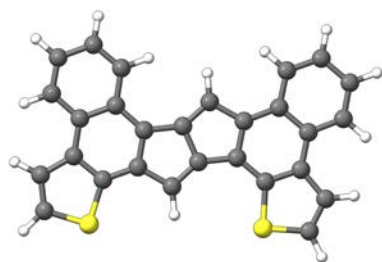
Sum of electronic and zero-point Energies = -2513.065675 Hartree

Sum of electronic and thermal Energies = -2513.045248 Hartree

Sum of electronic and thermal Enthalpies = -2513.044304 Hartree

Sum of electronic and thermal Free Energies = -2513.115124 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	0.725190	-1.589013	0.000105	H	0.644990	-2.670567	-0.000020
C	-0.283607	-0.678826	0.000274	C	0.283605	0.678827	0.000254
C	-0.725191	1.589014	-0.000021	H	-0.644987	2.670568	-0.000203
C	-1.737013	-0.541527	0.000101	C	-2.003950	0.852782	-0.000033
C	1.737015	0.541526	0.000146	C	2.003948	-0.852783	-0.000029
C	3.328731	-1.299833	-0.000175	C	2.799617	1.446880	0.000210
C	4.418485	-0.387714	-0.000150	C	4.150278	1.004414	0.000039
C	-2.799615	-1.446882	0.000165	C	-4.150281	-1.004414	0.000048
C	-3.328731	1.299838	-0.000152	C	-4.418480	0.387714	-0.000116
C	5.071469	2.108185	0.000164	H	6.149780	1.993158	0.000104
C	4.454618	3.319197	0.000136	H	4.918327	4.296828	0.000093
C	-5.561648	2.423232	-0.000444	H	-6.355460	3.158519	-0.000589
C	-5.684057	1.069709	-0.000314	H	-6.642231	0.561851	-0.000344
C	5.561648	-2.423235	-0.000530	H	6.355465	-3.158518	-0.000704
C	5.684055	-1.069711	-0.000316	H	6.642232	-0.561859	-0.000318
C	-5.071469	-2.108186	0.000083	H	-6.149780	-1.993149	-0.000004
C	-4.454619	-3.319197	0.000244	H	-4.918323	-4.296829	0.000299
S	-2.707374	-3.200165	0.000425	S	3.896378	-2.962913	-0.000324
S	-3.896378	2.962915	-0.000353	S	2.707374	3.200164	0.000475



IV

#p opt freq b3lyp/6-31g (d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1871.80007319 Hartree

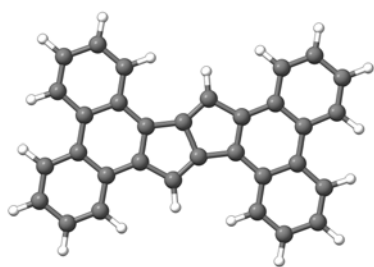
Sum of electronic and zero-point Energies = -1871.474056 Hartree

Sum of electronic and thermal Energies = -1871.452983 Hartree

Sum of electronic and thermal Enthalpies = -1871.452039 Hartree

Sum of electronic and thermal Free Energies = -1871.524010 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	-0.693677	-1.535629	-0.000040	H	-0.601933	-2.616395	-0.000051
C	0.299804	-0.612642	-0.000009	C	-0.286194	0.735968	-0.000009
C	0.725297	1.645972	-0.000025	H	0.639365	2.726092	-0.000030
C	1.753944	-0.457821	-0.000001	C	2.015444	0.922402	-0.000013
C	-1.744443	0.577995	-0.000008	C	-1.979972	-0.808120	-0.000010
C	-3.300643	-1.283523	-0.000004	C	-2.813827	1.514973	-0.000000
C	-4.396081	-0.395083	0.000001	C	-4.166984	1.021591	0.000003
C	2.824808	-1.360597	0.000010	C	4.161991	-0.906574	0.000013
C	3.343237	1.433005	-0.000009	C	4.441251	0.500574	0.000007
C	-5.229442	1.960171	0.000011	H	-6.254091	1.600581	0.000013
C	-4.986539	3.317121	0.000019	H	-5.817583	4.017051	0.000026
C	-3.658768	3.801534	0.000017	H	-3.473981	4.872301	0.000023
C	-2.601469	2.919609	0.000009	H	-1.583681	3.295933	0.000009
C	3.630999	2.824513	-0.000013	H	2.808333	3.533613	-0.000020
C	4.928914	3.284124	-0.000009	H	5.127453	4.352384	-0.000012
C	6.006099	2.368914	-0.000001	H	7.028259	2.737221	0.000002
C	5.763521	1.011964	0.000007	H	6.599735	0.319175	0.000013
C	-5.500261	-2.455748	0.000034	H	-6.279508	-3.206255	0.000057
C	-5.646990	-1.103360	0.000002	H	-6.618468	-0.622731	0.000006
C	5.088211	-2.005717	0.000015	H	6.166169	-1.892894	0.000016
C	4.480301	-3.222728	-0.000044	H	4.950633	-4.197066	-0.000083
S	2.735762	-3.110623	0.000051	S	-3.825879	-2.956648	-0.000031



#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1230.27432059 Hartree

Sum of electronic and zero-point Energies = -1229.881445 Hartree

Sum of electronic and thermal Energies = -1229.859652 Hartree

Sum of electronic and thermal Enthalpies = -1229.858708 Hartree

V

Sum of electronic and thermal Free Energies = -1229.932291 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	X	Y	Z		X	Y	Z
C	0.698101	1.590446	-0.000031	H	0.603229	2.669642	-0.000040
C	-0.300384	0.670873	-0.000018	C	0.300384	-0.670886	-0.000011
C	-0.698102	-1.590452	-0.000002	H	-0.603242	-2.669651	0.000007
C	-1.760455	0.499182	-0.000011	C	-1.994750	-0.871869	-0.000003
C	1.760458	-0.499187	-0.000018	C	1.994747	0.871859	-0.000013
C	3.317311	1.409100	-0.000003	C	2.833750	-1.438126	-0.000016
C	4.423803	0.497456	0.000018	C	4.178845	-0.939834	-0.000013
C	-2.833753	1.438125	-0.000013	C	-4.178842	0.939834	-0.000008
C	-3.317320	-1.409094	0.000010	C	-4.423806	-0.497453	0.000017
C	5.229420	-1.888480	-0.000033	H	6.260069	-1.551795	-0.000048
C	4.983388	-3.248326	-0.000044	H	5.815765	-3.946670	-0.000056
C	3.660033	-3.729867	-0.000045	H	3.468568	-4.799426	-0.000057
C	2.609250	-2.837332	-0.000032	H	1.588195	-3.203762	-0.000041
C	-3.563991	-2.804753	0.000026	H	-2.721535	-3.489919	0.000017
C	-4.850101	-3.302528	0.000049	H	-5.020527	-4.375629	0.000059
C	-5.941748	-2.413302	0.000063	H	-6.957214	-2.799723	0.000090
C	-5.727731	-1.047829	0.000046	H	-6.590703	-0.391385	0.000060
C	3.563984	2.804752	0.000004	H	2.721528	3.489923	-0.000033
C	4.850091	3.302534	0.000050	H	5.020534	4.375631	0.000072
C	5.941743	2.413306	0.000085	H	6.957208	2.799727	0.000125
C	5.727724	1.047833	0.000066	H	6.590697	0.391388	0.000100
C	-2.609240	2.837328	-0.000028	H	-1.588185	3.203757	-0.000030
C	-5.229415	1.888488	-0.000026	H	-6.260066	1.551809	-0.000031
C	-4.983376	3.248332	-0.000040	H	-5.815754	3.946673	-0.000055
C	-3.660019	3.729868	-0.000041	H	-3.468558	4.799427	-0.000049



VI

#p opt freq rb3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

Electronic Energy = -1257.18910128 Hartree

Sum of electronic and zero-point Energies = -1257.050512 Hartree

Sum of electronic and thermal Energies = -1257.040359 Hartree

Sum of electronic and thermal Enthalpies = -1257.039415 Hartree

Sum of electronic and thermal Free Energies = -1257.086946 Hartree

Atoms	Cartesian Coordinates			Atoms	Cartesian Coordinates		
	<i>X</i>	<i>Y</i>	<i>Z</i>		<i>X</i>	<i>Y</i>	<i>Z</i>
C	0.219021	-0.706079	-0.000031	C	-0.219027	0.706060	0.000044
C	0.869958	1.522006	-0.000023	C	2.051991	0.648195	-0.000146
C	1.687529	-0.695564	-0.000130	H	0.891386	2.605820	-0.000069
C	-1.687525	0.695554	0.000110	C	-2.051998	-0.648189	0.000116
C	-0.869971	-1.522018	0.000020	H	-0.891408	-2.605832	0.000021
S	3.765469	0.846295	0.000000	S	-3.765467	-0.846285	-0.000054
C	2.802435	-1.569864	0.000013	H	2.748076	-2.653173	0.000009
C	3.994729	-0.882144	0.000105	H	4.997479	-1.287232	0.000162
C	-2.802421	1.569860	0.000034	H	-2.748045	2.653177	0.000054
C	-3.994728	0.882158	-0.000001	H	-4.997485	1.287223	0.000018

#p opt freq b3lyp/6-31g(d)

Charge = 0, Multiplicity = 1, Point group = C_1

Number of imaginary frequencies = 0

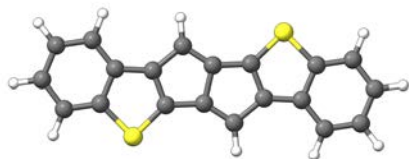
Electronic Energy = -1564.49205739 Hartree

Sum of electronic and zero-point Energies = -1564.259322 Hartree

Sum of electronic and thermal Energies = -1564.243697 Hartree

Sum of electronic and thermal Enthalpies = -1564.242753 Hartree

Sum of electronic and thermal Free Energies = -1564.302723 Hartree



VII

Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
C	0.167931	0.718560	0.000201	C	-0.167929	-0.718563	0.000188
C	0.977601	-1.456472	0.000278	C	2.115737	-0.518177	0.000166
C	1.622367	0.777527	0.000125	H	1.064833	-2.537619	0.000349
C	-1.622362	-0.777535	0.000244	C	-2.115737	0.518172	0.000292
C	-0.977599	1.456464	0.000399	H	-1.064828	2.537612	0.000484
C	3.546615	-0.582692	0.000018	C	6.329071	-0.178952	-0.000260
C	4.111247	0.725700	-0.000130	C	4.420569	-1.686871	0.000030
C	5.793138	-1.479056	-0.000114	C	5.490930	0.932867	-0.000264
H	4.017192	-2.695992	0.000129	H	6.464403	-2.333464	-0.000089
H	5.902748	1.938172	-0.000391	H	7.406081	-0.038544	-0.000383
C	-3.546613	0.582682	0.000064	C	-6.329070	0.178981	-0.000339
C	-4.111253	-0.725704	-0.000112	C	-4.420553	1.686872	0.000048
C	-5.793124	1.479078	-0.000163	C	-5.490942	-0.932848	-0.000305
H	-4.017161	2.695987	0.000188	H	-6.464382	2.333491	-0.000171
H	-5.902757	-1.938155	-0.000433	H	-7.406078	0.038564	-0.000493
S	2.863869	1.986006	-0.000064	S	-2.863881	-1.986021	-0.000023

NMR and HRMS Spectra of Newly Synthesized Compounds

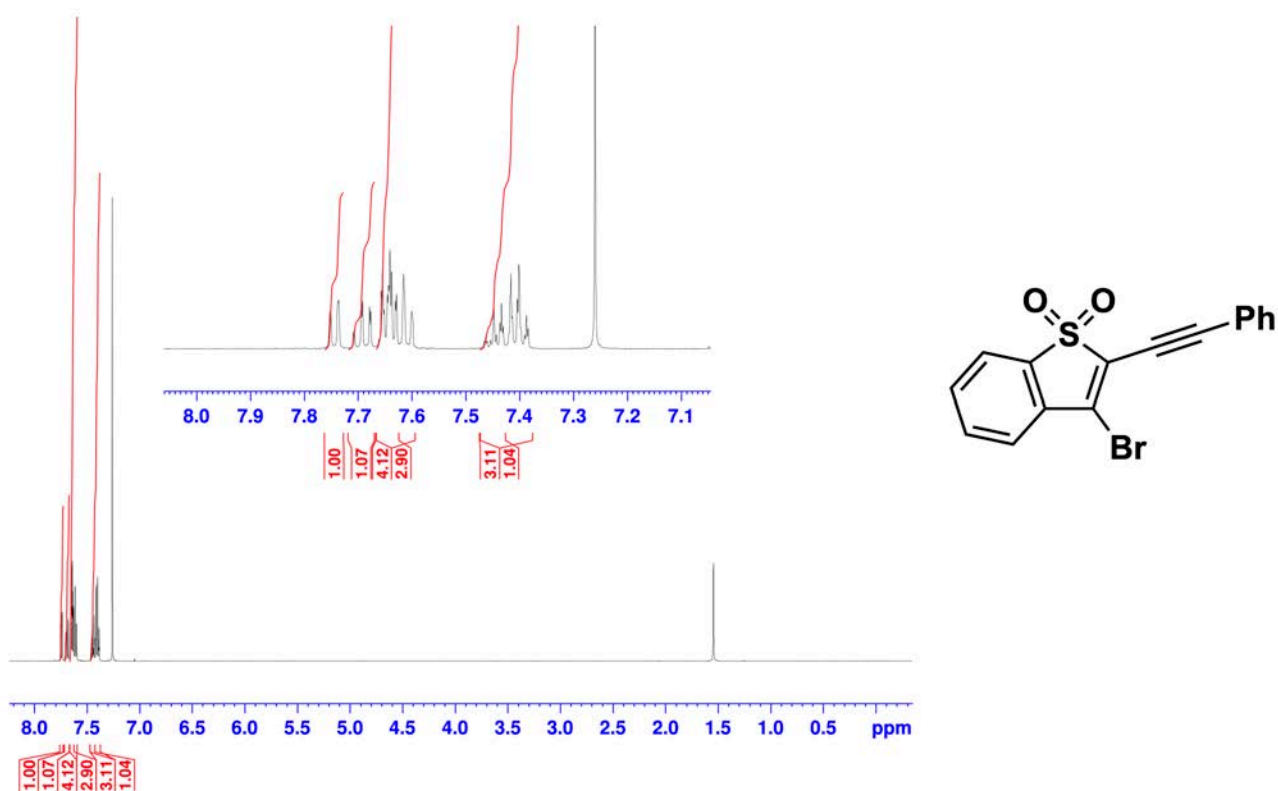


Figure S18. ¹H NMR of 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide (500 MHz, CDCl₃).

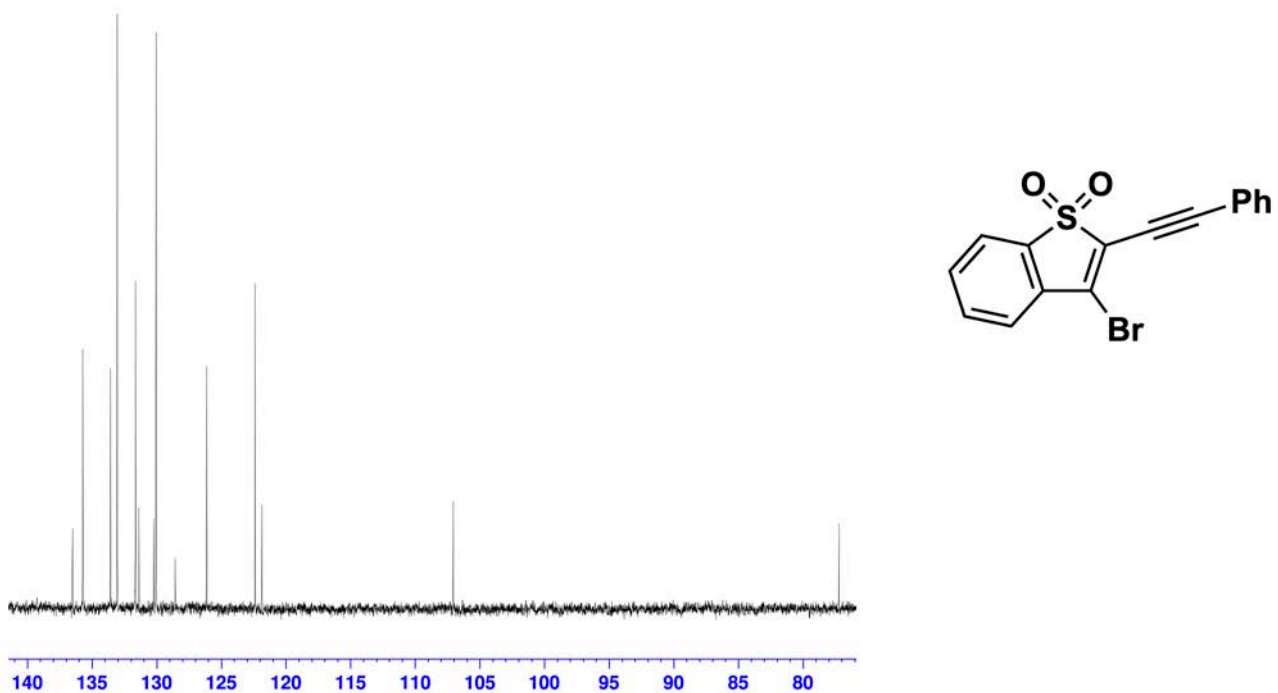


Figure S19. ¹³C{¹H} NMR spectrum of 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide (125 MHz, acetone-*d*₆).

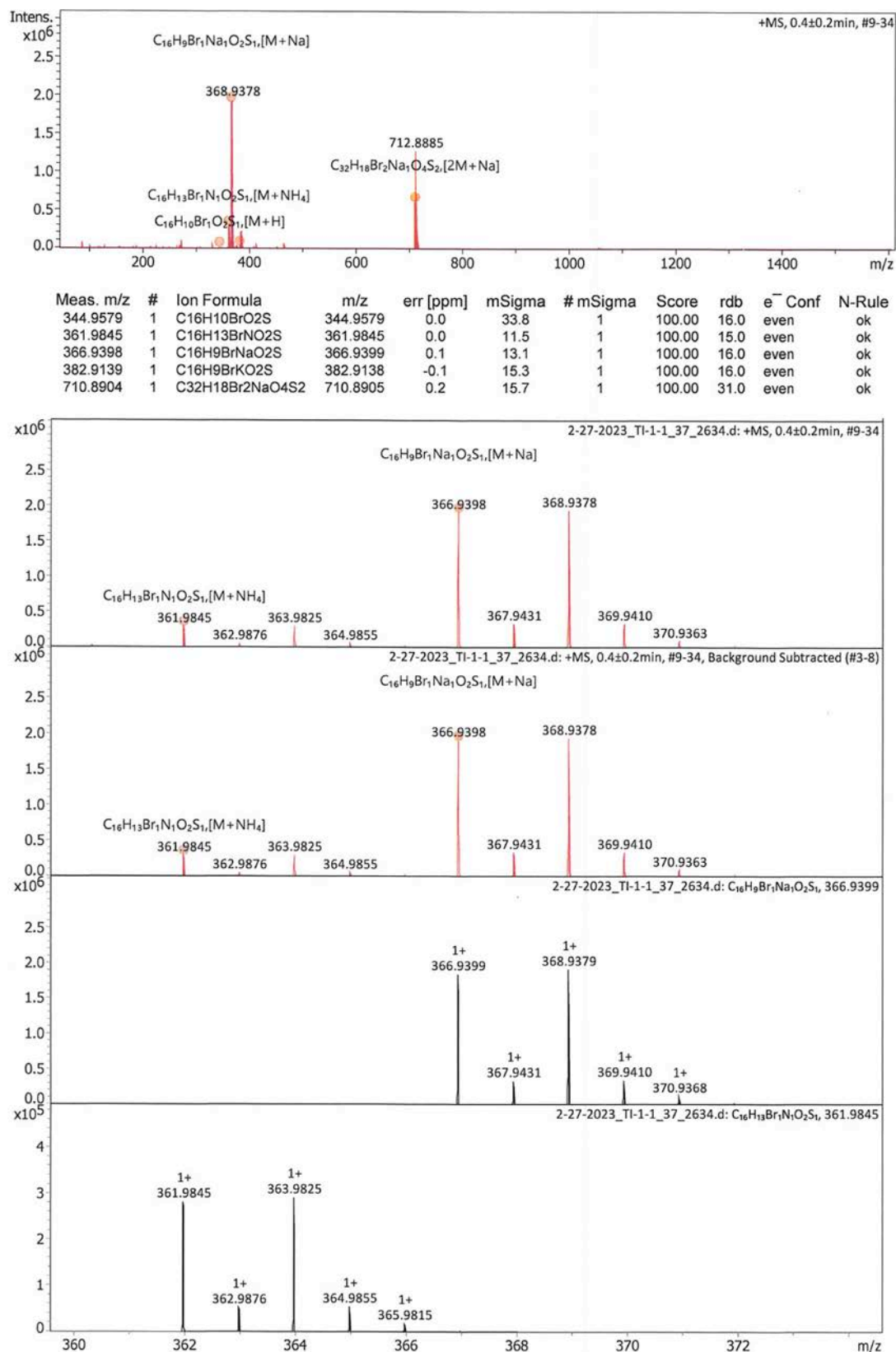


Figure S20. High-resolution mass spectrum of 3-bromo-2-(phenylethynyl)benzo[*b*]thiophene-1,1-dioxide (ESI). Top: full experimental spectrum; middle: enlarged view of the $[M+Na]^+$ and $[M+NH_4]^+$ signal; bottom: simulated spectrum.

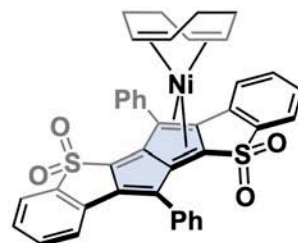
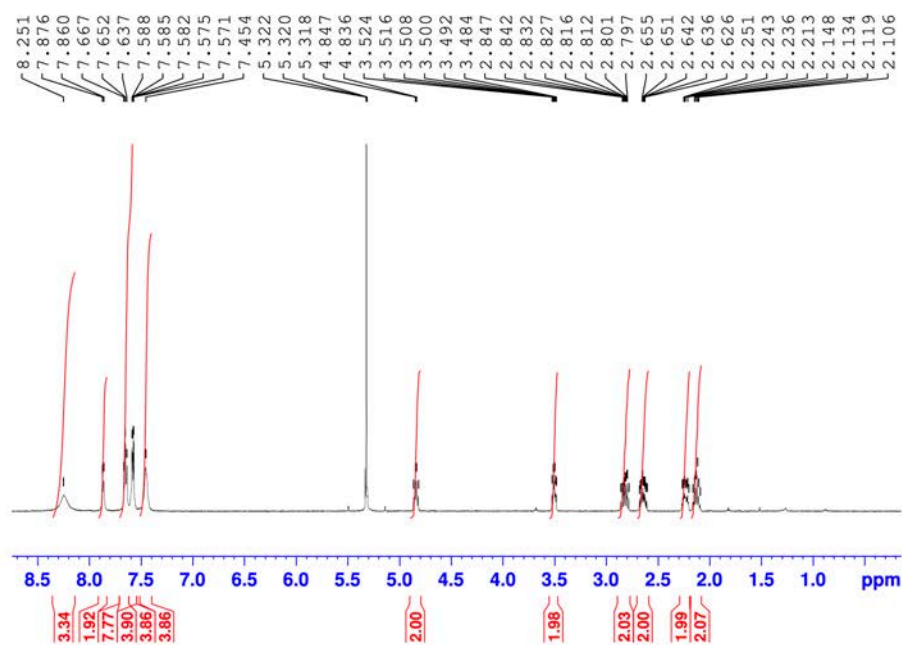


Figure S21. ¹H NMR spectrum of [Ni(cod)(1)] (500 MHz, CD₂Cl₂).

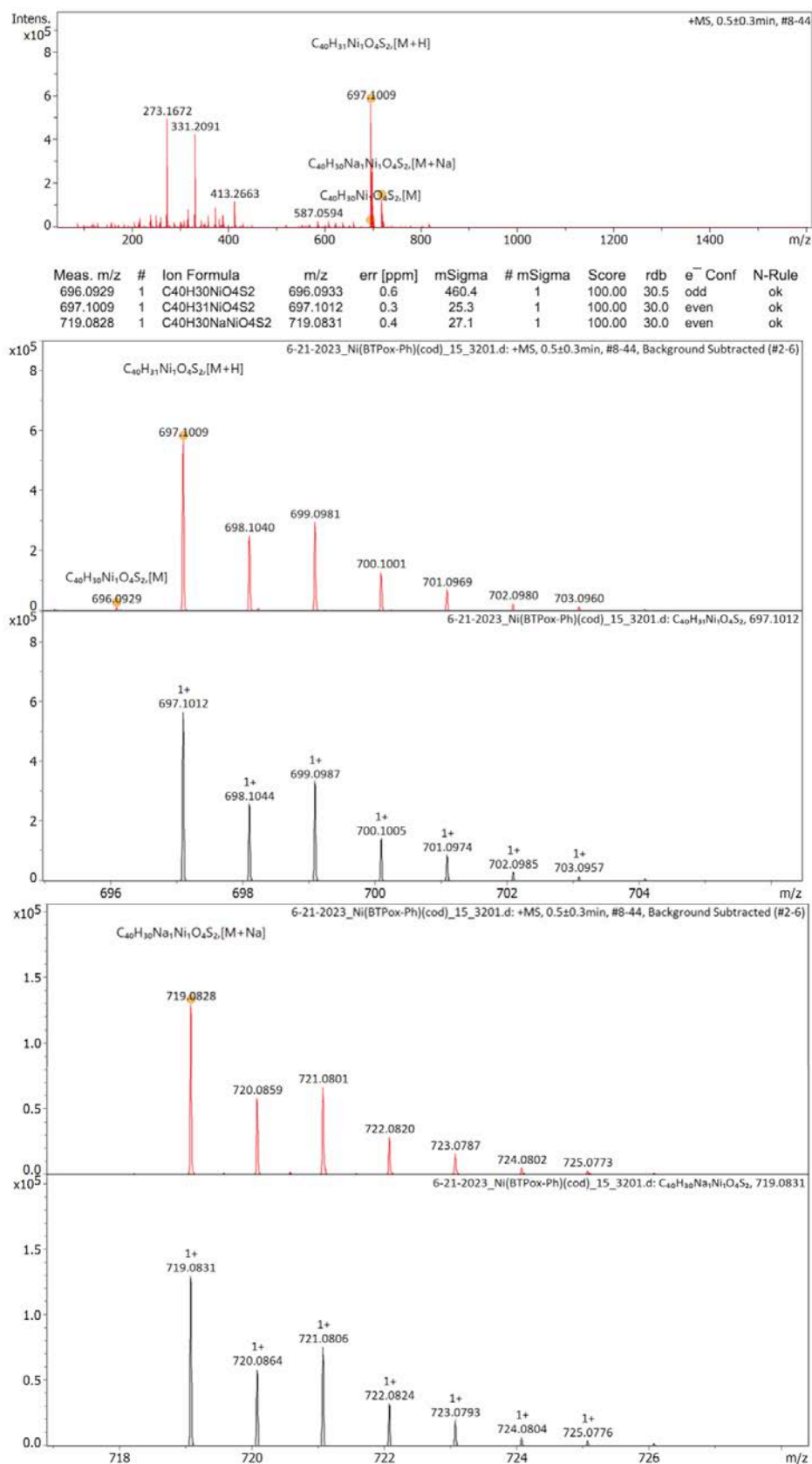


Figure S22. High-resolution mass spectrum of [Ni(cod)(1)] (ESI). Top: full experimental spectrum; middle: enlarged view and simulated spectrum of the $[M+H]^+$ signal; bottom: enlarged view and simulated spectrum of the $[M+Na]^+$ signal.

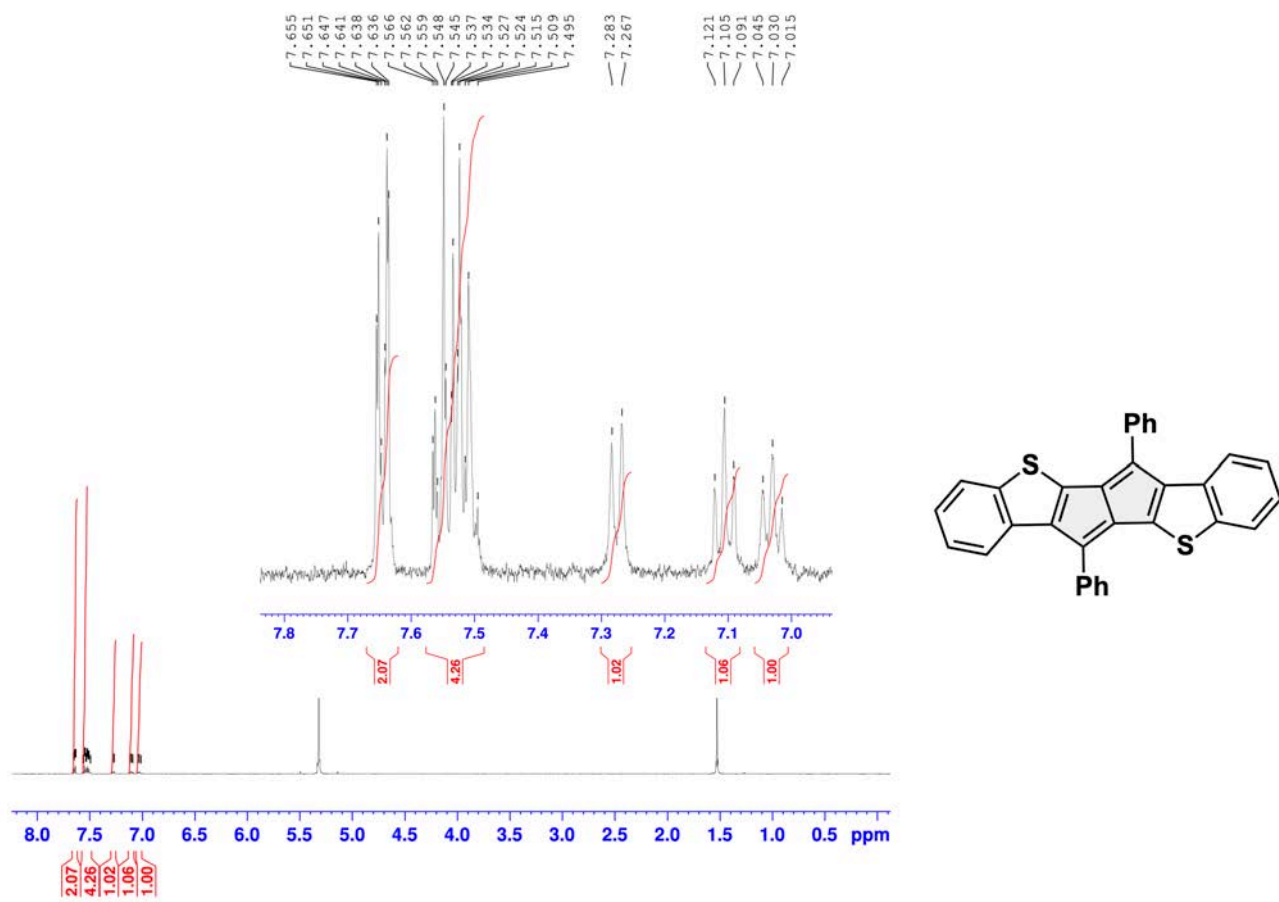


Figure S23. ¹H NMR spectrum of **2** (500 MHz, CD₂Cl₂).

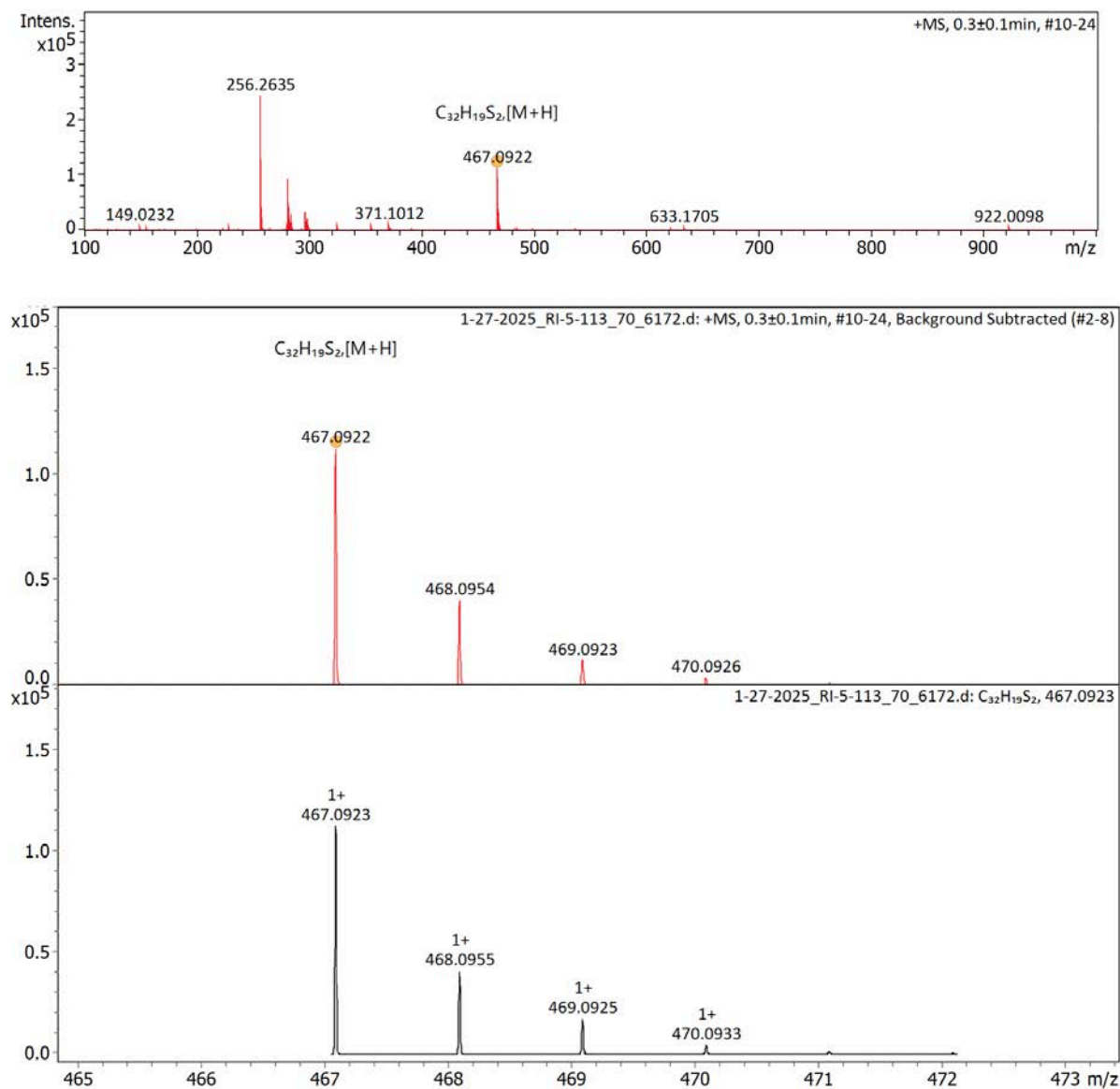


Figure S24. High-resolution mass spectrum of **2** (APCI). Top: full experimental spectrum; middle: enlarged view of the $[M+H]^+$ signal; bottom: simulated spectrum.

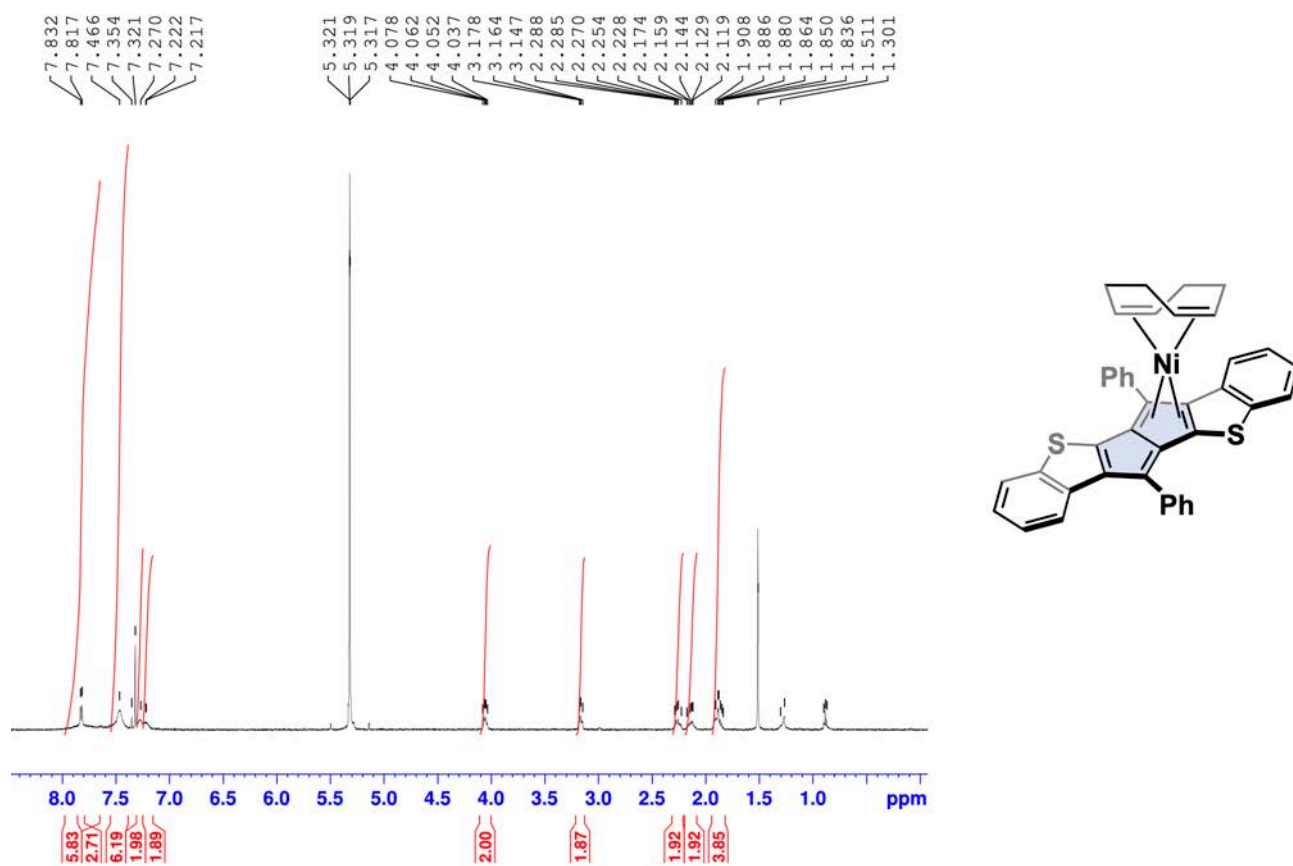


Figure S25. ¹H NMR spectrum of [Ni(cod)(**2**)] (500 MHz, CD₂Cl₂).

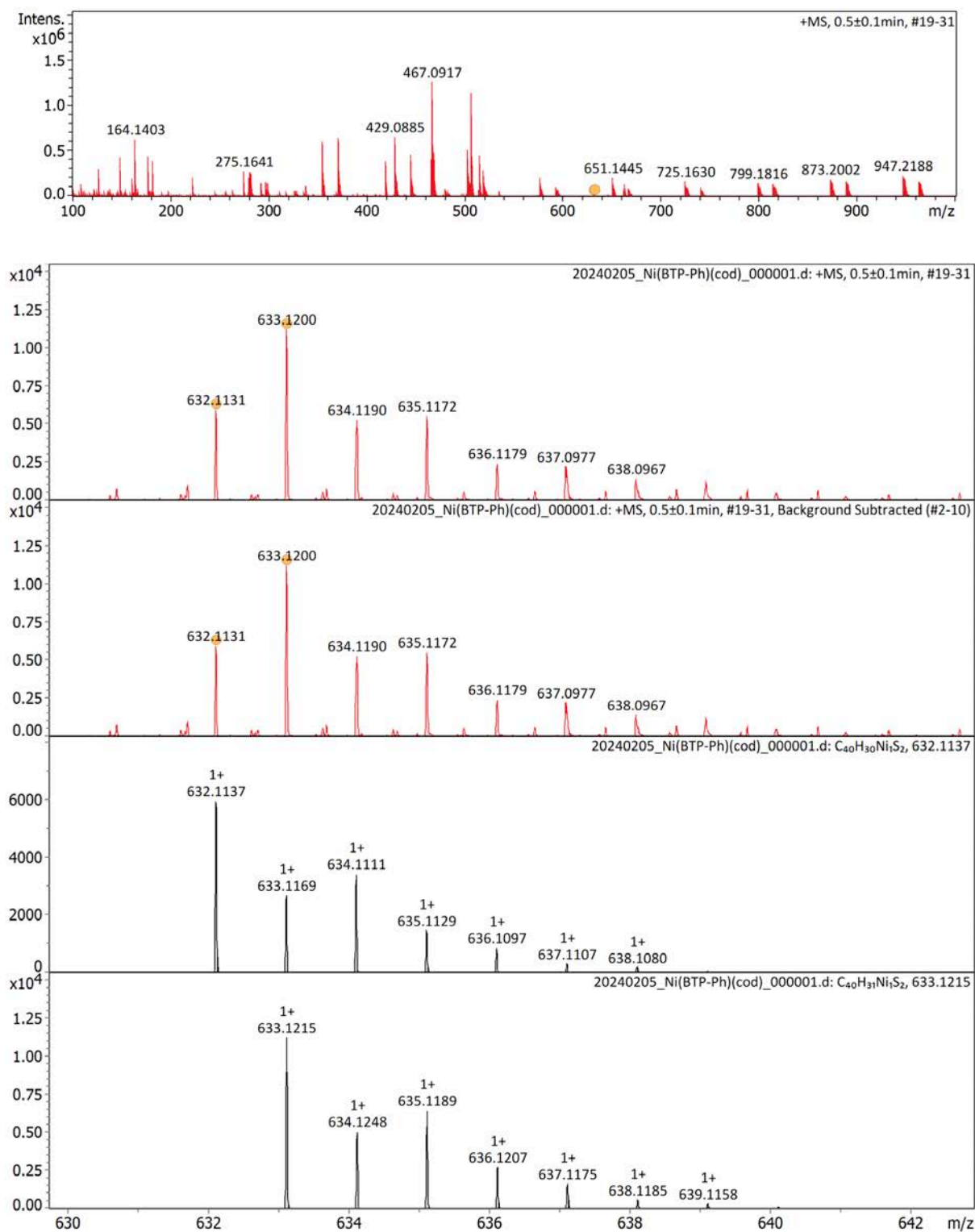


Figure S26. High-resolution mass spectrum of [Ni(cod)(2)] (APCI). Top: full experimental spectrum; middle: enlarged view of the [M]⁺ and [M+H]⁺ signals; bottom: simulated spectrum.

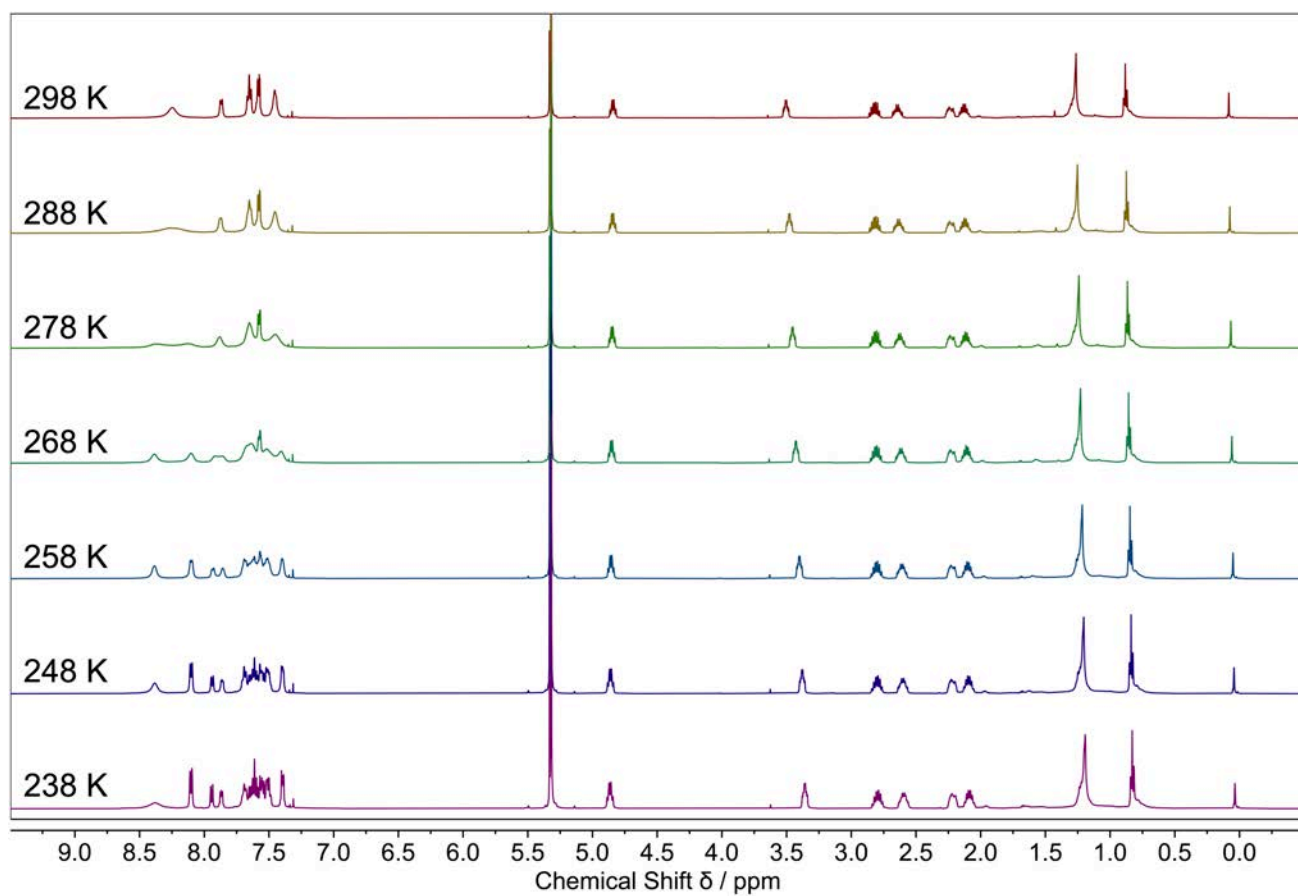


Figure S27. Variable-temperature ^1H NMR spectra of $[\text{Ni}(\text{cod})(1)]$ recorded from 298 to 238 K (500 MHz, in CD_2Cl_2).

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